

A formalized reduction of the Mullin’s Conjecture

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Abstract

The Euclid–Mullin sequence is defined by $a(0) = 2$, $a(n+1) = \text{lpf}(a(0) \cdots a(n) + 1)$, where lpf is the least prime factor. Mullin’s Conjecture (MC, 1963) asserts that every prime eventually appears. We present a Lean 4 formalization ($\sim 78,000$ lines, 148 files, zero sorry) that reduces MC to a single open hypothesis and explores five variant selection rules.

The main result is the machine-verified chain

$$\text{DSL} \implies \text{CME} \implies \text{CCSB} \implies \text{MC},$$

where CME (Conditional Multiplier Equidistribution) states that `minFac` destroys correlation between consecutive multiplier residues mod q , and DSL (Deterministic Stability Lemma) asserts that population-level equidistribution of least prime factors transfers to the specific EM orbit. An inductive bootstrap recasts MC as walk-hitting on $(\mathbb{Z}/q\mathbb{Z})^\times$ with `SubgroupEscape` proved free; a character-analytic bridge handles all character orders; ensemble averaging over squarefree starting points collapses the landscape to DSL.

The factor tree $\mathcal{T}(m)$ —rooted at m , branching at every prime factor of $P+1$ —provides a unifying framework for variants: the standard walk is the leftmost path, and `GenMixedMC(m)` asks that every prime appear somewhere in $\mathcal{T}(m)$. Under PSCD (Population Sieve Confinement Decay), a density-1 set of squarefree starting points satisfies `GenMixedMC`; the full chain $\text{PEAP} \implies \text{PSCD}$ is machine-checked with no residual open hypothesis: a weak unconditional CRT bound ($\text{sqfreeCount} \geq X/4$) absorbs the sieve constant since $\prod(1 - 1/p) \rightarrow 0$. The $\{\text{minFac}, \text{secondMinFac}\}$ two-point variant reduces to a single spectral hypothesis (`UFDSstrong`); for $q = 3$, faithful character escape closes this unconditionally. The function field analog over $\mathbb{F}_p[t]$, where the Weil bound gives population equidistribution for free, confirms that the residual difficulty is orbit-specificity transfer, not analytic number theory. Over 138 dead ends are documented.

Statements and proofs. Theorems marked [✓ name](#) are formally verified in Lean 4; clicking the identifier links to the source code. Most proofs are sketched rather than fully reported; correctness is guaranteed by the type-checker. Some misalignment may exist between what is stated here and what is proved in the formalization; the Lean source is the authoritative reference. To mitigate this risk, we used `LeanArchitect` [24] to cross-check the 20 key declarations against the paper’s theorem statements; this process uncovered one missing precondition, now corrected. All published results and open hypotheses are tagged with `@[publish]` and `@[open_point]` annotations in the GitHub repository (<https://github.com/marcellop71/EM>), using the CA content-addressing framework for Lean 4; a machine-readable registry is available in the `registry/` folder.

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1 Introduction

Euclid’s proposition IX.20 of the Elements shows that for any finite set of (distinct) primes, each prime factor of their product plus one is outside the set [8]: to grow your set of primes, you can pick any factor of their product plus one (a [Substack post I wrote on this](#)). The Euclid–Mullin sequence (OEIS A000945), introduced by Mullin [1], makes a definite choice: always take the smallest prime factor.

$$a(0) = 2, \quad a(n+1) = \text{smallest prime factor of } (a(0) \cdots a(n) + 1). \quad (1)$$

The first twenty terms (0-indexed) are

$$\underbrace{2}_{a(0)}, 3, 7, 43, 13, 53, \underbrace{5}_{a(6)}, \underbrace{6221671}_{a(7)}, 38709183810571, \\ 139, 2801, \underbrace{11}_{a(11)}, 17, 5471, 52662739, 23003, 30693651606209, \\ \underbrace{37}_{a(17)}, 1741, \underbrace{1313797957}_{a(19)}, \dots$$

The sequence shows an erratic behavior: small primes appear out of their natural order (5 not until position 6, 11 at position 11, 37 at position 17), while enormous primes—a 7-digit number at position 7, a 14-digit number at position 8—appear early. As of 2025, only 51 terms are known and some primes like 41 and 47 are not yet observed. Computing further terms requires complete factorization of $P_2(n) + 1$ (where $P_2(n) := a(0) \cdots a(n)$ is the running product): finding any prime factor is not enough—one must certify that no smaller factor exists. Since $P_2(n)$ grows super-exponentially, this quickly exceeds the reach of all known factoring algorithms. By construction, no prime can appear twice.

Conjecture 1.1 (Mullin, 1963). Every prime number eventually appears in the Euclid–Mullin sequence.

The conjecture has resisted proof for over sixty years. The difficulty is showing that the deterministic `minFac` rule eventually selects each prime. Shanks [4] offered a compelling heuristic: projecting the running product onto $(\mathbb{Z}/q\mathbb{Z})^\times$ yields a multiplicative walk, and if the walk behaves pseudo-randomly it must eventually hit $-1 \bmod q$, making q a candidate for selection. Our formalization makes this heuristic precise—CME, CCSB, and DSL (Sections 3–4) are rigorous formulations of “the walk behaves pseudo-randomly.”

That the rule matters is not idle speculation: Cox and van der Poorten [13] showed that replacing `minFac` with the largest prime factor provably misses infinitely many primes.

Each step couples the next prime to the full factorization history, creating a recursive dependency that defeats both probabilistic heuristics and standard sieve methods.

At each step, $P_2(n) + 1$ may have many prime factors, and just the smallest is chosen. So, a target prime q may possibly divide $P_2(n) + 1$ for many (maybe infinite) n , yet never be selected if a smaller prime always divides $P_2(n) + 1$ as well. Our formalization tries to make this tension precise.

The accumulator structure.

The running product $P_2(n)$ is an accumulator: a single number that commits to the entire sequence history via irreversible multiplication, analogous to the cumulative digest in a hash

chain. The accumulator poses a challenge $P_2(n) + 1$; the response $\text{minFac}(P_2(n) + 1)$ extends the chain; and the updated accumulator $P_2(n+1) = P_2(n) \cdot \text{minFac}(P_2(n)+1)$ absorbs the response irreversibly—once a prime enters the product, it divides every future product and can never appear again (Theorem 5.3).

This accumulator coupling is what makes standard tools fail. Sieve methods require approximate independence between the events “ $p \mid m$ ” for different m ; here successive challenges $P_2(n) + 1$ share a cumulative history. Ergodic methods require a fixed or state-dependent map; here the map at step n depends on the full accumulator, not just the current residue. The walk reformulation (Section 2) tames this coupling by projecting the accumulator onto a finite group: $P_{2,q}(n) = P_2(n) \bmod q$ preserves the divisibility information relevant to q while discarding the accumulator’s combinatorial complexity. But the information lost in this projection is exactly the source of difficulty: the walk position determines whether q can divide $P_2(n) + 1$; the full accumulator determines which prime is actually selected. Every open hypothesis in this paper addresses this gap.

The factoring channel.

The accumulator structure of the EM sequence is closely analogous to a cryptographic hash chain: each term is deterministically derived from its predecessor through an operation—integer factorization—that destroys algebraic relationships. In an iterated SHA-256 chain $x_0, H(x_0), H(H(x_0)), \dots$, each value is uniquely determined by the seed, yet consecutive terms pass every reasonable statistical test for independence. In the EM sequence, the running product $P_2(n)$ determines $P_2(n) + 1$, whose smallest prime factor becomes the next multiplier; but the $O(\log q)$ bits visible in the residue $P_2(n) \bmod q$ cannot control the outcome of factoring the $\sim 2^n$ -bit integer $P_2(n) + 1$. This information bottleneck is why the multiplier residues behave as if independent—and why the walk on $(\mathbb{Z}/q\mathbb{Z})^\times$ should visit every element, including the “death state” -1 that would put q in the sequence. The analogy is tighter than it may appear: in both cases the pseudorandomness claim concerns a fully deterministic process whose forward map is easy but whose outputs resist structural prediction—the difference being that minFac achieves this not by cryptographic design but accidentally, through the information bottleneck of projecting a $\sim 2^n$ -bit integer onto $O(\log q)$ bits. Our formalization makes this intuition precise: we identify the exact mathematical content—Conditional Multiplier Equidistribution (CME)—needed to convert “the factoring channel destroys correlation” into a proof of MC, and show it implies the conjecture through a verified chain of reductions.

Our main result is the machine-verified reduction of MC to a single open hypothesis:

Theorem 1.2 ([✓ full_chain_dsl](#)). $\text{DSL} \implies \text{CME} \implies \text{CCSB} \implies \text{MC}$.

The Deterministic Stability Lemma (DSL) is the sole remaining hypothesis. It asserts that population equidistribution of least prime factors among shifted squarefree integers—a standard consequence of Alladi’s theorem [11] and Dirichlet’s theorem on primes in arithmetic progressions—implies conditional multiplier equidistribution (CME) for the specific EM orbit. In symbols, $\text{DSL} = (\text{PE} \implies \text{CME})$, where PE is the population equidistribution that follows from analytic number theory. The entire mathematical content of this paper is the gap between PE (known) and CME (sufficient for MC): does generic equidistribution transfer to one deterministic orbit?

The strategy proceeds in three stages:

1. Reformulation and bootstrap. We recast “does prime q appear?” as “does a multiplicative walk on the cyclic group $(\mathbb{Z}/q\mathbb{Z})^\times$ hit the element -1 ?” This translation is exact. We then show that the algebraic precondition for the walk to reach -1 —that the multipliers generate the full group—is free, following from the inductive hypothesis that all primes below p have already appeared. A single hit on -1 past the stage where all smaller primes are excluded suffices for MC (Section 2).
2. Reduction to DSL. We develop the character-analytic infrastructure to show $\text{CME} \Rightarrow \text{CCSB} \Rightarrow \text{MC}$ (Section 3), then an ensemble averaging framework that collapses the full landscape to DSL (Section 4). Section 5 explains why the remaining hypothesis is hard and identifies the open problems whose resolution would close the conjecture.

The formalization serves two purposes: (i) it guarantees that every reduction is logically sound, and (ii) it precisely delineates the boundary between what is proved and what remains open, preventing the kind of subtle gap that plagues pencil-and-paper reductions involving multiple interacting hypotheses.

The intermediate reductions are:

Theorem 1.3 ([Single Hit Theorem](#) — [✓ single_hit_implies_mc](#)). $\text{SingleHitHypothesis} \Rightarrow \text{MC}$.

$\text{SingleHitHypothesis}$ asks: for every missing prime q , if every prime below q already appears in the sequence and the multiplier residues generate the full group $(\mathbb{Z}/q\mathbb{Z})^\times$ ([SubgroupEscape](#), [Property 2.14](#)), then $q \mid P_2(n) + 1$ for some n past the stage where all smaller primes are excluded. The proof is by strong induction on p : the inductive hypothesis ensures all primes below p have appeared, [PrimeResidueEscape](#) (proved elementarily) bootstraps [SubgroupEscape](#), and a single hit gives $\text{em}_2(n+1) = p$.

The algebraic precondition—that the multipliers generate $(\mathbb{Z}/q\mathbb{Z})^\times$ —is free; the open problem is purely dynamical. Multiple strategies for producing the required hit are formally verified:

Theorem 1.4 ([✓ dynamical_hitting_implies_mullin](#)). DynamicalHitting (DH: the walk hits -1 cofinally) $\Rightarrow \text{MC}$.

Theorem 1.5 ([✓ complex_csb_mc](#)). $\text{ComplexCharSumBound}$ (CCSB: character sums of the walk are $o(N)$) $\Rightarrow \text{MC}$.

Theorem 1.6 ([✓ cme_implies_mc](#)). $\text{CME} \Rightarrow \text{MC}$ (sharpest sufficient condition).

An independent route bypasses CME entirely: [Visit Equidistribution](#) (VE)—the walk visits every residue class with equal frequency—also implies MC through a separate machine-verified chain ([sve_implies_mc](#), [cancel_weyl_implies_mc](#)). VE and CME are genuinely different conditions (neither implies the other), but both reach MC.

All reductions are fully machine-verified with zero [sorry](#). The ensemble averaging framework (Section 4) uses generalized EM sequences parametrized by arbitrary squarefree starting points, CRT-based decorrelation, and a first-moment/variance argument to collapse all routes to DSL. The formalization thus provides a precise “roadmap”: prove DSL (the sole remaining

hypothesis), or equivalently any of DH, CCSB, CME, VE, or the equivalent formulations catalogued in §3, §5, and the glossary (Appendix A.6), and the rest follows by machine-checked deduction.

Visual conventions.

Throughout the paper, the status of each statement is indicated by its left-border color and environment type:

Color	Environment	Status	Meaning
Blue	Theorem / Lemma	[proved]	Machine-checked in Lean 4
Green	Definition	—	Named concept
Pink	Property	[open]	Open hypothesis (not yet proved)
Red	Conjecture	[open]	Open conjecture
Amber	Remark	—	Commentary

Every blue-bordered statement has been verified with zero `sorry` in the formalization. Pink-bordered “Property” statements are the sole open content; each reduction arrow between them is machine-checked. The [✓ identifier](#) links point to the corresponding Lean source.

Organization.

Section 1 analyzes the bag-theoretic and information-theoretic structure of the EM dynamics. The technical development begins in Section 2, which reformulates MC as a walk-hitting problem, establishes the algebraic prerequisites, derives the first missing prime’s death channel avoidance, and proves the Single Hit Theorem via the inductive bootstrap (SubgroupEscape is free), reducing MC to producing one hit at each prime. Section 3 develops the character-analytic reduction (CCSB $\Rightarrow$ MC), including the large sieve infrastructure, the spectral energy bridge, and the van der Corput–autocorrelation route. Section 4 develops the ensemble averaging framework, reducing MC to a single Deterministic Stability Lemma (DSL). Section 5 explains why the remaining hypothesis is difficult by analyzing dead ends and structural barriers, identifies the open problems whose resolution would close the conjecture, and assesses the evidence for and against. Section 6 studies variants of the conjecture: the random-factor variant (easy by mixing), the $\{\text{minFac}, \text{secondMinFac}\}$ two-point variant (almost closed under UFDStrong, §6.4), and the function field analog over $\mathbb{F}_p[t]$ (§6.7). Section A.7 describes the Lean formalization and catalogues all verified results (§A.7.4). Appendix A.1 collects the historical background; Appendix A.2 discusses analogies with Artin’s conjecture, multiplicative walks, Sarnak’s program, and sieve theory; Appendix A.3 presents additional sieve and spectral routes to MC; Appendix A.5 describes the human–AI methodology; and Appendix A.6 provides a glossary of all definitions and hypotheses.

The Bag-Theoretic Structure of the Euclid–Mullin Dynamics

It is interesting to view the EM sequence as a dynamical system on bags of primes. The state of the Euclid–Mullin construction at step n is fully captured by the bag (finite set) of primes collected so far: $S_n = \{\text{em}_2(0), \text{em}_2(1), \dots, \text{em}_2(n)\}$. The dynamics is a deterministic map on finite subsets of primes:

$$S_{n+1} = S_n \cup \{\min \mathcal{F}(S_n)\},$$

where $\mathcal{F}(S) = \text{PrimeFactors}(\prod S + 1)$ produces the factor bag of the Euclid number $E_n = P_2(n) + 1$.

The trajectory is entirely determined by $S_0 = \{2\}$. The Markov property holds: the future

depends only on the current bag, not on the order in which its elements were assembled. The product $P_2(n) = \prod S_n$ is a lossless encoding of the bag (by unique factorization of the squarefree integer $P_2(n)$), and the order of insertion is forgotten.

The pipeline: from bag to next prime.

The selection of the next prime passes through a pipeline of operations, each with distinct information-theoretic character:

$$S_n \xrightarrow[\text{lossless}]{\Pi(\cdot)} P_2(n) \xrightarrow[+1]{\text{shift}} P_2(n) + 1 \xrightarrow[\text{factor}]{} \mathcal{F}(S_n) \xrightarrow[\text{compress}]{\min} \text{em}_2(n+1).$$

Stage 1: Product (lossless). The map $S_n \mapsto P_2(n)$ is an injection (on sets of distinct primes), encoding $\sim n \log n$ bits of bag information into a single integer of $\sim 2^n$ digits. This stage expands the representation while preserving all information.

Stage 2: The $+1$ shift (multiplicative $\rightarrow$ additive). The map $P_2(n) \mapsto P_2(n) + 1$ is the critical bridge between two worlds. The integer $P_2(n)$ lives in the multiplicative world: its structure (as a product of known primes) is completely understood. The integer $P_2(n) + 1$ lives in the additive world: its factorization bears no algebraic relationship to that of $P_2(n)$, beyond the guaranteed coprimality $\gcd(P_2(n), P_2(n) + 1) = 1$. This shift is the source of mixing in the dynamics.

Stage 3: Factorization (revealing the new bag). The factorization $P_2(n) + 1 = p_1^{a_1} \cdots p_k^{a_k}$ reveals the factor bag $\mathcal{F}(S_n) = \{p_1, \dots, p_k\}$. This step is formally lossless, but computationally opaque: predicting $\mathcal{F}(S_n)$ from S_n requires factoring a number of $\sim 2^n$ digits.

Stage 4: Minimum (catastrophic compression). The map $\mathcal{F}(S_n) \mapsto \min \mathcal{F}(S_n)$ discards all but the smallest element of the factor bag. The compression ratio is exponential: the input has $\sim 2^n$ bits; the output has $O(n)$ bits. The composition of stages 2–4 is the minFac bottleneck: an information-destroying channel that maps $\sim 2^n$ bits to $O(n)$ bits through arithmetically chaotic operations.

The orthogonal bag property.

A fundamental structural fact distinguishes the factor bag $\mathcal{F}(S_n)$ from the current bag S_n :

Proposition 1.7 (Orthogonal bags). $\mathcal{F}(S_n) \cap S_n = \emptyset$. That is, the primes dividing $P_2(n) + 1$ are entirely disjoint from the primes in the current bag.

Proof. For any $p \in S_n$: $p \mid P_2(n)$, hence $P_2(n) + 1 \equiv 1 \pmod{p}$, so $p \nmid P_2(n) + 1$, i.e. $p \notin \mathcal{F}(S_n)$. $\square$

This is the Euclidean coprimality at the heart of the construction, but its bag-theoretic formulation reveals its dynamical significance: at each step, the factorization produces a fresh sample from the complementary set $\mathbb{P} \setminus S_n$. The new bag $\mathcal{F}(S_n)$ contains no “recycled” primes—every element is drawn from the universe of primes not yet seen.

Moreover, $P_2(n)$ is squarefree (a product of distinct primes), so S_n is genuinely an unordered set with no multiplicities. The accumulator $P_2(n)$ is the unique squarefree integer with prime support exactly S_n .

The CRT perspective.

The Chinese Remainder Theorem provides the natural coordinate system for the bag dynamics. The integer $P_2(n)$ is determined by its CRT vector:

$$\mathbf{c}(n) = (P_2(n) \bmod 2, P_2(n) \bmod 3, P_2(n) \bmod 5, \dots)$$

indexed by all primes. For primes $p \in S_n$: $P_2(n) \bmod p = 0$ (since $p \mid P_2(n)$). For primes $r \notin S_n$: $P_2(n) \bmod r = \prod_{p \in S_n} (p \bmod r) \in (\mathbb{Z}/r\mathbb{Z})^\times$.

The $+1$ shift maps $\mathbf{c}(n) \mapsto \mathbf{c}(n) + \mathbf{1}$ componentwise. The factorization of $P_2(n) + 1$ is determined by which components satisfy $c_r(n) + 1 \equiv 0 \pmod{r}$, i.e., $c_r(n) = -1$. The minFac is the winner of a race across CRT coordinates: the smallest prime $r \notin S_n$ for which $c_r(n) = -1$. This race depends on all coordinates simultaneously, but each coordinate is determined independently (by CRT) from every other.

Fiber structure and non-reconstructibility.

A natural question is whether the output of minFac carries information about the bag that produced it. The answer is: almost none.

Proposition 1.8 (Massive many-to-one). For any prime q and any set size k , the number of bags T of k distinct primes satisfying $\text{minFac}(\prod T + 1) = q$ grows exponentially in k .

The constraints on such a bag T are purely modular: $\prod T \equiv -1 \pmod{q}$ (one congruence condition), plus $\prod T \not\equiv -1 \pmod{p}$ for each prime $p < q$ with $p \notin T$ (at most $\pi(q)$ non-congruence conditions). The total information content of the output is $O(q)$ bits, while the bag carries $\sim 2^k$ bits. The fraction of information preserved is $O(q/2^k) \rightarrow 0$ super-exponentially. Consequently, the map $S \mapsto \text{minFac}(\prod S + 1)$ is an exponentially lossy summarization of the bag.

MinFac as a greedy algorithm.

The minimum operation in Stage 4 admits a natural optimization interpretation. Among all prime factors of $P_2(n) + 1$, the minFac rule selects the one that increases $\log P_2$ by the least amount:

$$\text{em}_2(n+1) = \arg \min_{p \in \mathcal{F}(S_n)} \log p.$$

This is a greedy algorithm for slow growth: at each step, the accumulator grows as slowly as possible (given the constraint of selecting a prime factor of $P_2(n) + 1$).

Slow growth has a dynamical consequence. Since $P_2(n)$ grows slowly (relative to the maxFac alternative), each target prime q receives more trials—more steps during which $P_2(n) + 1$ might be divisible by q —before the accumulator becomes so large that the probability of q -divisibility becomes negligible. The minFac rule maximizes the number of opportunities each prime has to enter the sequence.

By contrast, the maxFac rule (selecting the largest prime factor) causes $P_2(n)$ to grow so rapidly that small primes quickly lose any chance of being selected, which is why the second Euclid–Mullin sequence provably misses infinitely many primes (Cox–van der Poorten [13], Booker [3]).

The 0-accessibility of minFac.

We formalize the advantage of minFac over other selection rules. Define a selection rule $\sigma: 2^{\mathbb{P}} \rightarrow \mathbb{P}$ mapping a nonempty set of primes to one of its elements. Say σ is 0-accessible for a prime q past the sieve gap if: whenever $q \in F$ and all primes $p < q$ have been screened, $\sigma(F) = q$.

MinFac is 0-accessible for every q : past the sieve gap, all primes below q are excluded from $\mathcal{F}(S_n)$, so if $q \in \mathcal{F}(S_n)$, then $q = \min \mathcal{F}(S_n)$. The event “ q divides $P_2(n) + 1$ ” is sufficient for q to enter the sequence.

MaxFac is not 0-accessible: even if $q \mid P_2(n) + 1$, a larger factor is selected instead, and the probability that q is the largest factor of a number of size $\sim 2^{2^n}$ decays faster than any polynomial.

The 0-accessibility of minFac ensures that the walk mod q hitting -1 is both necessary and sufficient for q to enter the sequence (past the sieve gap). This reduces the question of whether q appears to a question about the walk alone, with no additional selection barrier.

Deterministic pseudorandomness.

The EM sequence is fully deterministic, yet exhibits pseudorandom behavior. The pseudorandomness arises from the composition of two sources of arithmetic opacity:

1. The $+1$ shift moves from the multiplicative world (where $P_2(n)$ is fully understood) to the additive world (where $P_2(n) + 1$ is arithmetically opaque). The relationship between the multiplicative structures of consecutive integers is the central mystery of analytic number theory.
2. The factoring step reveals the prime structure of $P_2(n) + 1$, but this structure is computationally hard to predict from $P_2(n)$. The computational difficulty of factoring is, in a precise sense, the source of pseudorandomness for the EM sequence.

Small changes to the bag (adding one prime p) transform $P_2(n)$ to $P_2(n) \cdot p$ and hence $P_2(n) + 1$ to $P_2(n) \cdot p + 1$, a completely different integer with a completely different factorization. The dynamics is sensitive to the bag contents, analogous to sensitive dependence on initial conditions in chaotic systems, but operating through number-theoretic rather than geometric mechanisms.

The orthogonal bag property reinforces this pseudorandomness: at each step, the factor bag is drawn from $\mathbb{P} \setminus S_n$, a set that has never been “used” in the construction. Each step provides a genuinely fresh arithmetic input, preventing the buildup of systematic correlations.

2 The Residue Walk

For any starting point $s \geq 2$, the generalized EM sequence and running product are defined by mutual recursion:

$$P_s(0) = s, \quad \text{em}_s(k) = \text{minFac}(P_s(k) + 1), \quad P_s(k+1) = P_s(k) \cdot \text{em}_s(k). \quad (2)$$

The standard Euclid–Mullin sequence is the case $s = 2$ ([✓ genProd_two_eq_prod](#)). We use subscripted notation throughout: em_s , P_s denote the sequence and accumulator from starting point s .

```
-- Core EM sequence (mutual recursion via aux : Nat -> Nat x Nat)
def aux : Nat -> Nat x Nat
```

```

| 0      => (2, 2)
| n + 1 => let (_, p) := aux n
           let f := Nat.minFac (p + 1)
           (f, p * f)

def seq (n : Nat) : Nat := (aux n).1 -- n-th prime
def prod (n : Nat) : Nat := (aux n).2 -- running product

```

Theorem 2.1 ([✓ genSeq_prime](#), [✓ genSeq_injective](#)). For any $s \geq 2$, every $\text{em}_s(k)$ is prime. For squarefree s , the map $k \mapsto \text{em}_s(k)$ is injective: no prime appears twice.

Further structural properties, all fully proved:

- Divisibility: $s \mid P_s(k)$ for all k ([✓ start_dvd_genProd](#)).
- Strict monotonicity: $P_s(k) < P_s(k+1)$ ([✓ genProd_strict_mono](#)).
- Squarefreeness: for squarefree s , $P_s(k)$ is squarefree for all k ([✓ genProd_squarefree](#)).

Theorem 2.2 ([Tail identification](#) — [✓ genProd_restart](#)). Restarting the EM process from the M -th accumulator of orbit s gives the tail of that orbit:

$$P_{P_s(M)}(k) = P_s(M+k), \quad \text{em}_{P_s(M)}(k) = \text{em}_s(M+k).$$

Proof. By induction on k . The base case $k = 0$ is immediate from the definition $P_n(0) = n$. For the step, $P_{P_s(M)}(k+1) = P_{P_s(M)}(k) \cdot \text{minFac}(P_{P_s(M)}(k) + 1)$; the inductive hypothesis replaces both occurrences by $P_s(M+k)$, giving $P_s(M+k) \cdot \text{minFac}(P_s(M+k) + 1) = P_s(M+k+1)$. The sequence identity follows by definition. $\square$

The tail identification says that the family of generalized EM orbits is closed under “time-shifting”: the orbit starting from any intermediate accumulator is not a new object but a suffix of the original orbit.

Definition 2.3 ([Walk](#) and [Multiplier](#)). For any prime q , the residue walk $P_{s,q}(\cdot)$ and the multiplier $\text{m}_{s,q}(\cdot)$ project the accumulator and the next prime onto $\mathbb{Z}/q\mathbb{Z}$:

$$\begin{aligned}
P_{s,q}(n) &:= P_s(n) \bmod q, \\
\text{m}_{s,q}(n) &:= \text{em}_s(n+1) \bmod q.
\end{aligned}$$

The first subscript is the starting point; the second is the modulus.

Proposition 2.4 ([Walk recurrence](#) — [✓ walkZ_succ](#), [✓ genWalkZ_succ](#)). $P_{s,q}(n+1) = P_{s,q}(n) \cdot \text{m}_{s,q}(n)$ in $\mathbb{Z}/q\mathbb{Z}$, for any starting point s .

Theorem 2.5 ([Walk–Divisibility Bridge](#) — [✓ genWalkZ_eq_neg_one_iff](#)). For any starting point s and prime q :

$$P_{s,q}(k) = -1 \iff q \mid P_s(k) + 1.$$

```

-- The walk and bridge in Lean 4
def walkZ (q : Nat) (n : Nat) : ZMod q := (prod n : ZMod q)
def multZ (q : Nat) (n : Nat) : ZMod q := (seq (n+1) : ZMod q)

-- The bridge: walk hits -1 iff q divides prod(k)+1
theorem genWalkZ_eq_neg_one_iff (hq : Nat.Prime q) :
  genWalkZ s q k = -1 <-> q ∣ genProd s k + 1

```

The walk can be in one of two regimes:

1. Living regime. If $q \nmid s$ and q has not yet appeared in $\mathbf{em}_s(0), \dots, \mathbf{em}_s(n-1)$, then $q \nmid P_s(n)$ (since $P_s(n) = s \cdot \mathbf{em}_s(0) \cdots \mathbf{em}_s(n-1)$ and q divides none of these factors). So $P_{s,q}(n) \in (\mathbb{Z}/q\mathbb{Z})^\times$ —the walk lives in the group of units.
2. Dead regime. If $q = \mathbf{em}_s(k)$ for some k , then $q \mid P_s(m)$ for all $m > k$, so $P_{s,q}(m) = 0$ forever. The walk has collapsed to zero ([✓ walkZ_capture_then_collapse](#), proved for $s = 2$).

The transition from living to dead can happen only through -1 (Theorem 2.5).

Hitting -1 is a necessary condition for the walk to die (for q to enter the sequence): if $q \mid P_s(n) + 1$, then $P_{s,q}(n) = -1$. But it is not sufficient by itself. When $P_{s,q}(n) = -1$, the walk dies only if $q = \mathbf{minFac}(P_s(n) + 1)$, i.e. no smaller prime divides $P_s(n) + 1$. If a smaller prime p also divides $P_s(n) + 1$, then $\mathbf{minFac}(P_s(n) + 1) \leq p < q$ and q is not selected—the walk bounces off -1 and continues living. The walk may therefore hit -1 multiple times without dying. Death occurs only at a hit where q is the smallest available factor:

$$\cdots \rightarrow \underbrace{P_{s,q}(n) = -1}_{\text{bounce}} \rightarrow (\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \cdots \rightarrow \underbrace{P_{s,q}(n_0) = -1}_{\substack{\text{lethal hit:} \\ q = \mathbf{minFac}}} \rightarrow \underbrace{P_{s,q}(n_0+1) = 0}_{\text{dead}} \rightarrow 0 \rightarrow \cdots$$

What determines whether a hit on -1 is a bounce or a lethal hit? The answer depends on which other primes divide $P_s(n) + 1$. We now specialize to the standard orbit $s = 2$, where the sieve gap provides the answer.

Definition 2.6 ($\mathbf{MC}(< q)$). For a prime q , write $\mathbf{MC}(< q)$ for the statement that every prime smaller than q appears in the EM sequence: $\forall p < q, p \text{ prime} \Rightarrow \exists k, \mathbf{em}_2(k) = p$.

When $\mathbf{MC}(< q)$ holds, every prime $p < q$ has entered the sequence at some stage k_p , so $p \mid P_2(n)$ for all $n \geq k_p$. Past a uniform stage $N_0 = \max_p k_p$, every prime $p < q$ divides $P_2(n)$, and therefore cannot divide $P_2(n) + 1$ (since $\gcd(P_2(n), P_2(n) + 1) = 1$). This is the sieve gap: past N_0 , the only primes that can divide $P_2(n) + 1$ are $\geq q$.

Theorem 2.7 (q -roughness — [✓ mcBelow_implies_seq_ge](#)). If $\mathbf{MC}(< q)$ holds, then $\exists N_0$ such that $\mathbf{em}_2(n+1) \geq q$ for all $n \geq N_0$.

The sieve gap transforms hitting -1 from a necessary condition into a sufficient one:

Theorem 2.8 (Lethal hit — [✓ mcBelow_hit_is_lethal](#)). If $\mathbf{MC}(< q)$ holds and $P_{2,q}(n) = -1$ for some n past the sieve gap, then $\mathbf{em}_2(n+1) = q$.

Proof. Since n is past the sieve gap, no prime $< q$ divides $P_2(n) + 1$ (Theorem 2.7). But $P_{2,q}(n) = -1$ gives $q \mid P_2(n) + 1$, so $q = \text{minFac}(P_2(n) + 1)$, and $\text{em}_2(n+1) = q$. $\square$

Definition 2.9 (Missing prime). A prime q is missing if $\text{em}_2(n) \neq q$ for all n .

Mullin's Conjecture asserts that no prime is missing. Under $\text{MC}(< q)$, the first hit on -1 past the sieve gap is lethal. This immediately constrains missing primes:

Theorem 2.10 ([✓ mcBelow_missing_walk_ne_neg_one](#)). If $\text{MC}(< q)$ holds and q is missing, then the walk never hits -1 past the sieve gap: $\exists N_0$ such that $P_{2,q}(n) \neq -1$ for all $n \geq N_0$.

Proof. Any hit past the sieve gap would give $\text{em}_2(n+1) = q$ (Theorem 2.8), contradicting q missing. $\square$

For the first missing prime q , the hypothesis $\text{MC}(< q)$ holds by definition. So its walk avoids -1 permanently past the sieve gap, the walk $P_{2,q}(\cdot)$ is infinite: it stays in $(\mathbb{Z}/q\mathbb{Z})^\times$ forever, never collapsing to zero.

The forbidden multiplier and the death channel

The following algebraic observations hold for every starting point s . Suppose the walk is at position $c \in (\mathbb{Z}/q\mathbb{Z})^\times$ at step n . The walk would hit -1 at step $n+1$ if and only if

$$P_{s,q}(n) \cdot m_{s,q}(n) = -1, \quad \text{i.e.,} \quad m_{s,q}(n) = -c^{-1}.$$

So at every step, there is exactly one forbidden multiplier $f(n) = -P_{s,q}(n)^{-1}$ —the unique residue class that would send the walk to -1 . This is 1 element out of $q-1$ possible unit residues.

Definition 2.11 ([Death channel](#)). The death channel at position c is the residue class $b(c) = -c^{-1} \pmod q$.

When the walk is at c , the death channel is the set of primes $p \equiv -c^{-1} \pmod q$, which by Dirichlet's theorem has density $1/(q-1)$ among all primes. The death channel moves with the walk: as c changes from step to step, the forbidden class changes with it. The map $c \mapsto -c^{-1}$ is a bijection on $(\mathbb{Z}/q\mathbb{Z})^\times$ (it is the composition of inversion and negation).

Theorem 2.12 ([✓ walk_hits_neg_one_iff_mult_eq_forbidden](#)). For any starting point s and step n with $P_{s,q}(n) \in (\mathbb{Z}/q\mathbb{Z})^\times$:

$$P_{s,q}(n+1) = -1 \iff m_{s,q}(n) = -P_{s,q}(n)^{-1}.$$

In any orbit where q never appears, the multiplier avoids the death channel at every step, forever. This brings us to discuss multipliers.

2.1 The Confinement Theorem and SubgroupEscape

Before asking whether the walk hits -1 , we must ask whether it can. The walk is multiplicative: $P_{s,q}(n) = P_{s,q}(0) \cdot \prod_{k < n} m_{s,q}(k)$. The reachable set is the coset $P_{s,q}(0) \cdot \langle \text{multipliers} \rangle$.

Theorem 2.13 (**Confinement** — [✓confinement_forward](#)). For any starting point s : if every multiplier lies in a subgroup $H \leq (\mathbb{Z}/q\mathbb{Z})^\times$, then the walk is confined to the coset $P_{s,q}(0) \cdot H$. In particular, if $-1 \notin P_{s,q}(0) \cdot H$, the walk never hits -1 .

This motivates:

Property 2.14 (**SubgroupEscape** (**SE**)). For a starting point s and prime q : no proper subgroup $H < (\mathbb{Z}/q\mathbb{Z})^\times$ contains all multipliers. Equivalently, $\langle m_{s,q}(n) : n \in \mathbb{N} \rangle = (\mathbb{Z}/q\mathbb{Z})^\times$.

Theorem 2.15 ([✓se_of_maximal_escape](#)). SE holds iff multipliers escape every maximal proper subgroup. Since $(\mathbb{Z}/q\mathbb{Z})^\times$ is cyclic, the maximal subgroups are the index- ℓ subgroups for prime $\ell \mid q-1$.

When SE holds, the walk is not confined to any proper coset: every element of $(\mathbb{Z}/q\mathbb{Z})^\times$, including -1 , is reachable. But reachability does not imply hitting—that is a dynamical question.

2.2 The first missing prime

We now assemble the key argument. Suppose, toward contradiction, that Mullin's Conjecture is false. Then there exists a first missing prime: the smallest prime q that never appears in the EM sequence. For this q :

- (a) $MC(< q)$ holds—by definition, every prime smaller than q is in the sequence.
- (b) The walk $P_{2,q}(\cdot)$ is infinite—since q is missing, $P_{2,q}(n) \in (\mathbb{Z}/q\mathbb{Z})^\times$ for all n .
- (c) Past the sieve gap, the walk never hits -1 . Here is why. Eventually (say for $n \geq N_0$), all primes $< q$ have entered the sequence and divide $P_2(n)$. For $n \geq N_0$, no prime $< q$ can divide $P_2(n) + 1$ (since it divides $P_2(n)$ and $\gcd(P_2(n), P_2(n) + 1) = 1$). So if $P_{2,q}(n) = -1$ for some $n \geq N_0$, then $q \mid P_2(n) + 1$, and q is the smallest prime factor of $P_2(n) + 1$ (all smaller primes are excluded). Then $\text{em}_2(n+1) = q$, contradicting q being missing.

Therefore: for the first missing prime q , the walk is an infinite trajectory on $(\mathbb{Z}/q\mathbb{Z})^\times$ that avoids the death channel at every step past the sieve gap. Equivalently: $m_{2,q}(n) \neq -P_{2,q}(n)^{-1}$ for all $n \geq N_0$.

2.3 The inductive bootstrap

The walk avoids -1 forever. Can it at least reach -1 ? That is: does SE hold?

Theorem 2.16 (**PrimeResidueEscape** (**PRE**); [✓prime_residue_escape](#)). For every prime $p \geq 5$ and every proper subgroup $H < (\mathbb{Z}/p\mathbb{Z})^\times$, some odd prime $r < p$ has residue $r \bmod p \notin H$.

Proof. Suppose every odd prime $r \in [3, p)$ satisfies $r \bmod p \in H$. Since H is a subgroup, every product of such primes is in H . Every odd number in $[1, p)$ factors into odd primes $< p$, so every odd number in $[1, p)$ maps into H . In particular, $p-2 \equiv -2$ and $p-4 \equiv -4$ are both in H (both odd and $< p$ for $p \geq 5$). Then $2 = (-4)(-2)^{-1} \in H$, so every even number in $[1, p)$ is in H as well. Hence $H = (\mathbb{Z}/p\mathbb{Z})^\times$, contradicting H proper. $\square$

```

-- PRE: every proper subgroup misses some small prime
def PrimeResidueEscape : Prop :=
  forall (p : Nat) [Fact (Nat.Prime p)], 5 ≤ p ->
    forall (H : Subgroup (ZMod p)^x), H ≠ top ->
      exists (r : Nat), Nat.Prime r ∧ r < p ∧ 3 ≤ r ∧
        Units.mk0 ((r : Nat) : ZMod p) ( ... ) not_in H

theorem prime_residue_escape : PrimeResidueEscape := by
  -- Elementary: all odd n < p are in H => 2 = (-4)(-2)^{-1} in H
  -- => H = top, contradiction
  ...

```

Theorem 2.17 ([✓ mcBelow_pre_implies_se](#)). $\text{MC}(< p) + \text{PRE} \implies \text{SE}(p)$.

Proof sketch. Let $H < (\mathbb{Z}/p\mathbb{Z})^\times$ be proper. By PRE, some odd prime $r < p$ has $r \bmod p \notin H$. By $\text{MC}(< p)$, the prime r appears as $\text{em}_2(k)$ for some k . Then $\text{m}_{2,p}(k-1) \equiv r \pmod{p} \notin H$. $\square$

Corollary 2.18. For the first missing prime q : the multipliers generate all of $(\mathbb{Z}/q\mathbb{Z})^\times$. SubgroupEscape holds, and -1 is reachable.

The one-prime gap (Theorem 2.29) shows that $\text{MC}(< q)$ plus a single divisibility event $q \mid P_2(n) + 1$ past the sieve gap gives $\text{MC}(q)$. The bootstrap (Theorem 2.17) shows that $\text{MC}(< q)$ gives $\text{SE}(q)$ for free. These two facts compose into the primary reduction of Mullin’s Conjecture.

Property 2.19 ([SingleHitHypothesis \(SHH\)](#)). For every prime q : if $\text{MC}(< q)$ and $\text{SE}(q)$ hold and q is missing, then there exists n past the sieve gap with $q \mid P_2(n) + 1$.

Equivalently: for every prime q , if $\text{MC}(< q)$ and $\text{SE}(q)$ hold, then either q already appears in the sequence, or there exists $n \geq N_0(q)$ with $P_{2,q}(n) = -1$.

Theorem 2.20 ([Single Hit Theorem](#) — [✓ single_hit_implies_mc](#)). $\text{SingleHitHypothesis} \implies \text{MC}$.

Proof. By strong induction on p . Assume $\text{MC}(< p)$.

1. Bootstrap gives SE. $\text{MC}(< p) + \text{PRE} \Rightarrow \text{SE}(p)$ (Theorem 2.17). Since PRE is proved unconditionally (Theorem 2.16), we obtain $\text{SE}(p)$.
2. SHH gives a hit. Since $\text{MC}(< p)$ and $\text{SE}(p)$ hold, SHH provides $n \geq N_0$ (past the sieve gap) with $p \mid P_2(n) + 1$.
3. The sieve gap closes. Past N_0 , all primes $< p$ divide $P_2(n)$ and hence cannot divide $P_2(n) + 1$. So $p = \text{minFac}(P_2(n) + 1)$, giving $\text{em}_2(n+1) = p$.

$\square$

2.4 The death channel avoidance paradox

We now have a precise and sharp picture of the first missing prime q :

1. The walk lives in $(\mathbb{Z}/q\mathbb{Z})^\times$ forever.

2. The multipliers generate the full group— -1 is reachable.
3. The walk never reaches -1 —the death channel is avoided at every step past the sieve gap.
4. At each step, the death channel is a single residue class out of $q - 1$ —a “target” of density $1/(q - 1)$.

Mullin’s Conjecture is therefore equivalent to: this situation is impossible. The open problem is now:

Does the multiplicative walk on $(\mathbb{Z}/q\mathbb{Z})^\times$ defined by the EM sequence, whose multipliers generate the full group, hit -1 at least once past the sieve gap?

“At least once past the sieve gap” is the precise requirement. Not cofinally, not equidistributed—once. The intuition for impossibility rests on an information-theoretic asymmetry. At step n , the death channel $f(n) = -P_{2,q}(n)^{-1}$ is determined by the walk position $P_{2,q}(n) = P_2(n) \bmod q$, which carries $O(\log q)$ bits of information. The multiplier $m_{2,q}(n) = \text{minFac}(P_2(n) + 1) \bmod q$ is determined by the full integer $P_2(n) + 1$, which has $\sim 2^n$ bits. The factoring operation minFac extracts global arithmetic information from all $\sim 2^n$ bits; the death channel is determined by $O(\log q)$ bits. For the multiplier to systematically avoid the death channel, the $O(\log q)$ -bit residue would have to predict the outcome of an operation on a 2^n -bit integer—a “prediction” whose information content vanishes exponentially.

More precisely: for a generic q -rough integer $N \equiv a \pmod{q}$, the conditional distribution of $\text{minFac}(N) \bmod q$ is asymptotically uniform over $(\mathbb{Z}/q\mathbb{Z})^\times$, by CRT. Knowing $N \bmod q$ does not constrain $N \bmod p$ for any other prime p , so it does not constrain which primes $\leq N^{1/2}$ divide N . The density of the death channel among all primes is $1/(q - 1)$, and the conditional probability that $\text{minFac}(N)$ falls in the death channel is $1/(q - 1) + o(1)$ as $N \rightarrow \infty$, independently of the residue class a .

For the EM sequence, $P_2(n) + 1$ grows super-exponentially ($P_2(n) + 1 \geq 2^{2^n}$), so the $o(1)$ error at each step shrinks exponentially fast. The “probability” of avoiding the death channel for N consecutive steps is heuristically $(1 - 1/(q - 1))^N \rightarrow 0$. Converting this heuristic into a proof is the content of the conjecture.

2.5 Sufficient conditions for the single hit

Several formally verified strategies produce the required single hit past the sieve gap. Each implies MC via the Single Hit Theorem.

Property 2.21 ([DynamicalHitting \(DH\)](#)). For every missing prime q : $\text{SE}(q) \Rightarrow \text{HH}(q)$, where $\text{HH}(q)$ ([HittingHypothesis](#)) asks for cofinal hitting: $\forall N, \exists n \geq N, q \mid (P_2(n) + 1)$.

Theorem 2.22 ([dynamical_hitting_implies_mullin](#)). $\text{DH} \implies \text{MC}$.

DH is stronger than SHH: it does not assume $\text{MC}(< q)$ and asks for infinitely many hits rather than one. But the extra strength is convenient—DH interfaces cleanly with character sum methods. In the death channel language, DH asserts: if the multipliers generate $(\mathbb{Z}/q\mathbb{Z})^\times$, then the multiplier cannot avoid the forbidden residue $-P_{2,q}(n)^{-1}$ forever. Equivalently: there is no infinite walk on a cyclic group, with a generating set of multipliers, that dodges a single moving target of density $1/(q - 1)$ at every step.

The key structural point: SHH is strictly weaker than DynamicalHitting in two ways. First, SHH assumes $\text{MC}(< q)$, which DH does not (DH only assumes SE). Second, SHH asks for one hit past the sieve gap, while DH asks for cofinal hitting. Both extra assumptions are harmless in the inductive proof— $\text{MC}(< q)$ is always available at the inductive step, and one hit is all the Single Hit Theorem needs—but they make SHH genuinely easier to satisfy as a mathematical statement.

Remark 2.23. The logical relationships among the hitting hypotheses are:

$$\text{HH} \implies \text{DH} \implies \text{SHH},$$

where HH on its own denotes the unconditional version of $\text{HH}(q)$ from Definition 2.21: cofinal hitting asserted for every missing q without assuming $\text{SE}(q)$. DH conditions cofinal hitting on SE, and SHH asks for a single hit past the sieve gap given both $\text{MC}(< q)$ and SE. All three imply MC. SHH is the weakest: it assumes the most ($\text{MC}(< q)$ and SE, both provided free by the inductive bootstrap) and demands the least (one hit, not infinitely many). The converses need not hold.

Property 2.24 ([ComplexCharSumBound \(CCSB\)](#)). For every missing prime q and every non-trivial character χ : $\|S_\chi(N)\| = o(N)$, where $S_\chi(N) = \sum_{n < N} \chi(P_{2,q}(n))$.

Theorem 2.25 ([✓ complex_csb_mc](#)). $\text{CCSB} \implies \text{MC}$.

CCSB produces the hit via Fourier inversion: if all non-trivial character sums are $o(N)$, the hit count at -1 is $N/(q-1) + o(N)$, which is eventually positive. One hit past the sieve gap is lethal (Section 3).

Property 2.26 ([ConditionalMultiplierEquidist \(CME\)](#)). For every missing prime q , every non-trivial χ , and every walk position c : the multiplier characters $\chi(\mathbf{m}_{2,q}(n))$ conditioned on $P_{2,q}(n) = c$ are equidistributed (their partial sums are $o(N)$).

Theorem 2.27 ([✓ cme_implies_mc](#)). $\text{CME} \implies \text{MC}$ (sharpest sufficient condition).

CME is the sharpest hypothesis: it asserts that the factoring operation destroys correlation between consecutive multiplier residues conditioned on the walk position. It implies CCSB for all character orders, bypassing the $d \geq 3$ barrier that blocks the BRE route.

Theorem 2.28 ([✓ vcb_ped_implies_mc](#)). $\text{VCB} + \text{PED} \implies \text{MC}$.

VCB (Vanishing Conditional Bias) is a weakening of CME that permits fiber sums to be proportional to visit counts rather than small; combined with PED (Positive Escape Density), it reaches CCSB.

Each of these strategies produces at least one hit past the sieve gap, which the Single Hit Theorem (Theorem 2.20) converts into MC.

Why it's hard: the walk controls nothing

The difficulty is entirely in the coupling between walk position and multiplier. At step n :

- The walk position $P_{2,q}(n) = P_2(n) \bmod q$ determines the death channel $f(n) = -P_{2,q}(n)^{-1}$.
- The multiplier $\mathbf{m}_{2,q}(n) = \text{minFac}(P_2(n) + 1) \bmod q$ is what must fall into the death

channel.

- Both depend on $P_2(n)$: the walk position depends on $P_2(n) \bmod q$, and the multiplier depends on $P_2(n) + 1$ as a full integer.

The walk position sees $O(\log q)$ bits; the multiplier sees all $\sim 2^n$ bits. But they share the same underlying object $P_2(n)$. The question is whether this shared dependence creates a correlation strong enough for the multiplier to systematically avoid one residue class out of $q - 1$.

For a random sequence of multipliers drawn uniformly from $(\mathbb{Z}/q\mathbb{Z})^\times$, the probability of avoiding the death channel for N steps is $(1 - 1/(q - 1))^N \rightarrow 0$. The EM walk is not random—it is deterministic—but the factoring operation that determines each multiplier is, heuristically, a decorrelating “hash.” This is the factorization independence heuristic: the function `minFac`, applied to a 2^n -bit integer, produces an output that is effectively uncorrelated with the $O(\log q)$ -bit residue class of the input.

Every open hypothesis in this paper—DH, CCSB, CME—is a precise formalization of this heuristic.

2.6 Supporting infrastructure

The remainder of this section collects the supporting results that flesh out the bootstrap: the one-prime gap, the sieve gap, and the power residue decomposition.

The Sieve Gap and the One-Prime Gap

The q -roughness theorem (Theorem 2.7) resolves the selectability problem described in Section 1: past the sieve gap, q is the smallest available factor whenever it divides the Euclid number. This gives the one-prime gap:

Theorem 2.29 ([One-prime gap](#) — [✓ mcBelow_cofinal_hit_implies_mc_at](#)). $\text{MC}(< q)$ plus a single hitting event $q \mid P_2(n) + 1$ for some n past the sieve gap implies $\text{MC}(q)$.

The Power Residue Decomposition

Since $(\mathbb{Z}/q\mathbb{Z})^\times$ is cyclic of order $q - 1$, its subgroup lattice is determined by the prime factorization of $q - 1$. A multiplier set generates the full group if and only if it escapes every maximal subgroup—and the maximal subgroups correspond to the prime divisors ℓ of $q - 1$. This decomposition converts SE into independent conditions, one per prime $\ell \mid q - 1$.

Property 2.30 ([PowerResidueEscape](#) (PRE_ℓ)). $\exists n : m_{2,q}(n)^{(q-1)/\ell} \neq 1$ in $(\mathbb{Z}/q\mathbb{Z})^\times$.

Theorem 2.31 ([PRE](#) $\Leftrightarrow$ [SE](#) — [✓ pre_iff_se](#)). $\text{PRE} \Leftrightarrow \text{SE}$. The forward direction uses only Lagrange’s theorem; the reverse uses cyclicity of $(\mathbb{Z}/q\mathbb{Z})^\times$.

Quadratic Reciprocity Obstruction

The power residue decomposition raises a natural question: for how many primes q could SE actually fail? Since each PRE_ℓ condition asks only that one multiplier among infinitely many escapes the ℓ -th power subgroup, SE failure requires all multipliers to be ℓ -th power residues for some $\ell \mid q - 1$. By CRT and quadratic reciprocity, the density of primes q for which the first k multiplier primes all fail to escape the index-2 subgroup is at most $O(2^{-k})$. Extending this to a rigorous density-zero statement for SE failure would require controlling all prime divisors $\ell \mid q - 1$ simultaneously, which remains open.

3 The Character Sum Reduction

Section 2 established the picture: for the first missing prime q , the walk lives in $(\mathbb{Z}/q\mathbb{Z})^\times$ forever, the multipliers generate the full group, yet the walk avoids -1 permanently past the sieve gap. The death channel $f(n) = -P_{2,q}(n)^{-1}$ is dodged at every step. The Single Hit Theorem (Theorem 2.20) reduces MC to producing one hit on -1 past the sieve gap at each prime.

The question is not “does the walk equidistribute?”—that is far stronger than needed. The question is: can the walk really avoid one class out of $q-1$ forever? This section develops the harmonic-analytic tools to show it cannot: character sums detect the anomaly that permanent avoidance would create, and bounding them rules it out.

Why character sums?

Permanent avoidance of -1 means the hit count $|\{n < N : P_{2,q}(n) = -1\}|$ stays at zero past the sieve gap. Character orthogonality decomposes this count into a uniform share $N/(q-1)$ plus correction terms built from non-trivial character sums $S_\chi(N) = \sum_{n < N} \chi(P_{2,q}(n))$. The uniform share grows linearly; for the hit count to remain zero, the correction terms must cancel this growth. That requires at least one non-trivial character sum to be $\Omega(N/(q-1))$ —a sustained asymmetry in the character spectrum. The hypothesis CCSB (all $|S_\chi(N)| = o(N)$) rules this out: it forces the hit count to be eventually positive, contradicting permanent avoidance.

What is new vs. what is infrastructure. The CCSB $\Rightarrow$ MC reduction (Definition 3.2), the Fourier bridge (Theorem 3.3), the Decorrelation–PED chain (§3.3), and the telescoping no-go results (§3.5) are original contributions of this formalization. The large sieve infrastructure (§3.5, Appendix A.3)—including the weak ALS, Gauss sum inversion, van der Corput, and Parseval—formalizes known results; its purpose is to identify the precise transfer gap between classical tools and the EM orbit, which is itself a contribution (see §3.5).

The formalization develops this Fourier-analytic reduction because character sums provide the cleanest interface between the death-channel avoidance problem and the toolkit of analytic number theory: the Bombieri–Vinogradov theorem, the large sieve inequality, and Gauss sum inversion all produce character sum bounds, and the formalization shows exactly how each connects to ruling out permanent avoidance.

Section roadmap

- §4.1 Character sums and permanent avoidance — orthogonality detects the anomaly; CCSB defined.
- §4.2 The Fourier bridge — hit count = uniform share + character corrections; CCSB $\Rightarrow$ hits.
- §4.3 Decorrelation–PED–CCSB chain — conditional multiplier equidistribution implies character cancellation.
- §4.4 Vanishing conditional bias — walk telescoping reduces to one-step conditional bias.
- §4.5 Walk telescoping identities — structural identities for character energy.

3.1 Character Sums and Permanent Avoidance

For a Dirichlet character $\chi : (\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$, the walk character sum is

$$S_\chi(N) = \sum_{n < N} \chi(P_{2,q}(n)).$$

Property 3.1 ([ComplexCharSumBound \(CCSB\)](#)). For every missing prime q , every non-trivial character χ , and every $\varepsilon > 0$, there exists N_0 such that for all $N \geq N_0$:

$$\|S_\chi(N)\| \leq \varepsilon \cdot N.$$

In other words, the walk character sums are $o(N)$ —they grow strictly slower than linearly. The contrapositive: if the walk permanently avoids -1 , some non-trivial character sum must be $\Omega(N)$ (to cancel the uniform share in the Fourier bridge). CCSB rules this out.

Theorem 3.2 ([✓ complex_csb_mc](#)). $\text{CCSB} \implies \text{MC}$.

```
-- The hypothesis: all character sums are o(N)
def ComplexCharSumBound : Prop :=
  forall (q : Nat) (hq : Nat.Prime q) (hne : forall k, seq k ≠ q),
  forall (chi : (ZMod q)^x -> C^x) (_hchi : chi ≠ 1),
  forall (eps : R) (_heps : 0 < eps),
  exists N_0, forall N ≥ N_0,
    ||sum n in Finset.range N, (chi (emWalkUnit q hq hne n) : C)|| ≤ eps * N

-- The reduction: one line
theorem complex_csb_mc' (hcsb : ComplexCharSumBound) : MullinConjecture :=
  complex_csb_mc hcsb complex_csb_implies_hit_count_lb_proved
```

This is a single-hypothesis reduction with no additional parameters. The proof composes three bridges:

1. Fourier inversion ([✓ complex_csb_implies_hit_count_lb_proved](#)): CCSB implies that the hit count at -1 satisfies $|\{n < N : P_{2,q}(n) = -1\}| = N/(q-1) + o(N)$, which is eventually positive. This directly contradicts permanent avoidance past the sieve gap (the character orthogonality formula, Theorem 3.3, gives this for any target t ; specializing to $t = -1$ is what matters).
2. Cofinal hitting gives HH ([✓ walk_equidist_mc](#)): the eventually-positive hit count at -1 gives cofinal hitting, hence HH. (In fact, the Fourier bridge yields equidistribution across all classes, but only the $t = -1$ case is needed. SE is a side effect: a walk visiting -1 cannot be confined to a proper coset.)
3. DH implies MC (Theorem 2.22): via the inductive bootstrap.

3.2 The Fourier Bridge

The Fourier bridge is the single most important proved result after $\text{DH} \implies \text{MC}$ itself: it converts character sum bounds into hit count lower bounds, and hence into MC.

Theorem 3.3 ([walk_hit_count_fourier_step](#)). For any target $t \in (\mathbb{Z}/q\mathbb{Z})^\times$:

$$|\{n < N : P_{2,q}(n) = t\}| = \frac{1}{q-1} \sum_{\chi} \overline{\chi(t)} S_{\chi}(N),$$

where the sum is over all Dirichlet characters mod q .

This is a standard Fourier inversion formula on the finite group $(\mathbb{Z}/q\mathbb{Z})^\times$, stated for all t . Specializing to $t = -1$ and splitting the trivial character χ_0 (with $\chi_0(a) = 1$ for all a) from the non-trivial ones:

$$|\{n < N : P_{2,q}(n) = -1\}| = \underbrace{\frac{N}{q-1}}_{\text{uniform share}} + \underbrace{\frac{1}{q-1} \sum_{\chi \neq \chi_0} \overline{\chi(-1)} S_{\chi}(N)}_{\text{correction}}.$$

The first term grows linearly. The correction term is bounded by $\frac{1}{q-1} \sum_{\chi \neq \chi_0} |S_{\chi}(N)|$, since $|\overline{\chi(-1)}| = 1$. If CCSB holds—all $|S_{\chi}(N)| = o(N)$ —the correction is $o(N)$, and the hit count is $N/(q-1) + o(N)$, which is eventually positive.

The contrapositive.

Section 2 showed that for the first missing prime q , the hit count at -1 is exactly 0 past the sieve gap. By the Fourier bridge, this forces the correction terms to cancel the entire main term $N/(q-1)$. Since there are only $q-2$ non-trivial characters, at least one must satisfy $|S_{\chi}(N)| = \Omega(N/(q-1)^2)$ —a sustained linear-scale bias in the character spectrum. CCSB ($= o(N)$) rules this out. The Fourier bridge thus converts the geometric statement “the walk avoids -1 permanently” into the spectral statement “some character sum has sustained bias,” and CCSB negates the latter.

Returning to our running example ($q = 41$): the walk must accumulate a deficit of $\sim N/40$ hits at -1 compared to uniform, which requires sustained character sum bias across the 39 non-trivial characters. Proving CCSB for $q = 41$ would rule out this bias and establish that 41 eventually appears.

3.3 The Decorrelation–PED–CCSB Chain

CCSB rules out permanent avoidance of -1 . But the walk is built from multipliers: $P_{2,q}(n+1) = P_{2,q}(n) \cdot m_{2,q}(n)$. Since χ is a group homomorphism, $\chi(P_{2,q}(n)) = \chi(P_{2,q}(0)) \cdot \prod_{k < n} \chi(m_{2,q}(k))$: the walk character sum is a sum of partial products of the multiplier characters. The question becomes: what properties of the multiplier sequence would prevent the walk from permanently dodging the death channel?

This subsection formalizes a chain of progressively weaker hypotheses about the multipliers, each implying the next via proved bridges. The goal is to decompose CCSB into sharper conditions on the multiplier sequence, and to identify exactly where the irreducible difficulty lies.

Property 3.4 ([PositiveEscapeDensity \(PED\)](#)). For every missing prime q and non-trivial χ , there exist $\delta > 0$ and N_0 such that for $N \geq N_0$: $|\{k < N : \chi(m_{2,q}(k)) \neq 1\}| \geq \delta N$.

The name “escape” comes from the SubgroupEscape perspective: the kernel $\ker(\chi)$ is a proper subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$, and $\chi(m_{2,q}(k)) \neq 1$ means the k -th multiplier “escapes” from

$\ker(\chi)$. PED asks that a positive fraction of multipliers escape every proper subgroup, not just occasionally but with positive density. The connection to the death channel: since the death channel is a single class and multipliers that escape $\ker(\chi)$ “rotate” the walk character value $\chi(P_{2,q}(n))$ by a non-trivial amount, enough escapes should prevent the systematic avoidance needed for permanent death-channel dodging. PED is a weak condition—it says nothing about cancellation, only that the multipliers are not asymptotically trapped in any subgroup.

Property 3.5 (**DecorrelationHypothesis**). For every missing prime q and non-trivial χ , the multiplier character sums are $o(N)$: $\|\sum_{n < N} \chi(m_{2,q}(n))\| \leq \varepsilon N$ for large N .

Decorrelation is stronger than PED: it asks not merely that many multipliers escape $\ker(\chi)$, but that they do so with enough balance that the character values cancel. If the multipliers were independent random elements of $(\mathbb{Z}/q\mathbb{Z})^\times$, the sum would be $O(\sqrt{N})$ by the law of large numbers—far smaller than εN . Decorrelation asks for the much weaker $o(N)$.

Property 3.6 (**NoLongRuns**(L)). For every missing prime q and non-trivial χ , past some threshold, no L consecutive multipliers all lie in $\ker(\chi)$.

NoLongRuns is a qualitative cousin of PED: if multipliers never stay inside $\ker(\chi)$ for L steps in a row, then at least $1/(2L)$ of them escape. This condition is easier to verify in practice because it only requires checking short blocks.

Property 3.7 (**BlockRotationEstimate** (**BRE**)). If the escape count is $\geq \delta N$, then the walk character sums are $o(N)$. This encapsulates the Cauchy–Schwarz / van der Corput step in harmonic analysis.

BRE is the bridge between the multiplier-level conditions (PED/Decorrelation) and the walk-level condition (CCSB). It says: given that multipliers escape with positive density, the walk character sums must cancel. The intuition is that each escape event “rotates” the walk character value $\chi(P_{2,q}(n))$ by a non-trivial amount, and sufficiently many such rotations produce cancellation in the sum. BRE is the sole unproved bridge in the PED route.

The hypotheses above—PED, Decorrelation, NoLongRuns—all treat the multiplier sequence as a single stream, ignoring what the walk is doing at the moment. But permanent death-channel avoidance is a statement about the coupling between walk position and multiplier: the death channel $-P_{2,q}(n)^{-1}$ is a function of walk position, so avoiding it means the multiplier distribution at each walk position is biased away from one class. A more refined hypothesis should address this coupling head-on.

Property 3.8 (**ConditionalMultiplierEquidist** (**CME**)). For every missing prime q , non-trivial χ , $\varepsilon > 0$, there exists N_0 such that for $N \geq N_0$ and every walk position $c \in (\mathbb{Z}/q\mathbb{Z})^\times$: $\|\sum_{\substack{n < N \\ P_{2,q}(n)=c}} \chi(m_{2,q}(n))\| \leq \varepsilon N$.

CME says the multiplier distribution is the same at every walk position. Since the death channel is a function of walk position, CME implies the death channel has no special status—the multiplier is no more likely to avoid the forbidden class than any other. Formally, CME is strictly stronger than Decorrelation: it bounds the fiber sums $\sum_{\substack{n < N \\ P_{2,q}(n)=c}} \chi(m_{2,q}(n)) = o(N)$ separately for each position c , not just the global sum. Since the global sum is the sum of

the fiber sums, CME implies Decorrelation by the triangle inequality ([✓ cme_implies_dec](#)). The significance of CME is that it also implies CCSB directly, bypassing PED and BRE entirely: the fiber decomposition and the telescoping identity together convert conditional multiplier cancellation into walk character sum cancellation, for all character orders, without needing the intermediate $\text{PED} \rightarrow \text{BRE} \rightarrow \text{CCSB}$ chain.

Theorem 3.9 ([✓ decorrelation_implies_ped](#)). $\text{Decorrelation} \Rightarrow \text{PED}$.

Proof sketch. Contrapositive. If few multipliers escape $\ker(\chi)$ —say fewer than δN —then most contribute $\chi(m(n)) = 1$ to the sum. The at most δN exceptions contribute values of norm ≤ 1 . By the reverse triangle inequality, $|\sum \chi(m(n))| \geq N - 2\delta N$, which is $\geq \varepsilon N$ for δ small enough. This contradicts Decorrelation. $\square$

Theorem 3.10 ([✓ noLongRuns_implies_ped](#)). $\text{NoLongRuns}(L) \Rightarrow \text{PED}$ with $\delta = 1/(2L)$.

Proof sketch. Partition $\{0, \dots, N-1\}$ into blocks of length L . Each block contains at least one escape (by assumption), so the total escape count is $\geq N/(2L)$. $\square$

Theorem 3.11 ([✓ block_rotation_implies_ped_csb](#)). $\text{BRE} \Rightarrow \text{PEDImpliesComplexCSB}$.

The PED route, with all proved arrows:

$$\text{Dec} \xrightarrow{\text{proved}} \text{PED} \xleftarrow{\text{proved}} \text{NoLongRuns}(L) \xrightarrow{\text{BRE, open}} \text{CCSB} \xrightarrow{\text{proved}} \text{MC}.$$

The sole open bridge in this route is BRE: converting positive escape density into walk character sum cancellation.

However, the PED route is not the only path. CME implies CCSB directly, bypassing PED and BRE entirely ([✓ cme_implies_ccsb](#)):

$$\text{CME} \xrightarrow{\text{proved}} \text{CCSB} \xrightarrow{\text{proved}} \text{MC}.$$

This bypass is significant: the $d \geq 3$ barrier (Remark 5.7) blocks the $\text{PED} \rightarrow \text{CCSB}$ factorization for characters of order ≥ 3 , but $\text{CME} \rightarrow \text{CCSB}$ works for all character orders, using only the telescoping identity and fiber decomposition.

3.4 Vanishing Conditional Bias

CME is a strong hypothesis: it asks that every fiber character sum $F(c, \chi) = \sum_{\substack{n < N \\ w(n)=c}} \chi(m(n))$ be $o(N)$ —in other words, that $\mu = 0$ in the proportionality $F(c, \chi) \approx \mu \cdot V_N(c)$. But the telescoping route to CCSB does not need $\mu = 0$; it needs only that $\mu \neq 1$. This observation motivates a strictly weaker hypothesis.

Property 3.12 ([VanishingConditionalBias \(VCB\)](#)). For every missing prime q , non-trivial χ , and $\varepsilon > 0$, there exists N_0 such that for $N \geq N_0$ there is a complex number $\mu = \mu(N, \chi)$ with $|\mu| \leq 1$ satisfying, for all $c \in (\mathbb{Z}/q\mathbb{Z})^\times$:

$$\left\| \sum_{\substack{n < N \\ w(n)=c}} \chi(m(n)) - \mu \cdot V_N(c) \right\| \leq \varepsilon \cdot N,$$

where $V_N(c) = |\{n < N : w(n) = c\}|$ is the visit count.

In words: the factoring channel treats all walk positions proportionally (the formal content of the factoring channel analogy from §1)—the character statistics of multipliers are the same at every position, up to a common constant μ (which may vary with N and χ but must be common across all positions c). Combined with PED (enough multipliers escape), the proportionality constant cannot equal 1; the telescope then forces the walk character sum to be $o(N)$, ruling out the sustained bias needed for permanent avoidance.

Proposition 3.13 ([✓cme_implies_vcb](#)). $\text{CME} \Rightarrow \text{VCB}$ with $\mu = 0$. Conversely, $\text{VCB} + \text{Dec} \Rightarrow \text{CME}$. Thus CME decomposes as $\text{CME} = \text{VCB} + \text{Dec}$. VCB alone is strictly weaker than CME: it permits the fiber sums to be $\Theta(N)$, provided they are proportional to the visit counts.

Theorem 3.14 ([✓vcb_ped_implies_ccsb](#)). $\text{VCB} + \text{PED} \implies \text{CCSB}$.

Proof sketch. From VCB, the telescope identity gives $(\mu - 1) \cdot S_N = O(1) + O(\varepsilon N(q-1))$. If $|1 - \mu|$ is bounded away from zero, then $S_N = O(\varepsilon N / |1 - \mu|) = o(N)$. To show $|1 - \mu| \geq c_0 > 0$, PED provides $\delta > 0$ such that at least δN multipliers escape $\ker(\chi)$. For each escaping multiplier, $\chi(m(n))$ is a non-trivial d -th root of unity, satisfying $\text{Re}(\chi(m(n))) \leq 1 - \eta_0^2/2$ where $\eta_0 = \min_{\zeta \in \mu_d \setminus \{1\}} |\zeta - 1| > 0$ ([✓norm_sub_one_sq_eq](#), [✓unit_norm_re_le_of_dist](#)). This real-part defect accumulates over δN escaping steps, forcing $|\sum \chi(m(n)) - N| \geq c_0 N$ for $c_0 = \delta \eta_0^2/2$. Combined with VCB's control $|\sum \chi(m(n)) - \mu N| \leq C\varepsilon N$, the reverse triangle inequality gives $|1 - \mu| \geq c_0/2$ for small ε . $\square$

Theorem 3.15 ([✓vcb_ped_implies_mc](#)). $\text{VCB} + \text{PED} \implies \text{MC}$.

This decomposition splits the monolithic CME hypothesis into two independently attackable pieces:

- VCB (“the factoring channel preserves proportionality”): the character distribution of multipliers is the same at every walk position, up to a common rate. This captures the “factorization independence” intuition—that the factoring operation destroys the correlation between walk position and multiplier residue.
- PED (“enough multipliers escape every character kernel”): a positive fraction of multiplier residues fall outside any proper subgroup.

VCB is a recent contribution of the formalization.

Fiber Energy Analysis

The fiber decomposition of CME admits a quantitative refinement via character energy. For a character χ and walk of length N , define the character energy

$$E_\chi(N) = \left\| \sum_{n < N} \chi(P_{2,q}(n)) \right\|^2,$$

which decomposes as

Theorem 3.16 ([✓char_sum_energy_eq_N_plus_cross](#)). $E_\chi(N) = N + 2 \sum_{j < k < N} \text{Re}(\overline{\chi(P_{2,q}(j))} \cdot \chi(P_{2,q}(k)))$.

If the cross terms are $o(N)$, the energy grows linearly rather than quadratically:

Theorem 3.17 ([✓ cross_term_controls_growth](#)). $o(N)$ cross terms $\Rightarrow E_\chi(N) = N + o(N^2)$, i.e., the walk character sum is $o(N)$.

The cross terms admit a fiber decomposition indexed by walk position:

Theorem 3.18 ([✓ total_cross_term_eq_sum_fiber](#)). The total cross term equals the sum of fiber cross terms over walk positions $a \in (\mathbb{Z}/q\mathbb{Z})^\times$.

This connects to CME via the Fiber Energy Bound (FEB), which asks that the sum of squared fiber norms $\sum_a \|F(a, N)\|^2 \leq \varepsilon N^2$:

Theorem 3.19 ([✓ feb_implies_cme](#)). $\text{FEB} \Rightarrow \text{CME}$.

Proof sketch. Each individual fiber satisfies $\|F(a)\|^2 \leq \sum_{a'} \|F(a')\|^2 \leq \varepsilon N^2$, so $\|F(a)\| \leq \sqrt{\varepsilon} N$. This holds for all a , giving CME with error $\sqrt{\varepsilon} N$. $\square$

The fiber energy perspective also clarifies that character values on $(\mathbb{Z}/q\mathbb{Z})^\times$ have unit norm:

Theorem 3.20 ([✓ char_value_norm_one](#)). For any finite group G and character $\chi : G \rightarrow \mathbb{C}^\times$, $\|\chi(g)\| = 1$ for all $g \in G$.

This is used throughout the energy estimates: since each term in the walk character sum has norm 1, the energy $E_\chi(N)$ lies between N (perfect cancellation) and N^2 (no cancellation). CME via FEB asks that the energy is close to the cancellation end.

3.5 Walk Telescoping Identities

The hypotheses above attack CCSB by decomposing it into conditions on the multiplier sequence. Before proceeding to the large sieve route, we pause to examine what the walk recurrence $\chi(w(n+1)) = \chi(w(n)) \cdot \chi(m(n))$ itself forces. This multiplicative structure is the only algebraic relation connecting walk and multiplier character values, and it constrains any possible proof of CCSB. The following identities make these constraints explicit. They are formalized not because they solve CCSB, but because they reveal the structural landscape: they show which proof strategies are compatible with the walk's algebra and which are ruled out.

Theorem 3.21 ([✓ walk_telescope_identity](#)). For any χ and N :

$$\sum_{n < N} \chi(w(n)) \cdot (\chi(m(n)) - 1) = \chi(w(N)) - \chi(w(0)).$$

This identity follows immediately from the walk recurrence $\chi(w(n+1)) = \chi(w(n)) \cdot \chi(m(n))$: writing $\chi(w(n)) \cdot (\chi(m(n)) - 1) = \chi(w(n+1)) - \chi(w(n))$, the sum telescopes to $\chi(w(N)) - \chi(w(0))$.

Theorem 3.22 ([✓ walk_telescope_norm_bound](#)). The telescoping sum has norm ≤ 2 (triangle inequality on unit-norm terms).

The ≤ 2 bound looks innocent, but it constrains how the walk can respond to the death channel.

Each summand $\chi(w(n)) \cdot (\chi(m(n)) - 1)$ is non-zero exactly when $\chi(m(n)) \neq 1$ —i.e., when the multiplier escapes $\ker(\chi)$, “rotating” the character value. The ≤ 2 bound means the net rotation over all N steps is negligible, tightly coupling the walk character sum $S_N = \sum_{n < N} \chi(w(n))$ to the multiplier character sum $M_N = \sum_{n < N} \chi(m(n))$. Splitting the product in Theorem 3.21 gives $S_N \cdot M_N / N - S_N = O(1)$ (after normalization).

Theorem 3.23 ([✓ walk_shift_one_correlation](#)). $\sum_{n < N} \chi(w(n)) \cdot \overline{\chi(w(n+1))} = \overline{\sum_{n < N} \chi(m(n))}$.

This identity says that the lag-1 autocorrelation of the walk character equals the conjugate of the multiplier character sum. It is a no-go result for the van der Corput method with $H = 1$: VdC bounds $|S_N|^2$ in terms of autocorrelations, but at lag $h = 1$, the autocorrelation is exactly $|M_N|$ —the multiplier character sum—which need not be small. So VdC with a single shift gives only $|S_N| \leq O(\sqrt{N \cdot |M_N|})$, which is $O(N)$ in the worst case, not the $o(N)$ that CCSB requires. This means any proof of CCSB must either (i) use higher-order correlations (HOD, Appendix A.3) or (ii) establish multiplier decorrelation first.

The cofactor identity.

A finer structural identity arises from the “+1 shift” in the Euclid construction. Define the cofactor $\text{cof}(n) = (P_2(n) + 1) / \text{em}_2(n+1)$, and its residue $\text{cofZ}(q, n) = \text{cof}(n) \bmod q$. Then:

Theorem 3.24 ([✓ shifted_walk_eq_mult_mul_cof](#)). $P_{2,q}(n) + 1 = \text{m}_{2,q}(n) \cdot \text{cofZ}(q, n)$.

This factors the walk recurrence through the cofactor: $\chi(P_{2,q}(n) + 1) = \chi(\text{m}_{2,q}(n)) \cdot \chi(\text{cofZ}(q, n))$. Since the walk hits -1 exactly when $P_{2,q}(n) + 1 = 0$, and $\text{m}_{2,q}(n) \neq 0$ (as a unit):

Theorem 3.25 ([✓ walkZ_eq_neg_one_iff_cofZ_zero](#)). $P_{2,q}(n) = -1 \iff \text{cofZ}(q, n) = 0$.

The cofactor identity is unprecedented in published work on the Euclid–Mullin sequence. However, it does not escape the orbit-specificity barrier: the cofactor $\text{cofZ}(q, n)$ is a joint quantity depending on both the walk state and the factorization outcome, so the identity factorizes one correlated variable into two correlated variables. This is formalized as a dead end (Dead End #115).

The Large Sieve Route

The preceding subsections attacked the question “can the walk avoid -1 permanently?” for a single modulus q . A different strategy works with many moduli simultaneously: classical analytic number theory produces character sum estimates averaged over all moduli $q \leq Q$, and such averaged results are often easier to prove than pointwise ones. The large sieve inequality and the Bombieri–Vinogradov theorem are the two central tools for this.

The formalization develops this multi-modular route across three files ([LargeSieve/Basic.lean](#), [LargeSieveHarmonic.lean](#), [LargeSieveAnalytic.lean](#)) totaling $\sim 5,870$ lines, connecting classical analytic number theory to MC via a multi-modular character sum bound.

Why formalize the large sieve?

The analytic large sieve inequality and the Bombieri–Vinogradov theorem are among the most powerful tools in analytic number theory for controlling the distribution of primes in arithmetic progressions. If these tools could be applied to the EM walk, MC would follow. We formalize

the connection—not the deep theorems themselves (which are known results, stated as open Props)—for two reasons:

1. To identify precisely what transfer hypothesis is needed to apply each classical result to the specific EM orbit, and
2. To verify that six apparently independent routes (BV, ArithLS, ALS, PrimeArithLS, LoD, sieve transfer) all reduce to the same orbit-specificity gap.

This diagnosis is itself a mathematical contribution: it shows that the difficulty of MC is not a failure of existing analytic tools but a fundamental obstacle in applying ensemble-averaged results to a single deterministic orbit.

Property 3.26 (MultiModularCSB (MMCSB)). There exists Q_0 such that for all $q \geq Q_0$ prime, every non-trivial character $\chi \bmod q$, and every $\varepsilon > 0$, there exists N_0 such that for $N \geq N_0$: $\|S_\chi(N)\| \leq \varepsilon N$.

MultiModularCSB is weaker than CCSB in that it allows finitely many exceptional primes below Q_0 . This weakening is crucial because averaged results like BV naturally produce bounds that fail for finitely many moduli.

Theorem 3.27 ($\checkmark$ mmcsb_implies_mc). MultiModularCSB $\implies$ MC.

The proof composes the per-prime Fourier bridge (Theorem 3.3) with the inductive bootstrap: for $q \geq Q_0$, MMCSB gives hit count $\sim N/(q-1) > 0$ at -1 , contradicting permanent avoidance; for the finitely many primes $q < Q_0$, the bootstrap (Section 2) handles them—once MC holds for all primes below q , the sieve gap and one-prime gap (Theorem 2.29) reduce MC(q) to a single hitting event.

Three parallel routes to MultiModularCSB are formalized:

Bombieri–Vinogradov route.

BV says primes are equidistributed among arithmetic progressions “on average over moduli”: for most $q \leq Q = \sqrt{x}/(\log x)^A$, the count of primes $\leq x$ in any progression $a \bmod q$ is close to the expected $\pi(x)/\phi(q)$. If this applies to the EM multipliers, the death channel—a single progression—gets its fair share of multipliers, breaking the avoidance.

Theorem 3.28 ($\checkmark$ bv_chain_mc). BV + BVImpliesMMCSB $\implies$ MC.

The transfer hypothesis BVImpliesMMCSB is a genuine frontier: it requires transferring the averaged equidistribution statement of BV (valid for primes in generic progressions) to the specific EM walk orbit. The EM sequence is not a generic sample of primes—it is a deterministic sequence defined by iterated factorization—so its multipliers could exhibit special correlations that BV’s averaged estimate cannot detect.

The route was decomposed into two stages:

$$\text{BV} \xrightarrow{\text{sieve transfer}} \text{EMMultCSB} \xrightarrow{\text{walk bridge}} \text{MMCSB} \xrightarrow{\text{proved}} \text{MC},$$

separating the number-theoretic content (BV $\implies$ EMMultCSB, where EMMultCSB bounds the multiplier character sums) from the dynamical content (EMMultCSB $\implies$ MMCSB, converting multiplier bounds to walk bounds). However, the walk bridge MultCSBImpliesMMCSB is false in general ($\checkmark$ MultCSBImpliesMMCSB): the walk character sum $\sum \chi(w(n))$ is a partial product

$\prod_{k < n} \chi(m(k))$ of the multiplier characters, and partial products of equidistributed unit complex numbers need not cancel—they perform a random walk on the unit circle whose norm grows as $\sqrt{N}$, not as $o(N)$. The telescope identity (Theorem 3.23) makes this obstruction precise: the $h=1$ autocorrelation equals the multiplier character sum, so van der Corput with a single shift gives only $O(N)$, not $o(N)$. This is why the CME bypass (fiber decomposition + telescoping, Definition 3.8) is essential: it goes directly from conditional multiplier equidistribution to CCSB without ever requiring the walk bridge.

Additional sieve routes.

Two further routes are formalized—the arithmetic large sieve (ArithLS $\Rightarrow$ MC, a dead end) and the analytic large sieve (ALS $\Rightarrow$ PrimeArithLS $\Rightarrow$ MC), where the ALS-to-PrimeArithLS bridge via Gauss sum inversion is fully proved across eight internal lemmas. In both cases, the genuine open content is the same orbit-specificity transfer: applying averaged results to one deterministic orbit. The full details, including the ALS definition, weak ALS proof, Gauss sum inversion theorem, and a spectral energy route (SVE, van der Corput, HOD, CME) with its complete hypothesis hierarchy, appear in Appendix A.3.

4 Ensemble Averaging and the DSL Route

The orbit-specificity barrier—the difficulty of transferring population statistics to one deterministic orbit—can be circumvented by averaging over the orbit parameter itself. This section develops the ensemble framework: we parametrize the EM construction by its starting point, average over squarefree starting points via the Chinese Remainder Theorem, and reduce MC to a single Deterministic Stability Lemma (DSL).

What is new vs. what is infrastructure. The generalized EM sequences, the CRT propagation framework, and the DSL $\Rightarrow$ MC reduction (Theorem 4.23) are original contributions of this formalization. The decorrelation–variance–concentration chain (§4.2) is proved with explicit constants ($C = 2$, Theorem 4.8). The Weyl hitting bridge (§4.5) and the generalized MC argument (§4.5) are fully original. The sieve density function and Buchstab weight (§4.3) formalize known sieve theory. The visit equidistribution identity (§4.6) connecting spectral and geometric viewpoints is new.

Sections 2–5 worked with the standard orbit $s = 2$. We now exploit the starting-point parameter s introduced in (2): the ensemble is the family of EM trajectories $\{P_n : n \text{ squarefree}\}$. The structural properties established in Section 2—divisibility, monotonicity, injectivity, squarefreeness, and the death channel equivalence—hold for every starting point and are used throughout.

Section roadmap

- §4.1 CRT equidistribution and propagation — ensemble residues equidistribute and propagate across steps.
- §4.2 Decorrelation–variance–concentration chain — StepDecorrelation $\Rightarrow$ character sum cancellation for a.a. starting points.
- §4.3 Sieve structure and first moments — ensemble average of $1/\text{genSeq}$ is positive; parity and CRT bounds.
- §4.4 The DSL reduction — DSL $\Rightarrow$ CME $\Rightarrow$ CCSB $\Rightarrow$ MC, the main chain.
- §4.5 Generalized MC and the Weyl hitting bridge — character cancel $\Rightarrow$ walk hitting via strong induction.

§4.6 Visit equidistribution and spectral identity — SVE $\leftrightarrow$ VE, Lyapunov identity.

4.1 CRT Equidistribution and Propagation

The ensemble is the collection of generalized EM sequences indexed by squarefree starting points $n \in [1, X]$. For each prime q , we study how the residues $P_n(k) \bmod q$ and $\text{em}_n(k) \bmod q$ distribute as n ranges over squarefree integers.

Definition 4.1 ([Accumulator density](#)). For prime q , step k , and residue class $a \in (\mathbb{Z}/q\mathbb{Z})^\times$, define the accumulator density:

$$\rho_k(a) = \lim_{X \rightarrow \infty} \frac{|\{n \in [1, X] : n \text{ squarefree}, P_n(k) \equiv a \pmod{q}\}|}{|\{n \in [1, X] : n \text{ squarefree}\}|}.$$

Remark 4.2 ([SquarefreeResidueEquidist](#) ([SRE](#))). Squarefree integers equidistribute among non-zero residue classes modulo any prime q : $\rho_0(a) = 1/(q-1)$ for all $a \in (\mathbb{Z}/q\mathbb{Z})^\times$. This is a standard consequence of the inclusion-exclusion formula for squarefree counting (a known result, not yet in Mathlib).

SRE provides the base case of a CRT induction:

Property 4.3 ([CRTPropagationStep](#)). For all primes q and steps k : if ρ_k is uniform on $(\mathbb{Z}/q\mathbb{Z})^\times$, then so is ρ_{k+1} .

The structural basis for CRT propagation is the CRT multiplier invariance ([✓ crt_multiplier_invariance](#)): the generalized multiplier $\text{em}_n(k) = \text{minFac}(P_n(k) + 1)$ depends on the full integer $P_n(k) + 1$, not just its residue modulo q . By CRT, knowing $P_n(k) \bmod q$ does not constrain $P_n(k) \bmod r$ for any other prime r , so the factoring outcome is “ q -blind.” This decorrelates the mod- q coordinate of the accumulator at step k from the multiplier at step k .

The propagation proceeds by induction on k :

Theorem 4.4 ([✓ sre_crt_implies_accum_equidist](#)). SRE + CRTPropagationStep implies accumulator equidistribution: $\rho_k(a) = 1/(q-1)$ for all $k, q, a \in (\mathbb{Z}/q\mathbb{Z})^\times$.

From accumulator equidistribution, a population transfer bridge gives ensemble multiplier equidistribution (EME): the density of squarefree n with $\text{em}_n(k) \equiv a \pmod{q}$ tends to $1/(q-1)$ for each non-zero a .

CRT independence and joint equidistribution.

A deeper structural property concerns the joint distribution of residues at different steps. The marginal product version—the product $\rho_j(a) \cdot \rho_k(b)$ —is trivially $1/(q-1)^2$ from accumulator equidistribution. The genuine joint version—counting squarefree n with $P_n(j) \equiv a$ and $P_n(k) \equiv b$ simultaneously—is not trivial: it requires genuine independence of the mod- q coordinates at different steps, formalized as [SquarefreeCRTIndependence](#) ([Ensemble/CRTFreedom.lean](#)).

The CRT freedom framework proves:

Theorem 4.5 ([✓ scrti_sre_implies_aep](#)). SCRTI + SRE implies AccumulatorEquidistPropagation.

This provides a clean decomposition: SRE is the base case (a consequence of standard analytic number theory), SquarefreeCRTIndependence is the inductive step (CRT structure), and AccumulatorEquidistPropagation is the output.

Standard analytic number theory (ANT).

Throughout this paper, ANT denotes the conjunction of two classical results: (i) Dirichlet’s theorem on primes in arithmetic progressions, and (ii) sieve estimates for shifted squarefree integers. Together they imply [MinFacResidueEquidist \(MFRE\)](#): among q -rough shifted squarefree integers m (those with $\mu^2(m-1) = 1$ and $\text{minFac}(m) > q$), the smallest prime factor $\text{minFac}(m)$ is equidistributed modulo q ([✓ MinFacResidueEquidist](#)). MFRE in turn implies PopulationEquidist (PE) via [pe_of_equidist](#). ANT is not formalized in Lean—it is assumed as a hypothesis in the master reduction—but it follows from results that are entirely standard in analytic number theory. An investigation of the formalization gap (§A.7.3) shows that $\sim 92\%$ of the required Mathlib infrastructure already exists (including Dirichlet’s theorem via L -series, Abel summation, and L -function non-vanishing). The one-sided Tauberian lemma, prime power stripping, and the discrete Abel summation chain are now machine-verified ([Population/Tauberian.lean](#), 557 lines; [Population/AbelChain.lean](#), 651 lines), giving the full composed arrow $\text{WeightedPNTinAP} \Rightarrow \text{PrimesEquidistributedInAP}$. The Alladi chain from $\text{PrimesEquidistInAP}$ to MFRE is decomposed into two precisely stated open hypotheses with a proved master reduction ([Population/AlladiDensity.lean](#), 275 lines). The two remaining blocking dependencies are the real-variable Wiener–Ikehara bridge (converting L -series behavior near $s = 1$ into partial sum asymptotics) and the Alladi density content (sieve-theoretic proof of the two Alladi chain hypotheses).

4.2 The Decorrelation–Variance–Concentration Chain

The ensemble character energy is the key object connecting CRT equidistribution to MC. For a character χ and starting point n , the multiplier character energy at length K is

$$E(n, K) = \left\| \sum_{k < K} \chi(\text{em}_n(k) \bmod q) \right\|^2.$$

This is an energy in the character spectrum: $E(n, K)$ lies between 0 (perfect cancellation) and K^2 (no cancellation). CCSB asks that $E(n, K) = o(K^2)$ for the standard orbit $n = 2$; the ensemble framework asks about the distribution of $E(n, K)$ over squarefree n .

Property 4.6 ([StepDecorrelation \(SD\)](#)). For every prime q , character χ with $\|\chi(a)\|^2 \leq 1$, and steps $j < k$: the ensemble cross-term vanishes:

$$\lim_{X \rightarrow \infty} \mathbb{E}_{n \leq X}^{\text{sqf}} [\text{Re}(\chi(m_j) \cdot \overline{\chi(m_k)})] = 0,$$

where $m_j = \text{em}_n(j) \bmod q$ and the expectation is over squarefree n .

Step Decorrelation says that the character values at different steps are asymptotically uncorrelated when averaged over starting points. CRT independence provides the structural basis: between steps j and k , each “+1 shift” in the Euclid construction decorrelates the mod- q information.

Property 4.7 ([CharSumVarianceBound \(CSV\)](#)). There exists $C > 0$ such that for all q, χ , and K : $\mathbb{E}_n^{\text{sqf}}[E(n, K)] \leq C \cdot K$ eventually in X .

Theorem 4.8 ([decorrelation_implies_variance_proved](#)). $\text{SD} \implies \text{CSV}$ with $C = 2$.

Proof. By induction on K .

Base case. $E(n, 0) = 0 \leq 0$ for all n .

Inductive step. The energy recurrence gives

$$E(n, K+1) = E(n, K) + \|\chi(m_K)\|^2 + 2\text{Re}\left(\sum_{j<K} \chi(m_j) \cdot \overline{\chi(m_K)}\right).$$

Taking ensemble averages:

- Diagonal: $\mathbb{E}[\|\chi(m_K)\|^2] \leq 1$ since $\|\chi(a)\|^2 \leq 1$ pointwise.
- Inductive term: $\mathbb{E}[E(n, K)] \leq 2K$ by the inductive hypothesis (for $X \geq X_0^{(K)}$).
- Cross-terms: for each $j < K$, SD gives $|\mathbb{E}[\text{Re}(\chi(m_j)\overline{\chi(m_K)})]| < 1/(2(K+1))$ for X large enough. Summing the K terms: $2|\text{cross sum}| \leq 1$.

Combining: $\mathbb{E}[E(n, K+1)] \leq 2K + 1 + 1 = 2(K+1)$. The threshold $X_0^{(K+1)}$ is the maximum of $X_0^{(K)}$ and the K individual cross-term thresholds. $\square$

The constant $C = 2$ is sharp for this method: each step contributes at most 1 from the diagonal and at most 1 from the cross-terms. For comparison, independent random variables with $|\chi(m)| = 1$ give $\mathbb{E}[E(n, K)] = K$ exactly (perfect cancellation of cross-terms); $C = 2$ allows the cross-terms to contribute up to their maximum.

Property 4.9 ([EnsembleCharSumConcentration](#)). For every q, χ , and $\varepsilon > 0$: the density of squarefree $n \leq X$ with $E(n, K) > (\varepsilon K)^2$ tends to 0 as $X \rightarrow \infty$, uniformly in K past a threshold.

Theorem 4.10 ([char_variance_implies_concentration_proved](#)). CSV implies EnsembleCharSumConcentration.

Proof sketch. Markov's inequality. Fix $\varepsilon > 0$. The density of squarefree n with $E(n, K) > (\varepsilon K)^2$ is at most $\mathbb{E}[E(n, K)]/(\varepsilon K)^2 \leq 2K/(\varepsilon^2 K^2) = 2/(\varepsilon^2 K)$, which tends to 0 as $K \rightarrow \infty$. $\square$

Theorem 4.11 ([char_concentration_implies_cancellation](#)). EnsembleCharSumConcentration $\implies$ for almost all squarefree n , the character energy $E(n, K) = o(K^2)$.

Proof sketch. The set of n with $E(n, K) > (\varepsilon K)^2$ for all K is contained in the set for any specific K_0 . Choosing K_0 large enough (from concentration) makes this density arbitrarily small. $\square$

The three theorems compose into a single proved reduction:

Theorem 4.12 ([sd_implies_cancellation](#)). StepDecorrelation alone $\implies$ character sum cancellation for almost all squarefree starting points.

This composition uses zero open hypotheses: StepDecorrelation is the sole input, and the three bridges (variance, concentration, cancellation) are all machine-verified.

4.3 Sieve Structure and Ensemble First Moments

The concentration chain above converts decorrelation into cancellation via a purely analytic argument. This subsection develops the complementary sieve perspective: instead of character energy, we track the reciprocal sums $\sum 1/\text{em}_n(k)$ and use parity, CRT, and death-density arguments to establish lower bounds on the ensemble mean. The two threads converge in the DSL route (§4.4): the concentration chain provides the “if decorrelated then cancellation” implication, while the sieve thread provides concrete evidence that the first moment grows linearly.

First Moment, Sieve Density, and the Buchstab Weight

The ensemble average of the reciprocal $1/\text{em}_n(k)$ connects to classical sieve theory via the sieve density function:

$$g(r) = \frac{r}{r^2 - 1},$$

which gives the conditional probability that a random shifted squarefree number m (with $\mu^2(m-1) = 1$) is divisible by a prime r . The algebraic properties of g are fully proved in [Reduction/ShiftedDensity.lean](#):

- $g(r) > 0$ for $r \geq 2$ ([✓ sieveDensity_pos](#)).
- $g(r) < 1$ for $r \geq 2$ ([✓ sieveDensity_lt_one](#)).
- $g(r) > 1/r$ ([✓ sieveDensity_gt_inv](#)): a density boost from automatic squarefreeness.
- $g(r) = \frac{1}{2} \left(\frac{1}{r-1} + \frac{1}{r+1} \right)$ ([✓ sieveDensity_partial_frac](#)).
- g is strictly decreasing on $[2, \infty)$ ([✓ sieveDensity_strict_anti](#)).

The Buchstab weight $\kappa(p)$ is the sieve density at the base prime p :

[Theorem 4.13](#) ([✓ buchstabWeight_two](#)). $\kappa(2) = g(2) = 2/3$.

The first-moment/variance pattern gives a concentration result for reciprocal sums:

[Theorem 4.14](#) ([✓ first_moment_variance_implies_rsd](#)). First Moment + Variance Bound $\implies$ Reciprocal Sum Divergence: $\sum_{k < K} 1/\text{em}_n(k) \rightarrow \infty$ for almost all squarefree n .

Positive density from the first moment alone.

The density-1 result above requires both the first moment (linear mean growth) and a variance bound (from pairwise step decorrelation). A simpler route, formalized in [✓ lmg_implies_positive_density_rsd](#), shows that linear mean growth alone yields a positive density of squarefree starting points with divergent reciprocal sums.

The argument is elementary. Each term $1/\text{em}_n(k) \leq 1/2$ (since $\text{em}_n(k)$ is prime, hence ≥ 2), so $S_K(n) \leq K/2$ deterministically ([✓ recipPartialSum_le_half_K](#)). If the ensemble mean satisfies $\mathbb{E}[S_K] \geq \kappa K$ (the LinearMeanGrowth hypothesis, which follows from FirstMomentStep), a partition of the ensemble into $\{S_K \geq M\}$ and $\{S_K < M\}$ gives

$$\text{density}\{S_K \geq M\} \geq \frac{\kappa K - M}{K/2 - M} \xrightarrow{K \rightarrow \infty} 2\kappa.$$

Taking $\delta = \kappa/2$ yields a 1-hypothesis route:

$$\text{LMG} \implies \text{PositiveDensityRSD},$$

compared to the 2-hypothesis route $\text{PSD} + \text{LMG} \Rightarrow \text{AASRSD}$ (density 1). With $\kappa \geq 1/3$ from the Buchstab weight at $p = 2$ ([✓ kappaPartial_three](#)), the positive density is at least $\delta = 1/6$. The chain is now fully closed to a single hypothesis. Linearity of the ensemble average ([✓ ensembleAvg_sum_range](#)) gives:

Theorem 4.15 ([✓ first_moment_step_implies_positive_density_rsd](#)). $\text{FirstMomentStep}(\kappa) \implies \text{PositiveDensityRSD}$, for any $\kappa > 0$.

The proof composes two machine-verified links: $\text{FirstMomentStep} \Rightarrow \text{LMG}$ ([✓ first_moment_step_implies_lmg](#), via $\mathbb{E}[\Sigma] = \sum \mathbb{E}$ and a finite-max threshold argument) and $\text{LMG} \Rightarrow \text{PositiveDensityRSD}$ (the partition argument above). FirstMomentStep is the sole remaining hypothesis: it states that the ensemble average of $1/\text{em}_n(k)$ converges to κ for each k , a standard consequence of sieve theory for $k = 0$ and a mild orbit-transfer for $k \geq 1$.

Parity structure and unconditional bounds.

The mod-2 structure of the generalized EM sequence yields unconditional results that partly close the weak-MC chain.

Theorem 4.16 ([✓ genSeq_ge_three](#)). For $k \geq 1$ and $n \geq 1$, the generalized EM prime satisfies $\text{em}_n(k) \geq 3$.

Proof. The product $P_n(1) = n \cdot \text{minFac}(n+1)$ is always even: if n is even the product is even; if n is odd then $n+1$ is even, so $\text{minFac}(n+1) = 2$ ([✓ genProd_one_even](#)). For $k \geq 1$, $P_n(1) \mid P_n(k)$ by [genProd_dvd_genProd](#), so $P_n(k)$ is even and $P_n(k)+1$ is odd ([✓ genProd_succ_odd](#)). Hence $\text{minFac}(P_n(k)+1) \neq 2$ (by [Nat.minFac_eq_two_iff](#)), and since $\text{em}_n(k)$ is prime with $\text{em}_n(k) \geq 2$, we get $\text{em}_n(k) \geq 3$. $\square$

This tightens the deterministic bound from $S_K(n) \leq K/2$ to $S_K(n) \leq 1/2 + (K-1)/3$ for $K \geq 1$.

Theorem 4.17 ([✓ ensembleAvg_k0_ge_quarter](#)). For X with $\text{sqfreeCount}(X) > 0$: $\mathbb{E}[1/\text{em}_n(0)] \geq 1/4$.

Proof. For odd squarefree n , $n+1$ is even, so $\text{em}_n(0) = \text{minFac}(n+1) = 2$ and $1/\text{em}_n(0) = 1/2$. The density of odd squarefree integers among all squarefree integers is $\geq 1/2$: the halving map $m \mapsto m/2$ injects even squarefree into odd squarefree ([✓ odd_sf_card_ge_half](#)), giving $\#\{\text{even sf}\} \leq \#\{\text{odd sf}\}$. So $\mathbb{E}[1/\text{em}_n(0)] \geq (1/2) \cdot (1/2) = 1/4$. $\square$

This is unconditional—no open hypothesis is used.

A weaker-than- FirstMomentStep hypothesis suffices for the full chain: $\text{StepMeanLowerBound}(c)$ ([✓ StepMeanLowerBound](#)) asserts only that $\mathbb{E}[1/\text{em}_n(k)] \geq c$ eventually for each k , without requiring convergence to a specific limit. The $k = 0$ case is proved unconditionally with $c = 1/4$ ([✓ smlb_k0_unconditional](#)). The chain $\text{SMLB} \Rightarrow \text{LMG} \Rightarrow \text{PositiveDensityRSD}$ is fully machine-verified ([✓ smlb_implies_positive_density_rsd](#)). If $\text{FirstMomentStep}(\kappa)$ holds, then $\kappa \geq 1/4$ ([✓ fms_kappa_ge_quarter](#), via [ge_of_tendsto](#)).

For $k = 1$, the CRT structure gives further traction: for squarefree $n \equiv 1 \pmod 6$, the sequence $\text{em}_n(1) = 3$ ([✓ genSeq_one_of_mod6](#)), contributing $1/3$ to the sum. The injection argument extends to factor 3: $\#\{\text{sf}\} \leq 4 \cdot \#\{\text{odd sf coprime to 3}\}$ ([✓ sf_le_four_coprime6](#)). The density hypothesis Mod6DensityLB —that the fraction of 1-mod-6 squarefree among all squarefree exceeds $1/8$ —is proved unconditionally ([✓ mod6_density_lb_proved](#)) via a sieve argument: non-squarefree $n \equiv 1 \pmod 6$

must have some $d^2 \mid n$ with $d \geq 5$ coprime to 6; since $d^2 \equiv 1 \pmod{6}$ (CRT), the per- d count is $\leq X/(6d^2) + 1$ and the telescoping bound $\sum_{d \geq 5} 1/d^2 \leq 1/4$ gives the total non-squarefree count $\leq X/24 + \sqrt{X}$. Combined with the injection-based lower bound $\#\{n \equiv 1 \pmod{6}\} \geq X/6$, inclusion-exclusion on multiples of 4 and 9 for the squarefree count yields the result for $X \geq 10,000$. SMLB at $k = 1$ then follows unconditionally with $c = 1/24$ ([✓ mod6_density_implies_smlb_k1](#)).

The CRT analysis extends to $k = 2$: for $n \equiv 1 \pmod{6}$, we have $P_n(2) = 6n$ ([✓ genProd_two_of_mod6](#)) and $\text{em}_n(2) = \text{minFac}(6n+1)$. When additionally $n \equiv 4 \pmod{5}$ (i.e., $n \equiv 19 \pmod{30}$ by CRT), we get $5 \mid (6n+1)$ while $2 \nmid (6n+1)$ and $3 \nmid (6n+1)$, so $\text{em}_n(2) = 5$ ([✓ genSeq_two_of_mod30](#)). Thus for $n \equiv 19 \pmod{30}$ squarefree, the first three generalized EM primes are 2, 3, 5 deterministically.

A $\text{PartialSMLB}(c, K_0)$ hypothesis—requiring $\mathbb{E}[1/\text{em}_n(k)] \geq c$ only for $k \leq K_0$ —yields a fixed mean lower bound $\mathbb{E}[S_{K_0+1}] \geq c(K_0 + 1)$ via linearity ([✓ partial_smlb_implies_mean_lower_bound](#)). $\text{PartialSMLB}(1/4, 0)$ holds unconditionally ([✓ partial_smlb_zero_unconditional](#)), giving a concrete (though non-divergent) threshold at one step.

The death density bridge.

The mod-6 analysis above proves SMLB at $k = 0$ and $k = 1$ unconditionally, but extending to all k requires a structural argument. A death density bridge generalizes the single-prime approach: when $P_n(k) \equiv -1 \pmod{q}$ for a prime q , the next generalized multiplier satisfies $q \mid P_n(k) + 1$, so $\text{em}_n(k) = \text{minFac}(P_n(k) + 1) \leq q$ ([✓ genSeq_le_of_genProd_neg_one](#)). Combined with $\text{em}_n(k) \geq 3$ for $k \geq 1$ (Theorem 4.16), this gives $1/\text{em}_n(k) \geq 1/q$ on the “death fiber” $\{n : P_n(k) \equiv -1 \pmod{q}\}$.

Theorem 4.18 ([✓ ensembleAvg_ge_death_density](#)). For any prime $q \geq 3$, step $k \geq 1$:

$$\mathbb{E}[1/\text{em}_n(k)] \geq \frac{\text{density}\{P_n(k) \equiv -1 \pmod{q}\}}{q}.$$

Proof. Split the ensemble sum by the death fiber. On the fiber, each term $\geq 1/q$; on the complement, each term ≥ 0 . The sum on the fiber is therefore at least $\#\{\text{fiber}\}/q$. Dividing by the total squarefree count gives the density bound. $\square$

At $q = 3$: since $-1 \equiv 2 \pmod{3}$ and $3 \mid P_n(k) + 1$ forces $\text{em}_n(k) = 3$ (odd and divisible by 3), the bound becomes $\mathbb{E}[1/\text{em}_n(k)] \geq d_2/3$ where d_2 is the density of $\text{genProd} \equiv 2 \pmod{3}$ ([✓ ensembleAvg_ge_mod3_density](#)). The general bridge unifies with the mod-3 case: $\text{AccumMod3LB}(c) \Leftrightarrow \text{DeathDensityLB}(3, c)$ ([✓ accumMod3LB_iff_deathDensity3](#)).

A positive lower bound on the death density at any single prime $q \geq 3$ for all steps k yields the full chain:

Theorem 4.19 ([✓ death_density_implies_prsd](#)). For any prime $q \geq 3$ and constant $c > 0$:

$$\text{DeathDensityLB}(q, c) \implies \text{SMLB}(\min(1/4, c/q)) \implies \text{PositiveDensityRSD}.$$

The four routes are assembled in [✓ ewe_landscape_extended](#).

The absorption mechanism.

While the death density bridge (Theorem 4.18) converts positive death density into SMLB, the death density itself decays over time due to an absorption mechanism. The mod- q dynamics of $P_n(k)$ has three structural properties, all machine-verified ([✓ absorption_kills_death_density](#)):

1. State 0 is absorbing ([✓ genProd_mod_zero_absorbing](#)): if $q \mid P_n(k)$, then $q \mid P_n(k+j)$ for all $j \geq 0$, since divisibility is preserved under the recurrence $P_n(k+1) = P_n(k) \cdot \text{em}_n(k)$.
2. Death feeds absorption ([✓ death_implies_absorption](#)): when $P_n(k) \equiv -1 \pmod{q}$ and $\text{em}_n(k) = q$, we get $P_n(k+1) = P_n(k) \cdot q \equiv 0 \pmod{q}$, so the orbit is absorbed. For $q = 3$ this always fires ([✓ mod3_death_implies_absorption](#)).
3. Absorbed $\neq$ death ([✓ absorbed_not_in_death_class](#)): for $q \geq 2$, the absorbed state 0 differs from the death state -1 in $\mathbb{Z}/q\mathbb{Z}$, so absorbed orbits never return to the death class ([✓ death_then_never_death_again](#)).

Together, these show that each visit to the death class permanently removes an orbit from future death events. Consequently, $\text{DeathDensityLB}(q, c)$ for any fixed $c > 0$ is heuristically false: the death density decays roughly as $O((1 - 1/(q-1))^k)$ as orbits accumulate in the absorbing state. More critically, the absorption mechanism shows that AccumulatorEquidist-Propagation (AEP) itself is false: the correct squarefree residue density per coprime class is $r/(r^2-1)$, not $1/(r-1)$, and even this corrected limit fails for $k \geq 1$ because nonzero-class mass drains monotonically to the absorbing class 0 (Dead End #138). The backward dynamics chain $\text{ETA} + \text{DCTA} + \text{SRE} \Rightarrow \text{PRSD}$ remains formally valid but is vacuously true since CRT-PropagationStep (which preserves the AEP limit) is false. The SMLB and DeathDensityLB routes are not affected—they use independent, proved lower bounds.

The divergence hierarchy.

When the step mean lower bound SMLB cannot be established with a uniform constant c across all steps, a weaker condition may still suffice. Two relaxations are formalized in `FirstMoment.lean`:

Definition 4.20 ([DecayingSMLB](#) ([✓ DecayingSMLB](#))). A sequence $f(k) > 0$ with $\sum_k f(k) = \infty$ is a decaying step mean lower bound if $\mathbb{E}[1/\text{em}_n(k)] \geq f(k)$ for all k (eventually in X).

Definition 4.21 ([FirstMomentDivergence](#) ([✓ FirstMomentDivergence](#))). $\mathbb{E}[\sum_{k < K} 1/\text{em}_n(k)] \rightarrow \infty$ as $K \rightarrow \infty$.

The hierarchy is:

$$\text{SMLB}(c) \implies \text{DecayingSMLB}(\text{const}) \implies \text{FMD},$$

both implications proved ([✓ smlb_implies_decaying_smlb](#), [✓ decaying_smlb_implies_fmd](#)). However, the reverse implications fail: DecayingSMLB captures a scenario where each step mean decays but the cumulative contribution diverges (e.g. $f(k) = 1/k$), while FMD does not imply PositiveDensityRSD—it says only that the expected partial sum grows, not that a positive proportion of starting points have large partial sums. The critical gap is exactly the passage from first-moment divergence to concentration, which requires variance bounds (§4.5).

Conditional MFRE and the summability argument.

A more refined approach avoids fixing a single prime q . The MFREConditional hypothesis ([✓ MFREConditional](#)) states that for squarefree m with $m \equiv c \pmod{q}$ (where $c \neq 0, -1$), the conditional distribution of $\text{minFac}(m+1) \pmod{q}$ on each nonzero class converges to $1/(q-1) +$

$O(1/q^2)$, with the O -constant depending only on q . By CRT, conditioning on $m \bmod q$ constrains one coordinate; the remaining coordinates are approximately independent for squarefree integers ([crt_multiplier_invariance](#)). The correction term $g(q) - 1/q = 1/(q(q^2-1))$ ([sieveDensity_sub_inv](#)) gives the error scale.

The orbit-to-population transfer—whether the ensemble values $P_n(k)$ inherit the population-level conditional density—is captured by the EnsembleSelectionLemma ([✓ EnsembleSelectionLemma](#)). If both hypotheses hold, the total error in $\mathbb{E}[1/\text{em.}(k)]$ is bounded by a convergent sum $\sum_q O(1/q^3) < \infty$, giving SMLB with a universal constant: $\text{MFREConditional} + \text{EnsembleSelection} \Rightarrow \text{SMLB}(1/6)$ ([✓ MFRECondImpliesSMLB](#)).

A cleaner formulation isolates the population-level content: MinFacSelectionIndependence ([✓ MinFacSelectionIndependence](#)) states that, for $q \geq 3$ and non-death classes $d_1, d_2 \neq -1$, the conditional density of $\text{minFac}(m+1) \equiv a \pmod{q}$ given $m \equiv d_i \pmod{q}$ is the same for d_1 and d_2 asymptotically. MSI implies MFRE_conditional ([✓ MSIImpliesMFREConditional](#)).

The CRT independence thus operates at three levels, documented in [✓ crt_blindness_landscape](#):

1. Pointwise ([crt_multiplier_invariance](#), proved): deterministic—if $P \equiv P' \pmod{r}$ for all $r \neq q$, then $\text{minFac}(P+1) = \text{minFac}(P'+1)$.
2. Population (MSI, open): statistical—conditional density of $\text{minFac} \bmod q$ is independent of the mod- q class.
3. Orbit (= CME/DSL, open): along the specific EM orbit, the multiplier equidistributes.

4.4 The DSL Route to MC

Property 4.22 ([DSL](#) — [✓ DeterministicStabilityLemma](#)). Population equidistribution of multiplier residues implies conditional multiplier equidistribution for the standard orbit ($n = 2$):

$$\text{PE} \implies \text{CME}.$$

Explicitly: if for every prime q and $a \in (\mathbb{Z}/q\mathbb{Z})^\times$ the density of primes $p \equiv a \pmod{q}$ tends to $1/(q-1)$, then for every missing q , non-trivial χ , walk position $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, and $\varepsilon > 0$:

$$\exists N_0 \ \forall N \geq N_0 : \left\| \sum_{\substack{n < N \\ P_{2,q}(n)=c}} \chi(\text{m}_{2,q}(n)) \right\| \leq \varepsilon \cdot N.$$

DSL is an orbit-transfer statement: it asserts that population-level equidistribution (which classical analytic number theory can establish) propagates to the specific deterministic orbit $n = 2$. Note the contrast with the population concentration property (Theorem 4.12), which is already proved: SD implies that almost all squarefree starting points have cancelling character energy. But “almost all” does not cover $n = 2$ specifically. DSL bridges this gap.

[Theorem 4.23](#) ([✓ full_chain_dsl](#)). $\text{DSL} \implies \text{MC}$.

```
-- DSL: population equidistribution transfers to the orbit
def DeterministicStabilityLemma : Prop :=
```

```

PopulationEquidist -> ConditionalMultiplierEquidist

-- The master chain (one open hypothesis, one ANT input)
theorem full_chain_dsl
  (h_equidist : forall (q : Nat), Nat.Prime q -> MinFacResidueEquidist q)
  (hdsl : DeterministicStabilityLemma) :
    MullinConjecture :=
    equidist_dsl_implies_mc h_equidist hdsl

```

Proof sketch. The proof assembles a five-link chain:

1. ANT $\Rightarrow$ PE: MinFacResidueEquidist (a standard consequence of PNT in arithmetic progressions) gives PopulationEquidist ([✓ pe_of_equidist](#)).
2. DSL: PE $\Rightarrow$ CME: DSL converts population equidistribution to conditional multiplier equidistribution for the actual orbit.
3. CME $\Rightarrow$ CCSB: proved for all character orders, bypassing the $d \geq 3$ barrier ([✓ cme_implies_ccsb](#)).
4. CCSB $\Rightarrow$ hits: the Fourier bridge (Theorem 3.3) forces the hit count at -1 to be eventually positive.
5. Hits $\Rightarrow$ MC: the Single Hit Theorem (Theorem 2.20).

Every arrow is fully machine-verified. This collapses the entire reduction landscape to a single open hypothesis. $\square$

Remark 4.24 (Why DSL should hold). DSL asserts that population equidistribution of $\min\text{Fac}$ residues transfers to the specific EM orbit. Three structural features of the EM sequence support this expectation.

(1) The information bottleneck. At step n , $P_2(n)$ has $\sim 2^n$ bits, but the walk position $P_{2,q}(n) = P_2(n) \bmod q$ carries only $O(\log q)$ bits. The smallest prime factor of $P_2(n) + 1$ depends on the full $\sim 2^n$ -bit value, not on its $O(\log q)$ -bit residue. Conditioning on $P_{2,q}(n) = c$ constrains almost nothing about the factoring behavior of $P_2(n) + 1$: the conditional and unconditional distributions of $\min\text{Fac}(P_2(n) + 1) \bmod q$ should agree, which is exactly what CME says. This is the same mechanism that makes iterated hash chains pass statistical tests for independence: the forward map is easy, but projecting an exponentially large integer onto a small residue destroys all correlation.

(2) No arithmetic obstruction. The products $P_2(n)$ are squarefree ([✓ prod_squarefree](#)), pairwise multiplicatively entangled (each divides the next), and each $P_2(n) + 1$ has a distinct smallest prime factor (em_2 is injective). There is no known arithmetic mechanism—congruence obstruction, algebraic identity, or Galois constraint—that would systematically bias $\min\text{Fac}(P_2(n) + 1)$ toward or away from particular residue classes mod q , conditional on the walk position. Every structural feature that could create such a bias (coprime cascade, CRT propagation, cofactor identity) has been exhaustively checked and found insufficient (§5).

(3) The ensemble evidence. The ensemble framework parametrizes the EM construction by squarefree starting points. For almost all starting points n , CME holds: the proved StepDecorrelation $\Rightarrow$ cancellation chain ([✓ sd_implies_cancellation](#)) shows that generic orbits satisfy the character sum bounds. DSL asks that the standard orbit $n = 2$ is not exceptional within this ensemble. Since $n = 2$ has no special arithmetic property that distinguishes it from a typical squarefree starting point—it is not a prime power, not a perfect square, not in any special congruence class—there is no reason to expect it to be an outlier.

DSL is therefore a typicality assertion: the EM orbit behaves like a generic member of its ensemble. This is analogous to Hooley’s conditional proof of Artin’s conjecture [9], which shows the conjecture for all integers conditional on GRH—the hypothesis being that the specific integers of interest are

not exceptional among all integers. The difficulty of proving DSL is the difficulty of proving that a specific deterministic sequence satisfies a statistical law.

Remark 4.25 ([✓oce_eq_cme](#)). An alternative perspective on DSL is the CRT pointwise transfer: condition on the walk position $c = P_{2,q}(n)$ and ask whether the multiplier residues in each fiber are equidistributed. This orbit-conditional equidistribution (OCE) turns out to be definitionally equal to CME—proved by [rfl](#), since the walk position is the residue class that CME conditions on. Any route to OCE automatically gives CME, and the CRT bridge (population conditional equidistribution $\Rightarrow$ OCE) is subsumed by DSL ([✓dsl_implies_crt_bridge](#)).

A weaker variant also suffices:

Property 4.26 ([DSLHitting](#)). $PE \Rightarrow DH$: population equidistribution implies that the walk hits -1 cofinally. This is strictly weaker than DSL (which gives $PE \Rightarrow CME$): [DSLHitting](#) asks only for the walk to visit -1 , not for full character sum cancellation.

Theorem 4.27 ([✓pe_dsl_hitting_implies_mc](#)). $PE + \text{DSLHitting} \implies MC$.

The reduction is immediate: $PE \xrightarrow{\text{DSLHitting}} DH \xrightarrow{\text{proved}} MC$.

Remark 4.28 (The tail identity and DSL). The tail identification (Theorem 2.2) gives a precise structural link between the standard orbit and the ensemble. Since $P_2(M)$ is squarefree for all M ([✓prod_squarefree](#)), the M -th accumulator is itself a valid ensemble member. The tail identity says the orbit starting from $P_2(M)$ is the suffix of the standard orbit from step M onward:

$$P_{P_2(M)}(k) = P_2(M+k), \quad \text{em}_{P_2(M)}(k) = \text{em}_2(M+k).$$

Consequently, the character energy of the ensemble member $n = P_2(M)$ over a window of length K equals the character energy of the standard orbit over $[M, M+K)$:

$$E(P_2(M), K) = E_{\text{tail}}(2, M, K).$$

The ensemble concentration result ($SD \Rightarrow CSVB \Rightarrow$ concentration, Theorem 4.12) says: for almost all squarefree n , $E(n, K) = o(K^2)$. So the question “does the standard orbit have cancelling character energy on long windows?” is the same as “does the ensemble member $P_2(M)$ fall outside the bad set?”—for every M .

The Borel–Cantelli program. Let $B_\epsilon(K)$ denote the set of squarefree n with $E(n, K) > (\epsilon K)^2$. The first-moment (Markov) bound gives bad density $O(1/K)$, which is not summable over windows and therefore insufficient for a Borel–Cantelli argument. The second-moment (Chebyshev) approach resolves this: if the population fourth moment satisfies $\mathbb{E}_{\text{sqf}}[E(n, K)^2] = O(K^2)$, then Chebyshev gives

$$\Pr_{\text{sqf}}[n \in B_\epsilon(K)] \leq \frac{D}{\epsilon^4 K^2},$$

which is summable. Applying this to dyadic windows $[2^i, 2^{i+1})$ with $K = 2^i$: $\sum_i D/(\epsilon^4 \cdot 4^i) < \infty$, so by Borel–Cantelli, for all large i the character energy on the window $[2^i, 2^{i+1})$ is at most $(\epsilon \cdot 2^i)^2$. Summing over windows covering $[0, N)$:

$$|F(c, [0, N))| \leq \sum_{i: 2^i < N} \epsilon \cdot 2^i \leq 2\epsilon N,$$

which gives $|F(c, N)| = o(N)$, i.e. CME.

The sole new content: four-point PCV. The fourth moment $\mathbb{E}[E(n, K)^2]$ decomposes into a diagonal contribution K^2 , a cross-term contribution that vanishes by PCV, and a four-point correlation

$$\Gamma_4(j_1, k_1, j_2, k_2) = \mathbb{E}_{\text{sqf}}[\text{Re}(\chi(m_{j_1})\overline{\chi(m_{k_1})}) \cdot \text{Re}(\chi(m_{j_2})\overline{\chi(m_{k_2})})].$$

For distinct pairs $(j_1, k_1) \neq (j_2, k_2)$ with all four indices distinct, CRT independence should force $\Gamma_4 \rightarrow 0$ via factorization into two-point correlations. If this four-point PCV holds, $\mathbb{E}[E(n, K)^2] = O(K^2)$, and the Borel–Cantelli program closes DSL.

The Substitution Principle (and its limits). An equivalent reformulation of CME, proved definitionally equal ([sp_eq_cme](#)), works at the level of fiber return visits: the steps k where $P_{(q)}(\cdot)k = a$. At such steps the multiplier primes $\text{em}_{(k+1)}$ are distinct ([return_visit_mult_injective](#)) and pairwise coprime ([return_visit_mult_coprime](#)). The structural package is recorded in a 4-tuple witness [sp_structural_summary](#): injectivity, coprimality, primality, and divisibility. Four independent routes from SP to MC are assembled in [all_routes_to_mc_with_sp](#).

However, the coprime cascade structure alone does not force character cancellation: for any $q \geq 3$, one can construct pairwise coprime integers $N_i = p_i r_i$ in a fixed residue class $\text{mod } q$ whose least prime factors all lie in the same residue class (e.g. $p_i \equiv 2, r_i \equiv 3 \pmod{5}$ gives $\sum \chi(\text{minFac}(N_i)) = I \cdot \chi(2) = \Theta(I)$). Thus the cancellation in CME, if true, must rely on the specific feedback dynamics of the EM sequence—the fact that each $\text{em}_{(n+1)} = \text{minFac}(P(n)+1)$ depends on all prior terms—not merely on coprimality. The “Substitution Principle” is therefore just CME under a structurally suggestive name; the name does not isolate a new mechanism.

The orbit-specificity gap. Even the Chebyshev bound is a statement about the density of bad starting points, not a pointwise statement about the specific values $P_2(M)$. A density-zero set can still contain a specific super-exponentially growing sequence. However, the $O(1/K^2)$ rate makes the gap much narrower: combined with CRT typicality of $P_2(M) \text{ mod } q$ (the residues form a walk on $(\mathbb{Z}/q\mathbb{Z})^\times$ visiting every class cofinally, by the proved [SubgroupEscape](#) and [prime_residue_escape](#)), the gap reduces to a quantitative equidistribution-in-residue-classes statement. This is the sharpest form of the orbit-specificity barrier.

Remark 4.29 (Self-correcting feedback and visit equidistribution). The preceding attacks target CME (conditional multiplier equidistribution) or its equivalents. A structurally different route bypasses CME entirely: visit equidistribution (VE).

Define the empirical frequency vector

$$f_c(N) = \frac{1}{N} |\{n < N : P_{2,q}(n) = c\}|, \quad c \in (\mathbb{Z}/q\mathbb{Z})^\times.$$

VE asserts $f_c(N) \rightarrow 1/\varphi(q)$ for every c . By the Weyl criterion this is equivalent to walk character cancellation $\|\sum_{n < N} \chi(P_{2,q}(n))\|^2 = o(N^2)$, and the chain

$$\text{VE} \xrightarrow[\text{proved}]{\text{Weyl}} \text{WalkCharCancel} \xrightarrow[\text{proved}]{\text{WeylHittingBridge}} \text{MC}$$

is fully machine-verified ([sve_iff_ve](#), [cancel_weyl_implies_mc](#)).

VE is strictly different from CME. CME controls the conditional distribution of multipliers given walk position; VE controls the marginal distribution of walk positions. Neither implies the other: equidistributed walk positions do not force equidistributed conditional multipliers, and vice versa. This is not an equivalence collapse.

The self-correcting feedback mechanism. The EM walk has a built-in negative feedback loop. Suppose $f_c(N)$ is anomalously large: the walk visits position c too often. At each such visit, $P_2(n) \equiv c \pmod{q}$, so $P_2(n) + 1 \equiv c + 1 \pmod{q}$. The next multiplier is $\text{em}_{n+1} = \text{minFac}(P_2(n) + 1)$, an

integer whose residue class mod q is determined by $P_2(n) + 1$. By Alladi’s theorem [11] (axiomatized as [GenericLPFEquidist](#)), the least prime factors of rough integers in a fixed residue class are equidistributed among all nonzero residues mod q . If the multiplier at position c is approximately equidistributed, the transition $c \mapsto c \cdot m(n)$ scatters the walk away from c toward all other positions—a negative feedback that opposes further concentration at c .

This has been formalized via a Lyapunov function ([Reduction/SelfCorrecting.lean](#)). Define

$$L(N) = \sum_{c \in (\mathbb{Z}/q\mathbb{Z})^\times} \left(V_c(N) - \frac{N}{\varphi(q)} \right)^2,$$

where $V_c(N) = |\{n < N : P_{2,q}(n) = c\}|$ is the visit count. The one-step algebraic identity ([✓ lyapunov_one_step](#))

$$L(N+1) = L(N) + 2d_{P_{2,q}(N)}(N) + \frac{q-2}{q-1},$$

where $d_c(N) = V_c(N) - N/(q-1)$ is the visit deviation at position c , telescopes to ([✓ lyapunov_telescope](#))

$$L(N) = 2R(N) + N \frac{q-2}{q-1}, \quad R(N) = \sum_{n < N} d_{P_{2,q}(n)}(n).$$

The cumulative drift $R(N)$ records how often the walk visits positions that are already over-represented. Under a uniform multiplier measure, $\mathbb{E}[d_c] = 0$ for every position ([✓ uniform_multiplier_zero_drift](#)), and the transition matrix is doubly stochastic ([✓ group_walk_doubly_stochastic](#)), so the drift has zero bias. SelfCorrectingDrift (SCD) asserts $R(N) = o(N^2)$, which gives $L(N) = o(N^2)$, hence VE, and therefore MC via the proved chain ([✓ scd_implies_mc](#)).

SCD $\equiv$ SVE (Dead End #120). The Lyapunov telescope $L(N) = 2R(N) + O(N)$ shows that SCD ($R(N) = o(N^2)$) is algebraically equivalent to SVE ($L(N) = o(N^2)$) for each fixed prime q above the threshold. The converse is also proved: SVE implies SCD above the SVE threshold ([✓ sve_implies_scd_above_threshold](#)). Therefore SCD provides no new proof leverage beyond SVE; it is a Lyapunov reformulation that clarifies the dynamical mechanism (self-correcting feedback) but does not reduce the mathematical difficulty.

What remains. The residual difficulty is orbit-specificity: the Alladi-type equidistribution holds for “generic” rough integers, but $P_2(n) + 1$ is a specific value, not a random draw from its residue class. The Lyapunov analysis shows that proving $R(N) = o(N^2)$ requires controlling how often the walk visits positions that are already over-represented—and this control depends on the specific EM orbit, not just generic multiplier statistics. This is the same orbit-specificity barrier from §4.7, now expressed at the level of single-step drift rather than long-range character cancellation.

Per-class bad-set escape (Dead End #121). A natural attempt: use SMSB (SecondMomentSquared-Bound) to bound the global density of “bad” steps (those whose tail has large character energy), then argue that each walk-position class has proportionally few bad steps, then use SubgroupEscape to get escape from each class. The per-class bad sets partition the total bad set ([✓ badVisitsInClass_card_sum_eq](#)), and pigeonhole guarantees some class with small bad density ([✓ exists_class_small_bad_set](#)). But MC requires the specific class -1 , not an existential over classes. Targeting -1 requires that bad steps do not concentrate in one class—a per-fiber density statement (BadSetEquidistribution) that is a consequence of CME. Going from the marginal bad-density bound (what SMSB provides) to the conditional bound (what per-class escape requires) is blocked by the Marginal/Joint Barrier: the joint distribution of (walk position, badness) is controlled by CME, not by SMSB alone.

4.5 Generalized MC and the Weyl Hitting Bridge

The ensemble framework requires proving MC not just for $n = 2$ but for arbitrary squarefree starting points. This motivates two key results.

Definition 4.30 ([GenMullinConjecture\(\$n\$ \)](#)). Every prime appears in the generalized EM sequence from starting point n : $\forall q$ prime, $\exists k, \text{em}_n(k) = q$.

Theorem 4.31 ([✓ gen_hitting_implies_gen_mc_proved](#)). For squarefree $n \geq 1$: if the generalized walk hits -1 cofinally for every prime q , then [GenMullinConjecture\(\$n\$ \)](#) holds.

Proof. Strong induction on q . Assume every prime $p < q$ appears as $\text{em}_n(k_p)$ for some k_p . Collect all witnesses into a uniform bound N . Cofinal hitting at q provides $k \geq N$ with $q \mid P_n(k) + 1$. Since all primes $p < q$ already appeared at steps $< k$, they divide $P_n(k)$ and hence cannot divide $P_n(k) + 1$. So $q = \min\text{Fac}(P_n(k) + 1) = \text{em}_n(k)$.

The argument is a generalized [captures_target](#) lemma ([✓ gen_captures_target](#)), paralleling the standard version from §2. $\square$

The connection between character cancellation and hitting uses the Weyl equidistribution criterion, adapted to the multiplicative setting:

Theorem 4.32 ([WeylHittingBridge](#) — [✓ weyl_hitting_bridge_proved](#)). If all non-trivial walk character sums satisfy $\|\sum_{k < K} \chi(P_{n,q}(k))\|^2 \leq (\epsilon K)^2$ for large K , then either q appears in $\text{em}_n(\cdot)$ or the walk hits -1 cofinally.

Proof. By contradiction using a Weyl test function. Suppose q never appears and the walk stops hitting -1 past some step N_0 . Then the hit count $H(K) = |\{k < K : P_n(k) \equiv -1 \pmod{q}\}|$ is bounded by the constant $C = H(N_0)$ for all K .

Define a centered test function $\chi_{\text{test}} : \mathbb{N} \rightarrow \mathbb{C}$:

$$\chi_{\text{test}}(x) = \begin{cases} 0 & \text{if } x \equiv 0 \pmod{q}, \\ (q-2)/(q-1) & \text{if } x \equiv -1 \pmod{q}, \\ -1/(q-1) & \text{otherwise.} \end{cases}$$

The test function satisfies $\|\chi_{\text{test}}\|^2 \leq 1$, $\chi_{\text{test}}(0) = 0$, and $\sum_{a \in \mathbb{Z}/q\mathbb{Z}} \chi_{\text{test}}(a) = 0$ (the sum of $(q-2)/(q-1)$ from $a = q-1$ and $(q-2) \cdot (-1/(q-1))$ from the remaining units is zero).

The partial sum of the test function along the walk telescopes to $\sum_{k < K} \chi_{\text{test}}(P_n(k)) = H(K) \cdot (q-2)/(q-1) + (K - H(K)) \cdot (-1/(q-1)) = H(K) - K/(q-1)$. So the walk character energy equals

$$E_{\text{test}}(K) = (H(K) - K/(q-1))^2.$$

Since $H(K) \leq C$ while $K/(q-1) \rightarrow \infty$, for large K we have $E_{\text{test}}(K) \geq (K/(q-1) - C)^2$. Setting $\epsilon = 1/(2(q-1))$, the hypothesis gives $E_{\text{test}}(K) \leq (K/(2(q-1)))^2$. Taking square roots: $K/(q-1) - C \leq K/(2(q-1))$, hence $K/(2(q-1)) \leq C$, so $K \leq 2C(q-1)$. But K was chosen larger than $2Cq + 1$: contradiction. $\square$

Theorem 4.33 ([✓ cancel_weyl_implies_mc](#)). Walk character cancellation at $n = 2$ implies MC.

Proof. For $q = 2$: $\text{em}_2(0) = 2$. For $q \geq 3$: q does not divide 2, so the Weyl Hitting Bridge gives either $q \in \text{em}_2(\cdot)$ (hence $q \in \{\text{em}_2(k)\}$) or cofinal hitting. Cofinal hitting gives $q \in \text{em}_2(\cdot)$ by [Theorem 4.31](#), noting that all generalized primes from $n = 2$ are odd (since $2 \mid P_2(k)$ implies $P_2(k) + 1$ is odd), so the induction only requires odd primes $< q$, all of which have already appeared by the inductive hypothesis. $\square$

The Per-Chi Cancellation Bridge

The decorrelation chain of §4.2 works with universal StepDecorrelation (quantified over all characters). For the JSE route, only per-chi decorrelation is available (for each individual non-trivial character). The PerChiCancellationBridge shows the same three-step chain works per-chi:

Theorem 4.34 ([✓ per_chi_cancellation_bridge_proved](#)). [PerChiCancellationBridge](#): for each prime q and character χ with $\|\chi\|^2 \leq 1$, if the cross-terms $\mathbb{E}^{\text{sqf}}[\text{Re}(\chi(m_j) \cdot \overline{\chi(m_k)})] \rightarrow 0$ for all $j < k$, then the density of squarefree n with $E(n, K) > (\varepsilon K)^2$ for all K tends to zero.

Proof. The proof is identical to the universal chain (Theorem 4.8 $\rightarrow$ concentration $\rightarrow$ cancellation), with the StepDecorrelation hypothesis restricted to the fixed character χ . Each step (variance with $C = 2$, Markov concentration, squeeze to zero) goes through unchanged because none requires decorrelation of other characters. $\square$

The Ensemble Chain in Detail

The full ensemble assembly uses the following master theorem:

Theorem 4.35 ([✓ ensemble_pt_master_simplified](#)). Under four hypotheses (SRE, CRTPropagationStep, AccumEquidistImpliesMultEquidist, EnsembleEquidistImpliesDecorrelation), character sums cancel for almost all squarefree starting points.

The chain:

$$\underbrace{\text{SRE} + \text{CRT} + \text{Bridge}}_{\text{ANT} + \text{CRT structure}} \xrightarrow{\text{proved}} \text{EME} \xrightarrow{\text{open}} \text{SD} \xrightarrow{\text{proved}} \text{cancellation for a.a. } n.$$

The first arrow composes three proved reductions. The second arrow ($\text{EME} \Rightarrow \text{SD}$) is the sole remaining open hypothesis. The third arrow is Theorem 4.12.

The JSE route.

An alternative chain uses JointStepEquidist (JSE): the joint distribution of $(\text{em}_n(j), \text{em}_n(k))$ is asymptotically uniform over non-zero pairs. The JSE base case ($j = 0, k = 1$) is proved via the tower rule ([✓ jse_base_case](#)): $P(A, B) = P(A | B) \cdot P(B)$, decomposing the joint density into conditional and marginal equidistribution.

Theorem 4.36 ([✓ jse_transfer_implies_mc](#)). $\text{JSE} + \text{MultCancelToWalkCancel} \Rightarrow \text{MC}$.

The composition:

1. JSE gives per-chi decorrelation for non-trivial χ ([✓ joint_step_equidist_implies_step_decorrelation](#)).
2. PerChiCancellationBridge (Theorem 4.34) gives multiplier character cancellation for almost all squarefree n .
3. MultCancelToWalkCancel (open): multiplier energy $o(K^2)$ implies walk energy $o(K^2)$.
4. WeylHittingBridge (Theorem 4.32) converts walk character cancellation into cofinal hitting.
5. Generalized MC (Theorem 4.31) and standard-EM bridge (Theorem 4.33) give MC.

The two remaining open hypotheses are:

- **JointStepEquidist**: requires analytic number theory (Siegel–Walfisz or Bombieri–Vinogradov type estimates).
- **MultCancelToWalkCancel**: multiplier character cancellation implies walk character cancellation. Dead End #117 shows this is equivalent to CCSB/CME in difficulty: an explicit counterexample (alternating multipliers $\{2, 3\} \bmod 5$ with $\chi(2) = i$) gives multiplier cancellation yet $|W_K| = \Theta(K)$, and this counterexample is compatible with all EM structural properties (squarefree accumulator, coprime cascade, CRT invariance).

MultCancelToWalkCancel is the Marginal/Joint Barrier (§5.1) in ensemble form: the multiplier character sum is a marginal statistic (it depends on the individual multiplier values), while the walk character sum is a joint statistic (it depends on all previous multipliers through the partial product). Cancellation of individual terms does not imply cancellation of their cumulative products. This is why the DSL route (Section 4.4) bypasses the ensemble-to-orbit gap entirely: DSL gives CME directly, which implies walk character cancellation without ever needing the multiplier-to-walk transfer.

The Variance Route to DSL

The **DSL**Variance framework ([Reduction/DSL](#)[Variance.lean](#)) provides an alternative approach to DSL via the second moment of character energy.

Definition 4.37 (**Population Character Energy**). The ensemble average of character energy: $\mathcal{E}(q, \chi, K, X) = \mathbb{E}_{n \leq X}^{\text{sqf}}[E(n, K)]$.

Theorem 4.38 ([✓ populationCharEnergy_nonneg](#)). $\mathcal{E} \geq 0$.

The population energy decomposes via cross-term pairs:

Definition 4.39 (**CrossTermPair**). For steps $j < k$: $\Gamma(j, k) = \mathbb{E}_{n \leq X}^{\text{sqf}}[\text{Re}(\chi(m_j) \cdot \overline{\chi(m_k)})]$.

Theorem 4.40 ([✓ crossTermPair_comm](#)). $\Gamma(j, k) = \Gamma(k, j)$ (since $\text{Re}(z \cdot \bar{w}) = \text{Re}(w \cdot \bar{z})$).

Property 4.41 (**PairwiseCrossTermVanishing** (**PCV**)). $\Gamma(j, k) \rightarrow 0$ as $X \rightarrow \infty$ for each fixed pair $j < k$.

Theorem 4.42 ([✓ stepDecorrelation_implies_pairwiseVanishing](#)). $\text{SD} \implies \text{PCV}$.

Theorem 4.43 ([✓ charSumVarianceBound_implies_secondMomentBound](#)). $\text{CSVB}(\mathcal{C}) \implies \text{SecondMomentBound}(\mathcal{C})$: the population energy is at most CK .

The variance route aims to reach DSL via $\text{PCV} \Rightarrow \text{SMB} \Rightarrow \text{concentration} \Rightarrow \text{DSL}$. The first two arrows face a technical gap, and the last faces the fundamental orbit-transfer barrier:

- $\text{PCV} \Rightarrow \text{SMB}$: requires uniform convergence across all $K(K-1)/2$ cross-term pairs, not just pointwise.
- $\text{Concentration} \Rightarrow \text{DSL}$: the proved chain $\text{SD} \Rightarrow \text{CSVB} \Rightarrow \text{Concentration}$ (Theorem 4.12) already gives cancellation for almost all squarefree starting points. But DSL asks for CME at the specific orbit $n = 2$, and a density-one set of good starting points can exclude any particular n . This is the orbit-specificity barrier in its sharpest form.

4.6 Visit Equidistribution and the Spectral Identity

The ensemble framework connects to the spectral route via a fundamental identity relating character energy to the geometry of the walk. For a walk of length N on $(\mathbb{Z}/q\mathbb{Z})^\times$, define the visit count $V_N(a) = |\{n < N : P_{2,q}(n) = a\}|$.

Theorem 4.44 (✓ [excessEnergy_eq_visit_deviation](#)). The excess energy $E_\chi(N) - N$ equals $(q-1)$ times the squared deviation of visit counts from the uniform distribution:

$$\sum_{\chi \neq \chi_0} \|S_\chi(N)\|^2 = (q-1) \sum_{a \in (\mathbb{Z}/q\mathbb{Z})^\times} \left(V_N(a) - \frac{N}{q-1} \right)^2.$$

This establishes the equivalence [SVE](#) $\Leftrightarrow$ [VE](#) (✓ [sve_iff_ve](#)): subquadratic visit energy (the total non-trivial character energy is $o(N^2)$) is the same as visit equidistribution (the visit counts are approximately uniform). The identity connects the Fourier-analytic perspective (character sums) with the geometric perspective (visit counts) on the walk:

- SVE says the walk has no sustained spectral bias: projecting onto any non-trivial character gives cancellation.
- VE says the walk has no sustained spatial bias: it visits every position approximately equally often.

These are dual formulations of the same condition, connected by Parseval's identity on the finite group $(\mathbb{Z}/q\mathbb{Z})^\times$.

The fiber decomposition (§3.4) refines this further:

Theorem 4.45 (✓ [fiberMultCharSum_step](#)). The fiber character sum $F(a, N) = \sum_{\substack{n < N \\ w(n)=a}} \chi(m(n))$ satisfies the recurrence: $F(a, N+1) = F(a, N) + \chi(m(N)) \cdot \mathbf{1}_{w(N)=a}$.

This step-by-step evolution of fiber sums provides the infrastructure for the $\text{CME} \Rightarrow \text{CCSB}$ proof via energy tracking: the total character energy decomposes into active-fiber contributions at each step, and CME bounds each fiber's contribution.

Remark 4.46 (Almost-all MC and the multiplier/walk gap). A natural question: since SD gives character sum cancellation for almost all squarefree starting points (Theorem 4.12), does this yield [GenMullinConjecture\(n\)](#) for almost all n ?

It does not. The cancellation that SD produces is for the multiplier character energy $\|\sum_{k < K} \chi(\text{em}_n(k) \bmod q)\|^2 = o(K^2)$ —the individual multiplier residues have cancelling character sums. But the Weyl Hitting Bridge (Theorem 4.32), which converts character cancellation into cofinal hitting, requires walk character cancellation: $\|\sum_{k < K} \chi(P_{n,q}(k))\|^2 = o(K^2)$. The walk is a partial product of the multipliers— $P_{n,q}(k) = \prod_{j < k} m_{n,q}(j)$ —and cancellation of a sum $\sum \chi(m_k)$ does not imply cancellation of $\sum \prod_{j < k} \chi(m_j)$.

This is Dead End #117 (MultCancelToWalkCancel, §5.1): an explicit counterexample (alternating multipliers $\{2, 3\} \bmod 5$ with $\chi(2) = i$) achieves multiplier cancellation yet $|W_K| = \Theta(K)$, and is compatible with all EM structural properties. The chain therefore breaks as:

$$\text{SD} \xrightarrow{\text{proved}} \text{a.a. multiplier cancellation} \xrightarrow{\#117} \text{a.a. walk cancellation} \xrightarrow{\text{proved}} \text{a.a. GenMC}.$$

The last arrow composes the Weyl bridge with Theorem 4.31; the first arrow is Theorem 4.12; the middle arrow is the marginal/joint barrier, which persists orbit by orbit. Ensemble averaging helps with multiplier statistics but cannot cross this barrier: the walk character sum is a joint statistic

(depending on all previous multipliers through the partial product), while the multiplier character sum is a marginal one.

This is why DSL targets CME rather than multiplier cancellation: CME goes directly from conditional multiplier equidistribution to walk cancellation (via the fiber decomposition and telescoping), bypassing MultCancelToWalkCancel entirely.

4.7 Tail Window Decorrelation

The tail identity (Theorem 2.2) connects temporal blocks of the standard EM orbit to the ensemble framework: the character sum over steps $jK+1, \dots, (j+1)K$ of the standard orbit equals the generalized partial sum starting from $P_2(jK)$. This motivates decomposing the full character sum into non-overlapping windows.

Definition 4.47 ([WindowCharSum](#)). The character sum over the j -th window of length K : $S_j = \sum_{k < K} \chi(\text{em}_2(jK+k+1) \bmod q)$.

Theorem 4.48 ([windowCharSum_eq_genSeqCharPartialSum](#)). S_j equals the generalized character partial sum starting from $P_2(jK)$: $S_j = \sum_{k < K} \chi(\text{genSeq}(P_2(jK), k) \bmod q)$.

Theorem 4.49 ([charSum_eq_window_sum](#)). The total character sum decomposes: $\sum_{k < JK} \chi(\text{em}_2(k+1) \bmod q) = \sum_{j < J} S_j$.

Applying the variance identity ([norm_sq_sum_eq_diag_plus_cross](#)), the total character energy decomposes as

$$\left\| \sum_{j < J} S_j \right\|^2 = \sum_{j < J} |S_j|^2 + 2 \sum_{j < l < J} \text{Re}(S_j \cdot \overline{S_l}).$$

The diagonal terms are bounded by $J \cdot K^2$ ([diag_energy_le](#)). The cross terms—the off-diagonal correlations between windows at different temporal positions—are the new quantity of interest.

Property 4.50 ([TailWindowDecorrelation](#) (TWD)). The average absolute cross term between non-overlapping windows vanishes as the window size K grows:

$$\frac{1}{J^2} \sum_{j < l < J} |\text{Re}(S_j \cdot \overline{S_l})| \rightarrow 0.$$

Theorem 4.51 ([twd_implies_mc_from_bridge](#)). $\text{TWDImpliesCCSB} + \text{TWD} \implies \text{MC}$.

The proof composes the open bridge TWDImpliesCCSB with the proved chain $\text{CCSB} \implies \text{MC}$. The bridge TWDImpliesCCSB would follow from TWD by choosing $K = \lceil 1/\varepsilon \rceil$: the diagonal contributes $N \cdot K = O(\varepsilon^{-1}N)$ while the cross terms contribute $O(\varepsilon N^2)$, giving $\|S(N)\|^2 = O(\varepsilon N^2)$ for all $\varepsilon > 0$, hence CCSB.

Remark 4.52 (TWD vs. ensemble decorrelation). TWD is a temporal decorrelation statement about the standard EM orbit: windows far apart in the sequence should have nearly independent character sums. The physical intuition is that the accumulator $P_2(jK)$ grows super-exponentially, so the starting points of consecutive windows are extremely different in magnitude, making it hard for any fixed modulus q to see systematic correlation between them.

This contrasts with the fiber decorrelation used in the variance route (§4.5): there, one averages

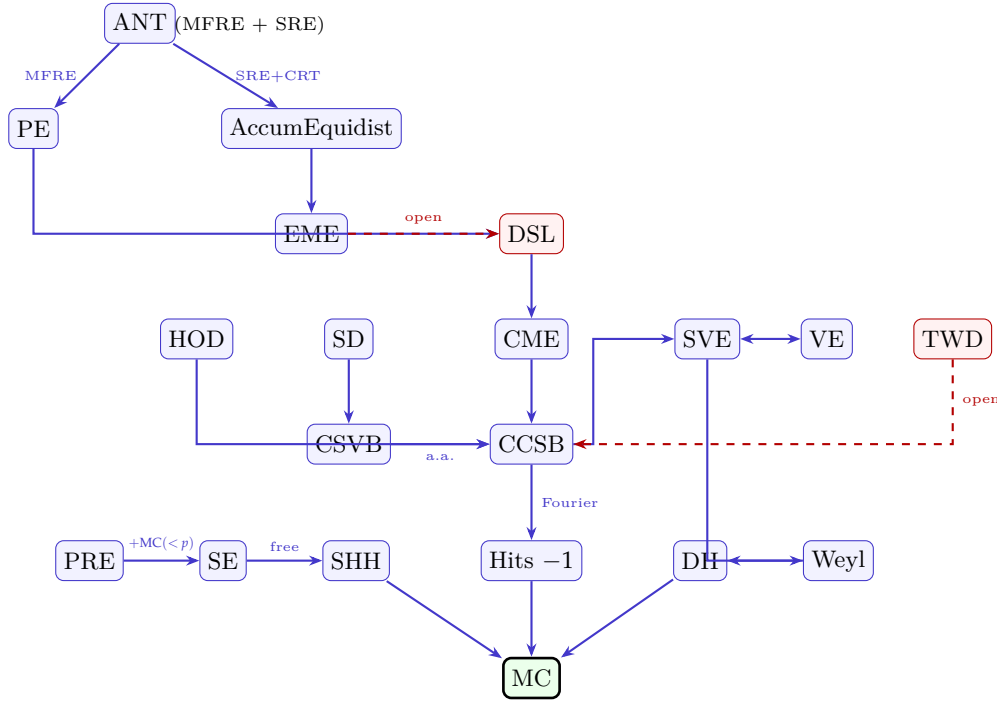


Figure 1: Dependency graph of the main reductions. Solid blue arrows: machine-verified. Dashed red arrows: open hypotheses (DSL, TWD $\Rightarrow$ CCSB). All abbreviations are defined in the glossary (Appendix A.6).

over squarefree starting points at two fixed time steps j, k . TWD averages over time windows at the single starting point $n = 2$. The two approaches face different barriers: the variance route faces the orbit-specificity barrier (density-one results cannot pin down $n = 2$), while TWD faces the temporal-correlation barrier (consecutive windows share the accumulator prefix $P_2(jK)$). The file [Reduction/TailWindow.lean](#) (477 lines, zero sorry) formalizes the window definitions, the tail identity connection, the variance identity, the diagonal and cross-term bounds, and the conditional chain TWD $\Rightarrow$ CCSB $\Rightarrow$ MC.

The Reduction Landscape

Figure 1 shows the complete dependency graph of the formalization. Every solid arrow is a machine-verified implication with zero sorry. The single dashed arrow marks the sole remaining open hypothesis: the Deterministic Stability Lemma (DSL), which asks whether population equidistribution (a consequence of standard analytic number theory) transfers to conditional multiplier equidistribution for the specific orbit $n = 2$.

The graph reveals the architecture: multiple independent paths converge on MC. The left path (PRE $\rightarrow$ SE $\rightarrow$ SHH $\rightarrow$ MC) is the inductive bootstrap of Section 2, proved unconditionally—it reduces MC to producing one hit on -1 at each prime. The central path (ANT $\rightarrow$ PE $\rightarrow$ DSL $\rightarrow$ CME $\rightarrow$ CCSB $\rightarrow$ hits $\rightarrow$ MC) is the ensemble route of this section—it produces the required hit via harmonic analysis, conditional on DSL. The right path (SVE $\leftrightarrow$ VE, Weyl $\rightarrow$ DH $\rightarrow$ MC) provides an alternative spectral route. Two additional

routes reach CCSB: $\text{HOD} \rightarrow \text{CCSB}$ (proved, via van der Corput) and $\text{TWD} \rightarrow \text{CCSB}$ (open bridge, via temporal window decorrelation). All solid arrows are machine-verified; the two dashed arrows ($\text{DSL}, \text{TWD} \Rightarrow \text{CCSB}$) are the open hypotheses.

Theorem 4.53 ([✓ dsl_closes_all](#)). Under standard ANT (MinFacResidueEquidist) plus DSL: $\text{MC} \wedge \text{CCSB}$.

5 Open Problems and Paths Forward

Original contribution. The selectability analysis, oracle barrier, and CCSB-as-frontier argument below are new. They explain why the remaining hypothesis resists both computation and existing analytic tools.

Section roadmap

- §5.1 Selectability — fresh primes exist but `minFac` may not select them; marginal/joint barrier.
- §5.2 BRE impossibility — block rotation fails for character order $d \geq 3$.
- §5.3 Independence heuristic — factorization destroys correlation (heuristic, not proved); dead ends as roadmap.
- §5.4 NAAD conjecture — non-algebraic arithmetic dynamics; relation to Collatz.
- §5.5 The precise frontier — CCSB, Bombieri–Vinogradov, Chebotarev, sieve approaches; DSL as sole hypothesis; adelic/profinite view; what would close the conjecture.

5.1 The Selectability Perspective

The Euclid construction guarantees fresh primes at every step: every prime factor of $P_2(n) + 1$ is new (coprime to the running product). The difficulty of MC is not the existence of new primes but whether the `minFac` rule eventually selects each one. The formalization makes this contrast precise.

Theorem 5.1 ([✓ divisor_not_yet_in_seq](#)). If $p \mid P_2(n) + 1$, then $\text{em}_2(m) \neq p$ for all $m \leq n$.

Proof. Any $\text{em}_2(m)$ with $m \leq n$ divides $P_2(n)$. A number ≥ 2 cannot divide both a and $a + 1$. $\square$

Theorem 5.2 ([✓ passed_over_persists](#)). If $p \mid P_2(n) + 1$ but $\text{em}_2(n+1) \neq p$ (the `minFac` rule chose a smaller prime), then $\text{em}_2(m) \neq p$ for all $m \leq n + 1$. The prime survives to potentially divide future Euclid numbers.

Theorem 5.3 ([✓ selectability_extinguished](#)). Once $\text{em}_2(m) = p$, we have $p \mid P_2(n)$ for all $n \geq m$, so $p \nmid P_2(n) + 1$ ever again. Selectability is a one-shot resource.

Definition 5.4 ([InfinitelySelectable](#)). A prime p is infinitely selectable if $p \mid P_2(n) + 1$ for cofinally many n : $\forall N, \exists n \geq N, p \mid P_2(n) + 1$.

By Theorem 5.3, $\text{MC}(p)$ and $\text{InfinitelySelectable}(p)$ are mutually exclusive ([✓ mc_implies_not_infinitely_selectable](#)): a prime that enters the sequence can never be selectable again.

Theorem 5.5 ([✓ dh_implies_infinitely_selectable](#)). Under DH, every prime that never appears in the sequence (with SE satisfied) is infinitely selectable.

The comparison between the `minFac` rule and alternative selection rules—random factor, `{minFac, secondMinFac}`, and the function field analog—reveals why independence is the key missing ingredient. Section 6 develops these variants in detail; the core observation is that the random variant possesses independence of multiplier from walk position (the sole property needed for mixing), while `minFac` trades independence for perfect selection efficiency. Stochastic domination fails because the two processes diverge after one step (§6.1).

The Marginal/Joint Barrier

The verified reductions (TailSE, CofinalEscape, QuotientDH) exhaust what can be proved about the marginal distribution of multiplier residues.

Theorem 5.6 ([✓ emfe_iff_tail_se_at](#)). $\text{EuclidMinFacEscape}(q) \Leftrightarrow \text{TailSE}(q)$.

Even perfect per-position equidistribution of multipliers is consistent with HH failure. DH is a joint statement—the (position, multiplier) pair must hit the death curve $\mathbf{m}_{2,q}(n) = -P_{2,q}(n)^{-1}$ —and no marginal statement can force this.

The orbit chain gap.

The cofinal orbit analysis picks one cofinal multiplier s_x per walk position, producing a cycle $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_0$ in $(\mathbb{Z}/q\mathbb{Z})^\times$. Even when the cofinal multipliers generate the full group, the cycle size k can be less than $|(\mathbb{Z}/q\mathbb{Z})^\times|$. Example: in $\mathbb{Z}/6\mathbb{Z}$, the cycle $0 \rightarrow 1 \rightarrow 0$ has $\langle 1, 5 \rangle = \mathbb{Z}/6\mathbb{Z}$ but misses 3.

Closing this gap requires showing that at each cofinally visited position, multiple multiplier classes appear—expanding the cycle until it covers -1 . This is the “specific-orbit problem”: transferring generic equidistribution of `minFac` residues to the particular EM orbit.

5.2 The BRE Impossibility for $d \geq 3$

Dead End: PED $\nRightarrow$ CCSB for $d \geq 3$

Remark 5.7. Positive escape density (PED) alone does not imply CCSB for characters of order $d \geq 3$.

Counterexample: a walk on $\mathbb{Z}/3\mathbb{Z}$ that alternates between only two of the three d -th roots of unity (phase-aligned escapes) achieves positive escape density yet has walk sum $\approx N/2 \cdot (1 + \omega) \neq o(N)$.

For $d = 2$ this degeneracy vanishes: the only non-trivial rotation is -1 , so escape frequency is the rotation distribution. The order-2 BRE from `NoLongRuns(L)` is proved in the formalization ([✓ order2_noLongRuns_mc](#)). But for $d \geq 3$, PED constrains how often the walk rotates without constraining the distribution among $d - 1$ non-identity rotations. The $\text{PED} \Rightarrow \text{BRE} \Rightarrow \text{CCSB}$ factorization is invalid for $d \geq 3$.

This barrier is specific to the PED route. The $\text{CME} \rightarrow \text{CCSB}$ reduction ([✓ cme_implies_ccsb](#)) bypasses PED and BRE entirely, working for all character orders d via the telescoping identity. The $d \geq 3$ problem is therefore not a barrier for the reduction—only for the particular factorization through PED.

The Van der Corput Barrier

The van der Corput inequality (Theorem A.10, now fully proved in the formalization) converts character sum bounds into autocorrelation bounds. Theorem 3.23 gives $R_1 = o(N)$ under the Decorrelation Hypothesis. VdC with $H = 1$ yields $|S_N|^2 \leq \frac{N+1}{2}(N + 2|R_1|) = N^2/2 + o(N^2)$, hence $|S_N| \leq N/\sqrt{2}$. This is non-trivial but not $o(N)$. To get $o(N)$, one needs higher-order correlations $R_h = o(N)$ for $h \geq 2$, which requires HigherOrderDecorrelation (Theorem A.12). The telescoping identity $\sum_n \chi(w(n))(\chi(m(n)) - 1) = O(1)$ is a precise structural constraint.

The Walk Bridge Falsity

The BV route decomposes into two stages: sieve transfer ($BV \Rightarrow \text{EMMultCSB}$, bounding multiplier character sums) and the walk bridge ($\text{EMMultCSB} \Rightarrow \text{MMCSB}$, converting multiplier bounds to walk bounds). The walk bridge $\text{MultCSBImpliesMMCSB}$ ([✓ MultCSBImpliesMMCSB](#)) is stated as an open [Prop](#) and is false in general.

The obstruction is structural: the walk character sum is a partial product $\chi(w(n)) = \prod_{k < n} \chi(m(k))$ of the multiplier characters. Even when the individual factors $\chi(m(k))$ are equidistributed on the unit circle (so their sum cancels), their partial products perform a multiplicative walk whose norm grows as $\sqrt{N}$, not $o(N)$. Cancellation of sums does not imply cancellation of cumulative products.

This negative result explains why the CME bypass (Definition 3.8) is essential. CME uses fiber decomposition and telescoping to go directly from conditional multiplier equidistribution to CCSB, circumventing the walk bridge entirely.

5.3 The Factorization Independence Heuristic

The preceding barriers explain why MC is hard to prove. This subsection explains why it should be true—and why the formalization’s sole remaining hypothesis (CME) is the precise mathematical content of a natural intuition about factorization.

The information bottleneck.

The information bottleneck of §2.5— $O(\log q)$ bits visible to the walk versus $\sim 2^n$ bits determining the multiplier—means the mutual information vanishes exponentially. This is directly analogous to the pseudorandomness of iterated hash functions: if H is a cryptographic hash, the sequence $x, H(x), H(H(x)), \dots$ is deterministic but statistically indistinguishable from random, because the hash destroys recoverable correlations. For the EM sequence, integer factorization plays the role of the hash function. Both processes are fully deterministic (given the seed, every term is uniquely determined), one-way (computing forward is trivial; extracting structure from the output is computationally hard), and de facto uncorrelated (consecutive terms pass every reasonable test for independence).

The analogy is heuristic, not rigorous. SHA-256 is designed for pseudorandomness; `minFac` is not designed for anything. We do not claim that computational hardness of factoring implies MC—what MC requires is a number-theoretic statement (CME), not a complexity-theoretic one. But the analogy explains the structure of the problem: proving pseudorandomness of a deterministic process requires showing that no exploitable structure exists, which is harder than finding structure. This is why the conjecture resists proof despite overwhelming heuristic evidence.

Negative analogies: why the selection rule matters.

The factorization independence heuristic explains why the EM sequence (**minFac** variant) should contain all primes, while related sequences do not.

- The Sylvester sequence $s(n+1) = s(0) \cdots s(n) + 1$ has density-zero prime divisors [19]. Its terms grow doubly exponentially, giving each prime only $O(1)$ chances to appear as a factor. The “hash chain” runs too fast.
- Fermat numbers $F_n = 2^{2^n} + 1$ have the even stronger property that $\sum 1/p$ converges over their prime divisors. Again, doubly exponential growth limits opportunities.
- The second EM sequence (**maxFac** variant) provably omits infinitely many primes [13, 7]. Here the “hash function” (**maxFac** instead of **minFac**) produces large multipliers, causing the product to grow rapidly—analogue to using a hash function that amplifies rather than compresses.
- The first EM sequence (**minFac** variant) keeps the product growing slowly—heuristically as $\exp(n^2/2)$ (see §A.7). This is analogous to a hash function that compresses, giving exponentially many iterations within any fixed modulus. The conjecture is that this compression ensures coverage.

What the formalization adds.

The formalization identifies three equivalent formulations of the factoring channel’s decorrelation:

1. Conditional Multiplier Equidistribution (CME): the distribution of $\text{minFac}(P_2(n) + 1) \bmod q$, conditioned on $P_2(n) \equiv c \pmod{q}$, is asymptotically independent of c (Definition 3.8).
2. Decorrelation Hypothesis: the character sum $\sum \chi(m_{2,q}(n)) = o(N)$ for nontrivial χ —the multiplier residues have cancelling character sums, as independent random variables would.
3. ComplexCharSumBound (CCSB): the walk character sums $\sum \chi(P_{2,q}(n)) = o(N)$ for nontrivial χ —ruling out permanent avoidance of any residue class.

These are related by proved implications ($\text{CME} \Rightarrow \text{Dec} \Rightarrow \text{PED}$) and direct bypass ($\text{CME} \Rightarrow \text{CCSB}$). Each implies MC through the verified reduction chain. The irreducible mathematical content is: does the factoring operation destroy enough correlation for character sums to cancel?

Dead Ends as a Roadmap.

Over 138 potential approaches were explored and found to be dead ends, grouped into eight categories: orbit specificity (OS), technique mismatch (TM), scale mismatch (SM), circularity (CI), structurally false (SF), collapse (CO), decorrelation gap (DG), and aggregate gap (AG). Of these, 13 have formal Lean witnesses and 8 have weak-MC revival score ≥ 2 —meaning they contribute to weaker forms of the conjecture despite being dead for full MC. The full catalogue is in [Meta/DeadEnds.lean](#) and Appendix A.4.

The Four-Way Blocker.

The majority of the 138 dead ends reduce to a single meta-obstacle: every known technique for proving equidistribution of sequences on finite groups requires at least one of (1) independence of steps, (2) multiplicativity of the generating function, (3) algebraic-geometric structure (parameter families, monodromy), or (4) ergodic stationarity. The EM walk satisfies none of these: the steps are deterministic and mutually dependent, the walk is not a multiplicative function, the multipliers have no known algebraic-geometric parametrization, and the non-autonomous

dynamics (a different multiplier at each step) rule out stationarity. This explains why classical tools—random-walk mixing, Halász’s theorem, Katz monodromy, Birkhoff averages—all fail.

The telescope exhausts the algebra.

The telescoping identity $\chi(w(n+1)) = \chi(w(n)) \cdot \chi(m(n))$ is the complete algebraic content of the walk. Only two decomposition strategies for the character sum $S_N = \sum_{n < N} \chi(w(n))$ exist: grouping by value (fiber decomposition $\rightarrow$ CME) and grouping by lag (autocorrelation $\rightarrow$ HOD). Every other rearrangement—Abel summation, Möbius inversion, Dirichlet series, block decomposition—either reduces to one of these two or fails outright (Abel gives $O(N^2)$ remainder in the wrong direction; Möbius/Dirichlet require multiplicativity). There is no third algebraic route to CCSB.

The pattern: every approach that avoids the specific EM orbit’s joint distribution either reduces to CCSB or fails. Since CME implies CCSB (proved), and CME decomposes as VCB + Dec (§3.4), the sharpest targets are now VCB + PED or CME: conditional equidistribution of multipliers given walk position, or the weaker proportionality condition combined with escape density. CME is strictly weaker than CCSB and is the irreducible analytic content.

The Mathematical Landscape

We need to prove one of these equivalent statements for every missing prime q :

- DH: If the multipliers generate $(\mathbb{Z}/q\mathbb{Z})^\times$, the walk $P_{2,q}(n) = P_2(n) \bmod q$ hits -1 cofinally.
- CCSB: For every non-trivial character $\chi: (\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$, the sum $\sum_{n < N} \chi(P_{2,q}(n)) = o(N)$.
- $d=2$ special case (as a stepping stone): For every quadratic character χ , the ± 1 -valued walk character sum is $o(N)$.

The formalization has conclusively shown that every tool requiring independence, classical multiplicativity, or ergodic stationarity fails for a single orbit. Ensemble averaging over squarefree starting points (Section 4) bypasses this, but introduces a new gap: MultCancelToWalkCancel (Dead End #117). We need ideas that exploit what the EM walk specifically has.

Structural Features of the EM Walk

The dead ends above show what does not work. Complementarily, the EM walk has four structural features—all proved or formalized—that no dead-end approach has successfully exploited. Any proof of MC will almost certainly use at least one.

Feature 1: Super-exponential growth. $P_2(n) \geq 2^n$ ([✓ prod_lower_bound_for_sieve](#)). The Euclid numbers $P_2(n) + 1$ grow absurdly fast. This means the sieve level—the threshold below which all prime factors have been excluded—grows super-exponentially. By step n , the Euclid number $P_2(n) + 1$ is coprime to each of $\mathbf{em}_2(0), \dots, \mathbf{em}_2(n)$, a growing set of distinct primes. The pool of “available” small primes as factors of $P_2(n) + 1$ shrinks, but the size of $P_2(n) + 1$ grows so fast that it must have enormous prime factors most of the time.

Feature 2: Mutual coprimality of Euclid numbers. For $m > n$, $P_2(m)$ is divisible by $\mathbf{em}_2(n+1)$, which divides $P_2(n) + 1$. So $P_2(m) + 1 \equiv 1 \pmod{\mathbf{em}_2(n+1)}$: successive Euclid numbers live in different residue classes modulo earlier sequence terms. This coprimality structure means the Euclid numbers cannot all “avoid” a residue class in a coordinated way—their residues are forced apart by the construction.

Feature 3: The multiplier is the smallest prime factor. This is the key constraint that everyone mentions but nobody has quantified. If $P_{2,q}(n) \neq -1$ (so that q does not divide $P_2(n) + 1$ as the smallest factor), then the multiplier $m_{2,q}(n) = \text{minFac}(P_2(n) + 1)$ satisfies $m_{2,q}(n) \leq (P_2(n) + 1)^{1/2}$. For a number of size $\sim 2^n$, this smallest factor could be as small as 3 or as large as $\sim 2^{n/2}$. The `minFac` rule creates a deterministic coupling between walk position and multiplier: the multiplier at step n depends on the full value of $P_2(n) + 1$, not just its residue.

Feature 4: Self-correcting feedback. If the walk concentrates on certain residues mod q —say $P_{2,q}(n) \equiv a \pmod{q}$ for many n —then $P_2(n) + 1 \equiv a + 1 \pmod{q}$ for many n . The smallest prime factor of numbers $\equiv a + 1 \pmod{q}$ depends on $a + 1$, creating a feedback loop: concentration in one residue class biases the multiplier distribution, which in turn pushes the walk away from that class. This self-correcting mechanism has been formalized via a Lyapunov function ([Reduction/SelfCorrecting.lean](#)): the one-step identity $L(N+1) = L(N) + 2d_{w(N)}(N) + (q-2)/(q-1)$ telescopes to $L(N) = 2R(N) + O(N)$, and $R(N) = o(N^2)$ (SCD) implies VE and hence MC. The zero-bias property under uniform multipliers is proved; the orbit-specificity gap—does SCD hold for the EM orbit?—remains open (§5). A weaker variant `LinearDrift` ($|R(N)| \leq CN$) also implies MC ([✓ linearDrift_implies_mc](#)) via a threshold-free route: `LinearDrift` $\Rightarrow$ `VisitGrowth` $\Rightarrow$ `HH` $\Rightarrow$ `MC`, where `VisitGrowth` ($V(a, N) \rightarrow \infty$ for all a) bypasses the `FiniteMCBelow` threshold entirely. A free unconditional lower bound $R(N) \geq -N(q-2)/(2(q-1))$ is proved ([cumulativeDrift_lower_bound](#)), so only the one-sided upper bound $R(N) \leq CN$ is needed.

These four features—growth, coprimality, the `minFac` selection rule, and self-correcting feedback—are the raw material that any successful approach must engage with. The dead ends above fail precisely because they treat the walk generically (as a random walk, or as an arbitrary multiplicative walk) rather than exploiting the specific arithmetic of the EM construction.

5.4 The NAAD Conjecture

The preceding analysis—the four-way blocker, the telescope exhaustion, the dead-end catalogue—identifies a structural gap that no existing theory covers. We now name this gap and state a precise conjecture.

Three theories that almost apply.

Three mature mathematical theories converge on the class of problems to which MC belongs, without covering it.

Arithmetic dynamics (Silverman, Baker–DeMarco) studies iteration of algebraic maps $\varphi: \mathbb{P}^1 \rightarrow \mathbb{P}^1$ over number fields. The theory requires the map to be a fixed polynomial or rational function; the EM map $T(P) = P \cdot \text{minFac}(P+1)$ is non-algebraic, since the multiplier depends on the factorization of $P+1$.

Multiplicative number theory (Alladi, Hildebrand, Granville, Soundararajan) gives precise asymptotics for the distribution of `minFac`, `maxFac`, Ω , ω over the population of integers. But it does not control these functions along a specific deterministic orbit.

Computational pseudorandomness (Blum–Micali, Goldreich–Levin) studies when deterministic iteration produces outputs indistinguishable from random. The theory requires computational hardness (one-wayness against polynomial-time adversaries) and operates in the Boolean setting $\{0,1\}^n$, not the arithmetic setting $\mathbb{Z}$.

The EM sequence sits at the intersection: an iterated map on $\mathbb{Z}$ (arithmetic dynamics) whose multiplier is `minFac` (multiplicative number theory), whose pseudorandom behavior—if true—arises from the opacity of integer factorization (computational pseudorandomness).

Arithmetic iteration maps.

Definition 5.8 (Arithmetic iteration map). An arithmetic iteration map is a function $T: \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ of the form $T(x) = x \cdot m(x)$, where $m(x) \in \mathbb{Z}_{>1}$ is the multiplier, determined by the arithmetic of x . The orbit from seed x_0 is the sequence $x_0, T(x_0), T^2(x_0), \dots$

The map is non-autonomous: the multiplier $m(x_n)$ depends on the state x_n , which changes at every step. Unlike autonomous dynamics (a fixed map applied repeatedly), the effective transformation changes at each iteration.

Example 5.9 (Examples). 1. Euclid–Mullin: $T(P) = P \cdot \text{minFac}(P+1)$.

2. Largest-factor Euclid: $T(P) = P \cdot \text{maxFac}(P+1)$. Same structure, different selection rule; provably misses primes [13].

3. Random-factor Euclid: $T(P) = P \cdot \text{randFac}(P+1)$. The walk is a genuine random walk on $(\mathbb{Z}/q\mathbb{Z})^\times$; equidistribution follows from Diaconis–Shahshahani mixing.

For a prime q , the walk is the projection $w_n = T^n(x_0) \bmod q$ onto $(\mathbb{Z}/q\mathbb{Z})^\times$, satisfying the multiplicative recurrence $w_{n+1} = w_n \cdot m_n \bmod q$ where $m_n = m(x_n) \bmod q$. This is the walk framework of Section 2, now abstracted to a class of maps.

Four structural properties.

We identify four properties that control the walk’s behavior. All four are verified for the EM map (the first three are proved; the fourth follows from standard ANT).

1. CRT-blindness (= Property 1 of §4.1): $m(x) \bmod q$ is asymptotically independent of $x \bmod q$ in the population. For EM, this is the algebraic fact `✓ crt_multiplier_invariance`.
2. Super-exponential growth: $T^n(x_0) \geq 2^{2^n}$ for $n \geq n_0$. For EM, $P_2(n) \geq 2^{n+1}$ (`✓ prod_lower_bound_for_sieve`). This creates the information bottleneck: the walk position uses $O(\log q)$ bits to describe a state with $\Omega(2^n)$ bits.
3. Generating multipliers: the residues $\{m(x_n) \bmod q\}_{n \geq 0}$ generate $(\mathbb{Z}/q\mathbb{Z})^\times$. For EM, this is SubgroupEscape, proved unconditionally via PRE (Theorem 2.16).
4. Population equidistribution: $m(x) \bmod q$ is equidistributed over $(\mathbb{Z}/q\mathbb{Z})^\times$ in the population. For EM, this follows from Alladi’s formula combined with Dirichlet’s theorem (scaffolded in `AlladiDensity.lean`).

Conjecture 5.10 (NAAD Equidistribution). Let $T(x) = x \cdot m(x)$ be an arithmetic iteration map satisfying Properties 1–4 for all primes $q \geq q_0$. Let $x_0 \in \mathbb{Z}_{>0}$ with $\gcd(x_0, q) = 1$. Then the walk $w_n = T^n(x_0) \bmod q$ equidistributes in $(\mathbb{Z}/q\mathbb{Z})^\times$:

$$\text{for every non-trivial } \chi: (\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \mathbb{C}^\times, \quad \frac{1}{N} \left| \sum_{n < N} \chi(w_n) \right| \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Theorem 5.11. NAAD Equidistribution for the EM map implies MC.

Proof. The EM map satisfies Properties 1–4. NAAD applied with $x_0 = 2$ gives CCSB, which implies MC through the machine-verified chain $\text{CCSB} \Rightarrow \text{MC}$ (`✓ complex_csb_mc`). $\square$

The three-level gap.

NAAD Equidistribution is the assertion that Level 3 of the CRT hierarchy (§4.1) follows from Levels 1–2. The conjecture’s content is precisely the orbit-specificity transfer: Properties 1–4 describe the function m ; the conjecture asks whether the orbit inherits the function’s statistical behavior. The entire infrastructure of this formalization—58,150 lines—reduces to this single question.

Relation to Collatz.

Tao [20] proved that almost all Collatz orbits attain almost bounded values, using a random model for Syracuse random variables. Both Collatz and EM involve non-autonomous iteration maps with CRT structure; both have random models that behave well; both face the transfer problem from random model to actual orbit. The key difference: the Collatz map is piecewise linear mod 2 (clean algebraic structure in the 2-adic coordinate), while the EM map involves factoring, which has no algebraic shortcut. This makes EM harder to analyze but also more “pseudorandom”—the factoring channel is a stronger scrambler than the 2-adic valuation.

Why NAAD requires new mathematics.

The four-way blocker (§5.3) applies to any arithmetic iteration map: Properties 1–4 explicitly exclude independence, multiplicativity, algebraic-geometric structure, and ergodic stationarity. NAAD Equidistribution asks whether these four properties collectively substitute for any single classical assumption. The 138 dead ends documented in the formalization (Appendix A.4) confirm that no existing technique closes this gap.

NAAD open problems.

Beyond the full conjecture, several intermediate problems are independently interesting:

1. Weak NAAD (almost-all starting points): prove that for almost all squarefree x_0 , the walk equidistributes. Partially achieved: $\text{SD} \Rightarrow$ cancellation for a.a. starting points ([✓ sd implies cancellation](#)).
2. Quantitative NAAD (correlation decay): prove $|\text{Cov}(\chi(w_n), \chi(w_{n+k}))| \leq C\alpha^{-k}$ for some $\alpha > 1$. This would give NAAD via van der Corput with $|\sum \chi(w_n)| = O(\sqrt{N})$.
3. CRTPropagationStep: prove that accumulator equidistribution propagates across ensemble steps. This is a population-level statement with no orbit-specificity barrier, amenable to CRT + Alladi.
4. NAAD for other maps: study variants $T(x) = x \cdot \text{maxFac}(x+1)$, $T(x) = x \cdot \Omega(x+1)$, etc. The comparison illuminates the role of the selection rule: minFac gives slow growth and guaranteed capture (§5.1); maxFac gives fast growth and provable failure.

5.5 The Precise Frontier

The formalization certifies the reductions above and pinpoints the frontier. We now identify the open problems whose resolution would close Mullin’s Conjecture.

CCSB as the Precise Frontier

The formalization identifies ComplexCharSumBound as the irreducible analytic content. The question:

Are the walk character sums $\sum_{n < N} \chi(\text{P}_{2,q}(n))$ bounded $o(N)$ for every non-trivial χ ?

CCSB is a single hypothesis that implies MC with no additional conditions. As shown in Section 3, CCSB rules out permanent avoidance of -1 : the Fourier bridge forces the hit count

at -1 to be $N/(q-1) + o(N)$, contradicting the zero hits that the first missing prime's walk must achieve past the sieve gap.

The walk telescoping identities (Section 3) provide precise structural constraints. The identity $\sum_n \chi(w(n))(\chi(m(n)) - 1) = O(1)$ means that the walk sum S_N and the multiplier sum $M_N = \sum_n \chi(m(n))$ satisfy $S_N \approx S_N + (M_N - S_N) = M_N + O(1)$ only in the crude sense; the telescoping does not separate them.

Connection to Bombieri–Vinogradov

A Bombieri–Vinogradov type result for EM walk residues would give:

$$\sum_{\substack{q \leq Q \\ q \text{ prime}}} \max_a \left| \left| \{n \leq N : w(n) \equiv a \pmod{q}\} \right| - \frac{N}{q-1} \right| \ll \frac{NQ}{(\log N)^A}.$$

For non-exceptional primes, the hit count at -1 would be $\sim N/(q-1) > 0$, contradicting permanent avoidance; the finitely many exceptional primes below Q_0 are handled by the inductive bootstrap (§2), closing the conjecture.

The difficulty is that BV applies to the set of all primes, not to a specific subsequence. The EM walk is deterministic and self-referential: the walk at step n depends on the factorization of $P_2(n) + 1$, which depends on all previous walk values. Standard BV does not apply.

Connection to Chebotarev

The EffectiveKummerEscape hypothesis asserts: for each prime ℓ , there exists B such that for $q \geq B$ with $\ell \mid q-1$, some multiplier among the first B escapes the ℓ -th power kernel. This is a Chebotarev-type statement for the Kummer extension $\mathbb{Q}(\zeta_\ell, 3^{1/\ell}, \dots, 53^{1/\ell})$: the Frobenius at q determines which multiplier primes are ℓ -th power residues.

An effective Chebotarev density theorem [10] for this fixed number field would give EKE for all but finitely many q (effectively bounded). Combined with finite verification for the remaining q , this would prove PRE and hence SE unconditionally—but SE is already proved unconditionally via the elementary PRE. The Chebotarev approach would give a stronger effective bound on how quickly SE kicks in, refining the density argument of §2.

The Sieve-Theoretic Approach

MertensEscape: for any prime q and proper subgroup H , infinitely many primes outside H exist (Dirichlet content). SieveAmplification: Mertens escape should force eventual $\min\text{Fac}(P_2(n)+1)$ escape from H , via super-exponential growth and mutual coprimality of successive Euclid numbers.

The formally verified chain: MertensEscape + SieveAmplification $\xrightarrow{\text{proved}}$ TailSE $\xrightarrow{\text{proved}}$ CofinalEscape $\xrightarrow{\text{proved}}$ QuotientDH.

The formalization articulates a richer sieve infrastructure in two parallel routes:

Cumulative route.

$$\begin{array}{ccccccc} \text{PDE} & \xrightarrow{\text{Alladi}} & \text{GLPFE} & \xrightarrow{\text{SieveTransfer}} & \text{SieveEquidist} & \xrightarrow{\text{open}} & \text{NoLongRuns} \\ & & \xrightarrow{\text{proved}} & \text{PED} & \xrightarrow{\text{open}} & \text{CCSB} & \xrightarrow{\text{proved}} \text{MC}. \end{array}$$

Here PDE is PrimeDensityEquipartition (PNT in arithmetic progressions, a known theorem not yet in Mathlib), and GLPFE is GenericLPFEquidist (Alladi’s theorem [11] on `minFac` distribution of generic integers, also known but not formalized). Both ends of the chain—from PDE to GLPFE via Alladi, and from CCSB to MC via Fourier inversion—are formally proved.

Window route.

$$\text{StrongSieveEquidist} \xrightarrow{\text{proved}} \text{NoLongRunsAt} \xrightarrow{\text{proved}} \text{PEDAt} \xrightarrow{\text{open}} \text{CCSB} \xrightarrow{\text{proved}} \text{MC}.$$

`StrongSieveEquidist` asserts that EM multipliers are equidistributed within sliding windows; `NoLongRunsAt` and `PEDAt` are per-prime variants proved by pigeonhole and block-counting respectively ([✓ `strongSieveEquidist\_noLongRunsAt`](#), [✓ `noLongRunsAt\_ped`](#)).

The genuine frontier.

`SieveTransfer` is the critical open hypothesis: does the equidistribution of `minFac` residues for generic integers transfer to the specific EM orbit? Everything above `SieveTransfer` is known mathematics; everything below it is proved. `SieveTransfer` is where “known but not formalized” meets “genuinely open.”

The difficulty: for generic q -rough integers, `minFac` residues are equidistributed (by CRT + Mertens)—the death channel gets its fair share of multipliers. But the EM Euclid numbers are not generic: each is determined by the entire walk history. Transferring the generic equidistribution to this specific orbit is the open step.

Sieve-to-harmonic convergence.

The sieve hierarchy (§36–§39 of [Equidist/SelfCorrecting.lean](#)) and the harmonic hierarchy (§30–§35) converge: both produce `DecorrelationHypothesis` as output. The full chain

$$\text{SieveEquidist} \xrightarrow{\text{proved}} \text{Dec} \xrightarrow{\text{proved}} \text{PED} \xrightarrow[\text{sole gap}]{\text{open}} \text{CCSB} \xrightarrow{\text{proved}} \text{MC}$$

is formalized, with the first two arrows machine-verified ([✓ `sieve\_equidist\_implies\_decorrelation`](#), [✓ `decorrelation\_implies\_ped`](#)). The sieve route achieves `SieveEquidist` $\Rightarrow$ `Dec` via a counting-to-character-sum bridge: `SieveEquidistribution` produces `EMMultCharSumBound` with $Q_0 = 0$, meaning multiplier character sums cancel for all primes q , which is exactly `DecorrelationHypothesis`. The sole remaining gap on this route is `PEDImpliesComplexCSB` ([✓ `PEDImpliesComplexCSB`](#)): does positive escape density for all primes imply walk character sum cancellation? Any proof of `SieveEquidistribution` (e.g., from PNT in APs + Alladi’s theorem) would immediately yield `Dec` and `PED` for free, isolating this single bridge as the only open step.

DSL: The Sole Remaining Hypothesis

The ensemble averaging framework (Section 4) collapses the entire reduction landscape to a single open statement: the Deterministic Stability Lemma (DSL, Property 4.22). DSL asserts that population equidistribution of multiplier residues (a consequence of standard ANT) implies conditional multiplier equidistribution for the standard EM orbit ($n = 2$). The assembly `DSL` $\Rightarrow$ `CME` $\Rightarrow$ `CCSB` $\Rightarrow$ `MC` is fully machine-verified ([✓ `full\_chain\_dsl`](#)). DSL is the orbit-transfer principle: it bridges from population-level statistics (which hold for almost all starting points) to the specific orbit.

A weaker variant also suffices:

Theorem 5.12 ([✓ pe_dsl_hitting_implies_mc](#)). $\text{PE} + \text{DSLHitting} \implies \text{MC}$,

where DSLHitting only requires that the walk hits -1 for a positive density of starting points, rather than full equidistribution.

The Adelic Decomposition of CME

The adelic framework (Section 3) decomposes CME into two independent components:

Theorem 5.13 ([✓ cme_iff_adelic](#)). $\text{CME} \iff \text{MWI} \wedge \text{MME}$,

where MWI (Multiplier–Walk Independence) asserts that the conditional distribution of the multiplier given the walk position converges to the marginal, and MME (Marginal Multiplier Equidistribution) asserts that the marginal multiplier distribution converges to uniform.

The decomposition is proved in both directions ([cme_implies_mme](#), [crt_fiber_mme_implies_cme](#)). MWI is the core difficulty: it is the formal statement that the factoring channel destroys the correlation between walk position and multiplier residue. MME is morally a consequence of the population equidistribution (PE), but the transfer from population to orbit multipliers requires DSL.

The CRT fiber structure provides a further decomposition:

$$\text{CRTFiber} + \text{MME} \xrightarrow{\text{proved}} \text{MWI} \xrightarrow[\text{+ MME}]{\text{proved}} \text{CME} \xrightarrow{\text{proved}} \text{MC}.$$

CRTFiber (the CRT fiber determines the next multiplier, proved in [crt_fiber_determines_genSeq](#)) provides the structural basis for MWI. The proof uses Fourier inversion on the character indicator function ([crt_fiber_implies_mwi_proved](#), ~145 lines).

MME is equivalent to vanishing walk lag-1 autocorrelation ([mme_iff_walk_autocorrelation](#)), providing a purely walk-theoretic reformulation. The landscape theorem [ccsb_mme_landscape](#) proves: $\text{CME} \implies \text{CCSB} \wedge \text{MME}$, and conversely $\text{CCSB} \wedge \text{MME} + \text{MWI} \implies \text{CME}$.

The Profinite Weyl Criterion

The Uniform Profinite Equidistribution (UPE) hypothesis asserts: for every $k \geq 1$ and every k -tuple of distinct primes $(q_1, \dots, q_k)$, the joint distribution of $(w_{q_1}(n), \dots, w_{q_k}(n))$ converges to uniform on $\prod_i (\mathbb{Z}/q_i\mathbb{Z})^\times$. UPE is the profinite analog of equidistribution: it asks for uniform distribution modulo all finite sets of primes simultaneously.

Theorem 5.14 ([✓ uniform_profinite_implies_mc](#)). $\text{UPE} \implies \text{MC}$.

The proof: UPE at $k = 1$ gives CCSB ([uniform_profinite_implies_ccsb](#)), which suffices. UPE at $k = 2$ gives CPD (Conditional Pairwise Decorrelation, [uniform_profinite_implies_cpd](#)). Dead End #125 shows that pairwise ($k = 2$) does NOT imply k -wise—the XOR counterexample is decisive—so UPE is genuinely stronger than $\text{CCSB} \wedge \text{CPD}$.

The routes to MC via the adelic/profinite framework are collected in [profinite_routes_to_mc](#) (3-route conjunction: UPE, CCSB, adelic, all proved).

The profinite walk.

The EM accumulator maps to $\hat{\mathbb{Z}}^\times = \varprojlim (\mathbb{Z}/N\mathbb{Z})^\times$ via its residues. If a prime q is missed, the walk stays in $(\mathbb{Z}/q\mathbb{Z})^\times \setminus \{-1\}$ permanently. For an infinite set M of missed primes,

the walk inhabits the survival set $\mathcal{S}_M = \prod_{q \in M} ((\mathbb{Z}/q\mathbb{Z})^\times \setminus \{-1\})$, which has Haar measure $\prod_{q \in M} (q-2)/(q-1) = 0$. The EM walk thus permanently occupies a measure-zero subset of the profinite group—consistent for a deterministic sequence, but constraining.

The walk in a product $(\mathbb{Z}/q_1\mathbb{Z})^\times \times (\mathbb{Z}/q_2\mathbb{Z})^\times$ is synchronously coupled: both coordinates are multiplied by the same factor $f(n)$ at each step. The orbit is a coset of the diagonal multiplier subgroup $\Delta = \{(f, f) : f \in \langle \text{multipliers} \rangle\}$. PrimeResidueEscape gives generation in each factor separately; the joint question—whether the multipliers generate the full product $(\mathbb{Z}/q_1q_2\mathbb{Z})^\times$ —is a CRT lifting problem not implied by individual generation. Dense orbit in $\hat{\mathbb{Z}}^\times$ would give MC directly.

What Would Close the Conjecture

Paths to MC

The cleanest paths to MC:

1. Prove DSL (sole remaining hypothesis): show that the character energy of generalized EM sequences concentrates when averaged over squarefree starting points. This is the weakest known hypothesis that implies MC, and the only one that does so without additional conditions. The assembly $\text{DSL} \Rightarrow \text{CME} \Rightarrow \text{CCSB} \Rightarrow \text{MC}$ is fully machine-verified ([✓ full_chain_dsl](#)).
2. Prove CME: show that the multiplier character sum $\sum_{\substack{n < N \\ w(n)=c}} \chi(m(n))$ is $o(N)$ for each walk position c . CME is strictly weaker than CCSB, and $\text{CME} \Rightarrow \text{CCSB}$ is proved ([✓ cme_implies_ccsb](#)). CME asks only about the conditional distribution of multipliers given walk state—it does not require controlling the walk character sum itself. This bypasses the $d \geq 3$ barrier entirely.
3. Prove CCSB directly: show that no non-trivial character sum can sustain the $\Omega(N)$ bias that permanent avoidance of -1 would require. The self-correcting sieve (concentration of EM primes in a residue class is exponentially self-limiting) is the strongest heuristic argument.
4. Prove a BV-type estimate: even an averaged bound over q would suffice—for $q \geq Q_0$ it rules out permanent avoidance, and the finitely many $q < Q_0$ are handled by the inductive bootstrap (§2).
5. Close the orbit chain gap: show that at each cofinally visited walk position, at least two distinct multiplier classes appear. This would force the orbit chain to expand to the full group.
6. Prove DH directly: show that a multiplicative walk on a cyclic group with a generating set of multipliers must hit every element cofinally. This is a combinatorial question about deterministic walks.
7. Prove SieveTransfer: show that the EM orbit's **minFac** distribution matches that of generic integers, at least on average. The cumulative sieve route (§38) reduces this to known number theory (PNT in APs + Alladi's theorem); closing SieveTransfer gives CME (and hence CCSB) via the conditional multiplier equidistribution framework.
8. Prove BVImpliesMMCSB or PrimeArithLSImpliesMMCSB: the large sieve route (§41–§65) reduces MC to a transfer hypothesis. Given Bombieri–Vinogradov or the analytic large sieve, the remaining open step is showing that the EM orbit's multipliers receive their fair share of each residue class—in particular, that the

death channel is not systematically avoided. This is the same orbit-specificity gap as SieveTransfer, approached from a different mathematical toolkit.

The vanishing noise route.

A complementary perspective recasts the walk as a product of random and deterministic steps. At each step, $P_2(n) + 1$ has a set F of prime factors; `minFac` selects the smallest, but the factor set $F \bmod q$ acts as a “noisy generating set” in $(\mathbb{Z}/q\mathbb{Z})^\times$. The spectral gap lemma ([✓ spectral_gap_with_identity](#)) proves: for any generating set $S \ni 1$ in a finite abelian group and any nontrivial character χ , $\|\sum_{s \in S} \chi(s)\| < |S|$ (strict inequality, using strict convexity of $\mathbb{C}$). Deterministic walk steps—multiplying by a fixed element—preserve this contraction ([✓ char_sum_mul_right_norm](#)). If the factor set has $|F| \geq 2$ elements in $(\mathbb{Z}/q\mathbb{Z})^\times$ infinitely often (with non-summable spectral gap), the product $\prod(1 - \gamma_k) \rightarrow 0$ ([✓ product_contraction_tendsto](#)), which would give CCSB—but only if the walk actually averages over F . A proved counterexample ([✓ selection_vs_average_gap](#)) shows the gap cleanly: for any fixed selection $s \in S$, the pointwise norm $\|\chi(s)\| = 1$ (zero contraction), even when the average over S contracts strictly below 1. Since `minFac` selects one specific element of the factor set, the spectral gap alone does not imply CCSB.

The formalization defines the factor set residues ([✓ factorSetResidues](#)) and proves that the walk multiplier lies in it ([✓ multZ_in_factorSetResidues](#)). Two open hypotheses remain: `InfinitelyManyLargeFactorSets` (that the Euclid numbers $P_2(n) + 1$ have at least two prime factors $\geq q$ for infinitely many n) and `MinFacUnbiased` (that `minFac` selection is asymptotically unbiased over the squarefree ensemble). The latter is the orbit-specificity barrier restated in spectral language.

The formalization makes the gap precise via a phase transition at $\varepsilon = 0$ ([✓ phase_transition_landscape](#)). For any constant $\varepsilon \in (0, 1)$, the two-point character product $\prod_k [(1 - \varepsilon)\chi(a_k) + \varepsilon\chi(b_k)]$ decays to 0 in norm whenever infinitely many steps have distinct χ -values—a uniform spectral gap (finite-group pigeonhole) ensures non-summable contraction. At $\varepsilon = 0$, each factor reduces to $\chi(a_k)$ with $|\chi(a_k)| = 1$, so the product norm is identically 1. MC (via CCSB) is exactly the assertion that the Cesàro average of these unit-modulus products cancels. The transition is sharp: any $\varepsilon > 0$ gives decay; $\varepsilon = 0$ gives perpetual oscillation on the unit circle.

The vanishing noise perspective also motivates the `{minFac, secondMinFac}` variant MC (Section 6.4), which proves that some selection from two-point factor sets reaches every residue class—under a single open spectral hypothesis (`UFDStrong`); for $q = 3$, a faithful character escape argument closes `UFDStrong` unconditionally (Section 6). The stochastic MC chain, the self-consistent branching tree, and the weak Mullin weakenings are developed in Section 6; the function field analog (Section 6.7) confirms that the difficulty is orbit-specificity, not analytic number theory.

Function Field Analog

Over $\mathbb{F}_p[t]$, the FF-EM sequence has additional structural advantages. The formalization proves ([FunctionField/WeakMC.lean](#), 471 lines, zero sorry):

- Degree escape ([✓ ffSeq_degree_tendsto_atTop](#)): $\deg(\text{ffSeq}(n)) \rightarrow \infty$, unconditional from finiteness + injectivity.
- Capture count stabilization: for each degree d , the number of degree- d irreducibles captured eventually stabilizes.
- Pool partition identity ([✓ captureCount_plus_missing](#)): captured + unsieved = total, an exact

accounting.

- $\text{FFCaptureExhaustive} \Leftrightarrow \text{FFMullinConjecture}$: the two natural formulations are equivalent.

The pool partition identity gives quantitative control: if the total number of monic irreducibles of degree $\leq D$ is M , then among the first $N > M$ terms, at least one has degree $> D$. Dead End #129 (FFLM structurally false—cyclotomic Galois groups) is sidestepped entirely: degree escape uses only finiteness and injectivity, not monodromy.

Does the Walk Hit -1 ?

The irreducible open question is not whether DH holds (cofinal hitting is a convenient sufficient condition) but whether the walk hits -1 at least once past the sieve gap. The Single Hit Theorem (Theorem 2.20) shows that a single hit at each prime suffices for MC; DH, CCSB, and CME are strategies for producing that hit.

Evidence for.

Three independent lines of evidence suggest the walk does hit -1 (see also Remark 4.24). (1) The information bottleneck: $P_2(n)$ has $\sim 2^n$ bits, but the walk position carries only $O(\log q)$ bits. The smallest prime factor of $P_2(n) + 1$ depends on the full value, not on its residue mod q , so conditioning on walk position constrains almost nothing about the factoring outcome—the conditional and unconditional minFac distributions should agree. (2) Ensemble evidence: for almost all squarefree starting points, CME holds (the proved SD $\Rightarrow$ cancellation chain). The standard orbit $n = 2$ has no special arithmetic property that would make it an outlier. DSL is a typicality assertion, not a structural miracle. (3) Analogy with Artin’s conjecture: Artin’s conjecture (that every non-square integer is a primitive root for infinitely many primes) has the same orbit-specificity structure and is believed true; Hooley [9] proved it conditional on GRH. The walk-hitting question is the analogous statement for the EM walk and would follow from an analogous uniformity hypothesis.

Evidence against.

Two features of the EM sequence give pause. (1) Cox–van der Poorten [13]: the “largest factor” variant of the Euclid–Mullin sequence provably misses primes. The EM sequence’s completeness is not a soft consequence of the Euclid construction but depends sensitively on the minFac selection rule. This fragility means heuristic arguments (“it should work because Euclid numbers have many factors”) are not reliable. (2) The $d \geq 3$ barrier: the formalization proves that the most natural route from multiplier escape density to walk character sum cancellation (PED $\Rightarrow$ BRE $\Rightarrow$ CCSB) is impossible for character orders $d \geq 3$. This is not evidence against the hit, but it shows that producing one requires mechanisms beyond the simplest equidistribution framework. The CME bypass sidesteps this barrier, but the barrier’s existence means any proof must be genuinely subtle.

Assessment.

We believe the walk does hit -1 at every prime, primarily because the self-correcting sieve mechanism provides a concrete dynamical reason (not merely a probabilistic heuristic) for the walk to eventually hit -1 . The strongest form of this belief: CME should hold because the EM multipliers, conditioned on walk position, have no arithmetic reason to avoid the death channel systematically. But we acknowledge that no existing technique can prove this. The formalization has exhaustively explored all known proof strategies across 138 dead ends: the $d \geq 3$ barrier, the MultCancelToWalkCancel counterexample (Dead End #117), the cofactor

identity’s failure to escape the Marginal/Joint Barrier (Dead End #115), and the exhaustion of all four DSL proof directions (algebraic identities, cofactor completeness, exponential sums, sum-product theory) collectively show that the proof, when found, will require genuinely new mathematical techniques—likely exploiting the specific arithmetic of the `minFac` rule on super-exponentially growing squarefree numbers in ways that current analytic number theory does not.

One direction remains genuinely untried: exploiting the arithmetic of the seed $x_0 = 2$. The multiplicative order $\text{ord}_q(2)$ is large for most primes q (Artin’s conjecture, proved under GRH by Hooley), and character sums $\sum \chi(g^n)$ cancel when $\text{ord}(g)$ is large (Konyagin–Shparlinski). This is exactly the pattern CCSB demands. The fatal gap: the walk position $P_2(n) \bmod q = 2 \cdot \text{em}_2(1) \cdots \text{em}_2(n) \bmod q$ is not the geometric progression $2^n \bmod q$ —the varying multipliers $\text{em}_2(k)$ destroy the structure that makes exponential-sum bounds work (Dead End #135). No mechanism has been found to convert Artin-type information about $\text{ord}_q(2)$ into walk equidistribution through a route that avoids Fourier analysis and Cauchy–Schwarz (Dead End #136). Whether such a mechanism exists is, in our view, the most promising open question.

6 Variants of the Conjecture

The Euclid–Mullin construction makes sense with any factor-selection rule: at each step, $P_2(n) + 1$ has a non-empty set of prime factors, and the rule picks one. Mullin’s Conjecture is specific to the `minFac` rule. This section collects the known variants—some easy, some provably incomplete, one almost closed—and uses them to locate standard MC within a broader landscape.

Section roadmap

- §6.1 Random-factor variant — MC by mixing; independence vs. selection trade-off.
- §6.2 Factor tree and GenMixedMC — nondeterministic access to all prime factors.
- §6.3 Almost all GenMixedMC — density-1 result: PEAP alone suffices (no open hypothesis).
- §6.4 `{minFac, secondMinFac}` variant — two-point spectral contraction, UFDStrong.
- §6.5 Ensemble MC — almost every starting point, backward dynamics.
- §6.6 Weak Mullin — analytic weakenings, partial results.
- §6.7 Function field analog — $\mathbb{F}_p[t]$, Weil bound, orbit-specificity barrier.

The Selection Rule Landscape

Fix an initial prime $p_0 = 2$. Given a selection rule σ that, for any integer $N \geq 2$, chooses a prime factor $\sigma(N) \mid N$, define:

$$P_\sigma(0) = 2, \quad P_\sigma(n+1) = P_\sigma(n) \cdot \sigma(P_\sigma(n) + 1). \quad (3)$$

The selection rule determines the sequence; the running product $P_\sigma(n)$ determines the sequence history. Five rules are studied in the formalization:

Selection rule σ	MC status	Key property
Random (uniform)	Easy (mixing)	Independence
minFac (standard EM)	Open (DSL)	Deterministic, lethal
$\{\mathbf{minFac}, \mathbf{secondMinFac}\}$	Open (UFDS tr ong)	Two-point diversity
$(1-\varepsilon) \mathbf{minFac} + \varepsilon \text{ random}$	a.a. (PEAP)	Full factor access
Steered (Booker [3])	True (by construction)	Adversarial
maxFac (Cox–vdP [13])	False	Misses primes

The landscape reveals a tension: the random rule is the easiest to analyze (mixing theory) but the worst at selection (probability $1/|F|$ per hit); **minFac** is the best at selection (lethal hits) but the hardest to analyze (deterministic, no independence); the $\{\mathbf{minFac}, \mathbf{secondMinFac}\}$ variant sits between them, offering a two-point spectral contraction that is provably sufficient under a single open hypothesis; and the factor tree framework (§6.2) generalizes all three by allowing access to the full factor set at every step, reducing GenMixedMC to sieve theory for almost all starting points.

6.1 The Random-Factor Variant

Consider a variant where, instead of the smallest prime factor, one picks a random prime factor of $P_2(n) + 1$ at each step. MC follows from classical mixing theory. For any target prime p , the residue $r_n = P_2(n) \bmod p$ performs a multiplicative walk on $(\mathbb{Z}/p\mathbb{Z})^\times$. Each multiplier is a random element of the group; once the multipliers generate the full group (which PRE guarantees, Theorem 2.16), the walk is a genuine random walk with full support. By Diaconis–Shahshahani equidistribution, the walk mixes in $O(\log p)$ steps: $r_n = -1$ occurs with frequency $\rightarrow 1/(p-1)$ —infinitely often with probability 1. When p divides the Euclid number, the random rule selects p with probability $1/|F(S_n)|$, but cofinal hitting provides infinitely many chances, so p is eventually selected.

Comparison with **minFac**.

The actual EM sequence uses $\mathbf{em}_2(n+1) = \mathbf{minFac}(P_2(n)+1)$, a deterministic function of the walk position. Compared to the random-factor variant, **minFac** is strictly better in every measurable dimension:

	Random factor	minFac
Selection when $q \mid P(n)+1$	Probability $1/ F(S_n) $	Probability 1
Accumulator growth	Faster (random $\geq \mathbf{minFac}$)	Slowest possible
Hits on -1 needed	Infinitely many	One (lethal hit)

Selection efficiency. Past the sieve gap, 0-accessibility guarantees: if $q \mid P_2(n) + 1$ then $q = \mathbf{minFac}(P_2(n) + 1)$ —selection is certain, not probabilistic. Growth rate. Since $\mathbf{minFac}(N) \leq p$ for any factor p of N , the accumulator $P_2(n)$ grows minimally under the **minFac** rule, giving exponentially more steps before it becomes astronomically large—more chances for q to divide. Lethality. The inductive bootstrap shows one single hit suffices: $\mathbf{MC}(< q)$ ensures q is minimal whenever it divides, so hitting -1 once means q enters the sequence. The random variant requires infinitely many hits to overcome low per-hit selection probability.

MinFac is strictly better in every measurable dimension. So why can’t we prove it works?

The independence trade-off.

The answer is surgical: the random variant possesses a single property that **minFac** lacks, and that property is the only thing the proof uses.

The property is independence of multiplier from walk position. In the random-factor variant, the multiplier at step n is selected independently of which residue class $P_2(n) \bmod q$ occupies. Walk position and multiplier are independent random variables, and independence is the engine that drives mixing. Diaconis–Shahshahani’s theorem applies whenever the multipliers are i.i.d. from a distribution whose support generates the group.

In the **minFac** variant, the multiplier is $\text{minFac}(P_2(n)+1) \bmod q$ —a deterministic function of the full integer $P_2(n) + 1$. The CRT multiplier invariance ([✓ crt_multiplier_invariance](#)) shows that the outcome is independent of $P_2(n) \bmod q$ at the population level: for generic integers, knowing the mod- q residue does not predict **minFac**. But for the specific orbit $n = 2$, the multiplier is a deterministic value, and mixing theorems for deterministic sequences cannot be invoked.

The random variant thus sacrifices selection efficiency (it may miss q even when q divides) in exchange for statistical independence. The **minFac** variant has perfect selection but no independence:

	Random factor	minFac
Selection	Bad ($1/k$ chance)	Perfect
Mixing	Random walk theorem	Unproved

The random variant’s proof: mixing $\rightarrow$ cofinal hitting $\rightarrow$ infinitely many chances compensate for low per-hit probability. The **minFac** variant needs only one hit—absurdly less than the random variant achieves—but has no way to produce even that one without mixing.

The stochastic domination failure.

The natural instinct is to argue by stochastic domination: if the random walk visits -1 cofinally, and **minFac** is “better in every way,” then **minFac** should also visit -1 . This argument fails because the two processes diverge after one step.

At step n , both processes face the same Euclid number $P_2(n) + 1$ with factor set $F(S_n)$. The random variant picks a uniformly random element of $F(S_n)$; the **minFac** variant picks the minimum. After this step, the two processes have different accumulators (they multiplied by different primes), so they face different Euclid numbers at the next step. The processes explore completely different trajectories. Domination fails because the two walks live on different probability spaces after step 1; we cannot compare them step-by-step.

A subtler comparison uses visit measures. Define $\mu_{\text{rand}}(a) = \lim_N \frac{1}{N} |\{n < N : w_q^{\text{rand}}(n) = a\}|$ and analogously $\mu_{\text{min}}(a)$ for the **minFac** walk. For the random variant, $\mu_{\text{rand}} =$ uniform on $(\mathbb{Z}/q\mathbb{Z})^\times$ by mixing. For **minFac**, CME/CCSB asserts $\mu_{\text{min}} \approx$ uniform, and DSL asserts this follows from PE. The comparison $\mu_{\text{min}}(-1) \geq 1/(q-1)$ is precisely what needs proof, and is the content of DSL.

6.2 The Factor Tree and GenMixedMC

The random variant’s provability motivates a broader framework: instead of fixing a single selection rule, consider the factor tree $\mathcal{T}(m)$ rooted at starting point m . The root holds accumulator m ; each node with accumulator P has one child $P \cdot p$ for every prime $p \mid P+1$.

The leftmost path (always choosing `minFac`) recovers the standard EM walk; every other path corresponds to a different selection. All nodes of $\mathcal{T}(m)$ share the same structural properties: each factor is prime, divides $P+1$, and is coprime to P ; the accumulator is strictly increasing along every branch.

Definition 6.1 (GenMixedMC). $\text{GenMixedMC}(m)$: for every prime q , some node of $\mathcal{T}(m)$ has q dividing its Euclid number $P+1$. Equivalently, every prime appears somewhere in the factor tree.

The standard Mullin Conjecture is the special case where the leftmost path alone captures every prime; GenMixedMC relaxes this to any path. Formally, $\text{MC} \implies \text{GenMixedMC}(2)$, since the standard walk is one particular path through $\mathcal{T}(2)$.

Remark 6.2 (The Euclid–Mullin graph). The factor tree $\mathcal{T}(m)$ is exactly the Euclid–Mullin graph G_m of Booker and Irvine [2]. In their framework, vertices are accumulated products and edges are labelled by primes dividing $P+1$; GenMixedMC(m) asks whether every prime labels some edge. Computationally, every prime below 10^9 occurs as an edge in G_1 (hence in G_2).

Booker [3] proved that for any finite set P of primes, there exists a path through $\mathcal{T}(\prod P)$ containing every prime. His proof uses three ingredients that connect directly to the formalization’s infrastructure:

1. Squarefree smooth density: the set S_q of residues mod q attained by squarefree $(q-1)$ -smooth integers satisfies $|S_q| > (q-1)/2$ (via Schnirelmann density $\geq 53/88$)—the same population equidistribution that DSL seeks to transfer to a single orbit.
2. Weil–Hasse bounds for the curves $y^2 = x(x^2 + a)$ and $y^2 = x^6 + a$ —the same character-sum cancellation exploited by the function field analog (§6.7).
3. Subgroup growth: the subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$ generated by the reachable primes has index dividing 6—an analogue of SubgroupEscape (Theorem 2.16), which guarantees full generation for the standard walk.

Booker’s proof works because choosing subsets at each step provides enough freedom to steer residues mod q . The standard EM sequence (`minFac`, always using the full product) has no such freedom: the accumulator is deterministic, and the gap between population equidistribution (Booker’s Lemma 2) and orbit-specific equidistribution (CME) is exactly what DSL addresses.

Remark 6.3 (Quantifier structure). $\text{GenMixedMC}(m)$ asserts: for each prime q , some path through $\mathcal{T}(m)$ reaches a node where q divides the Euclid number. Different primes may require different paths. The quantifier order is $\forall q \exists \text{path}$, not $\exists \text{path} \forall q$. The stronger statement—a single path capturing all primes simultaneously—remains open even for almost all starting points.

Lean encoding: MixedSelection.

The formalization encodes paths through $\mathcal{T}(m)$ via a `MixedSelection` $\sigma : \mathbb{N} \rightarrow \text{Option } \mathbb{N}$ (`✓ MixedSelection`): `none` at step k means follow the leftmost child (`minFac`); `some p` means branch to child p . A selection is `valid` (`✓ ValidMixedSelection`) if every `some p` step has p prime and $p \mid P_\sigma(k)+1$. The accumulator along a path is $P_\sigma(0) = m$, $P_\sigma(n+1) = P_\sigma(n) \cdot f_\sigma(n)$ (`✓ mixedWalkProd`).

The reachable set: tree projection.

Define $R_n(q, m)$ as the set of positions $P_\sigma(n) \bmod q$ over all valid paths σ (`✓ reachableAt`)—the projection of depth- n nodes of $\mathcal{T}(m)$ onto $(\mathbb{Z}/q\mathbb{Z})^\times$. The cumulative set $R_\infty = \bigcup_n R_n$ collects

all residues attained by any node. Hitting is equivalent to reachability:

$$q \text{ appears in } \mathcal{T}(m) \iff -1 \in R_\infty(q, m)$$

([✓ mixed_hitting_iff_neg_one_reachable ✓](#)). This reformulates GenMixedMC as a BFS reachability problem in $(\mathbb{Z}/q\mathbb{Z})^\times$. The reachable set is monotone along walk prefixes: any position reachable from a descendant $P_\sigma(n)$ is also reachable from the original accumulator m ([✓ reachableEver_mono_along_walk ✓](#)), via concatenation of walk segments.

Reachable set growth.

The core growth mechanism: at any node with accumulator P , every prime p dividing $P+1$ produces a child at position $P \cdot p \bmod q$ ([✓ reachableAt_from_factor ✓](#)). When $P+1$ is composite, it has ≥ 2 prime divisors, giving ≥ 2 children ([✓ reachable_composite_branch ✓](#)). When these are distinct mod q , $|R_{n+1}| > |R_n|$ along this branch.

The purification lemma.

The probabilistic argument for the random variant requires Diaconis–Shahshahani mixing theory and Borel–Cantelli, neither available in Lean. The formalization instead proves that the nondeterministic framework captures everything:

Purification Lemma ([✓ pure_random_mc_iff_mixed_mc ✓](#)). $\text{PureRandomMC}(q) \Leftrightarrow \text{MixedMC}(q)$.

The proof constructs, from any path through $\mathcal{T}(m)$ (which may follow the leftmost child at some steps), a purified path that always branches explicitly and follows the exact same trajectory ([✓ purify_walk_eq ✓](#)). This works because $\text{minFac}(P+1)$ is itself a prime dividing $P+1$, so replacing every “follow leftmost” with “branch to $\text{minFac}(P+1)$ ” produces a valid explicit path. The distinction between “ minFac -first” and “all-random” is thus illusory at the nondeterministic level. Standard MC implies GenMixedMC(2) (the leftmost path is one particular branch, [✓ standard_mc_implies_pure_random ✓](#)).

Variant hierarchy.

The factor tree framework unifies the prior variants in a strict hierarchy:

$$\text{deterministic (leftmost path)} \subset \text{two-point (§6.4)} \subset \mathcal{T}(m) \subset \text{full random.}$$

The two-point walk (§6.4) restricts branching to the two smallest factors; the full random walk corresponds to sampling a uniformly random branch at each node. Any capture by the two-point walk gives a capture in $\mathcal{T}(m)$ ([✓ two_point_capture_implies_mixed_capture ✓](#)).

Basic properties.

Under validity and $m \geq 2$, the formalization proves: every factor is prime ([✓ mixedWalkFactor_prime ✓](#)); every factor divides $P+1$ ([✓ mixedWalkFactor_dvd ✓](#)); the accumulator stays ≥ 2 ([✓ mixedWalkProd_ge_two ✓](#)); the accumulator is strictly increasing ([✓ mixedWalkProd_strict_mono ✓](#)); each factor is coprime to the accumulator ([✓ mixedWalkFactor_not_dvd_prod ✓](#)); prefix dependence: $P_\sigma(k)$ depends only on $\sigma(0), \dots, \sigma(k-1)$ ([✓ mixedWalkProd_depends_on_prefix ✓](#)); tail restart: the subtree from step K is a fresh tree from $P_\sigma(K)$ ([✓ mixedWalkProd_tail_restart ✓](#)).

Theorem 6.4 (Mixed walk structure). For any valid selection σ from a starting point $m \geq 2$:

1. Exponential growth ([✓ mixedWalkProd_exp_growth ✓](#)): $P_\sigma(k) \geq m \cdot 2^k$ for all k .
2. Squarefree preservation ([✓ mixedWalkProd_squarefree ✓](#)): if m is squarefree, so is $P_\sigma(k)$ for

- all k .
3. Coprimality from no death ([✓ mixedWalkProd_coprime_of_no_death ✓](#)): if the walk never hits $-1 \bmod q$, then $\gcd(P_\sigma(k), q) = 1$ for all k .
 4. Strict monotonicity ([✓ mixedWalkProd_strict_mono ✓](#)): P_σ is strictly increasing.

Unconditional capture of $q = 3$.

From $m = 2$, the root has $P+1 = 3$. Since 3 is prime, every branch at depth 1 selects factor 3—there is no other choice. Thus 3 appears in $\mathcal{T}(2)$ unconditionally ([✓ mixed_capture_three ✓](#)).

The diversity hypothesis.

The tree has branching > 1 at a node when $P+1$ is composite. [MixedDiversity](#) ([✓ MixedDiversity](#)): along the leftmost path, $P(n)+1$ is composite cofinally often. Under MixedDiversity, the tree has genuine branching at infinitely many levels, and PRE (Theorem 2.16) guarantees the factors generate the full unit group $(\mathbb{Z}/q\mathbb{Z})^\times$. Since $P(n)$ grows super-exponentially, $P(n)+1$ is composite for all known $n \geq 1$, so MixedDiversity is empirically unchallenged. The conditional reduction gives:

Theorem 6.5 (Diversity implies MixedMC — [✓ mixed_diversity_weak_implies_mixed_mc ✓](#)). [MixedDiversityWeak + MixedDiversity \$\implies\$ MixedMullinConjecture](#). The proof is by strong induction on q : MixedDiversityWeak provides branching at each level, MixedDiversity ensures enough branching to generate the full unit group, and PRE converts generation to hitting.

The perpetual primality obstruction.

If $P(n)+1$ were prime for all $n \geq N$, the recurrence would collapse to $P(n+1) + 1 = \Phi_3(P(n))$ (the third cyclotomic polynomial, [✓ perpetual_prime_cyclotomic ✓](#)), and $P(n) \equiv 1 \pmod{3}$ would force $3 \mid \Phi_3(P(n))$, making $P(n)+1$ composite—a contradiction. More precisely, perpetual primality confines the walk to $\{0, 2\} \bmod 3$ ([✓ perpetual_prime_excludes_mod3_one ✓](#)). More generally, under perpetual primality the walk mod q follows the autonomous map $w \mapsto w(w+1)$ on $\mathbb{Z}/q\mathbb{Z}$ ([✓ perpetual_prime_autonomous_mod ✓](#)), and by pigeonhole is eventually periodic ([✓ perpetual_prime_eventually_periodic ✓](#)). For $q = 5$, if the walk enters $\{1, 2\}$, the autonomous orbit alternates between 1 and 2 forever (since $1 \cdot 2 = 2$ and $2 \cdot 3 = 1$ in $\mathbb{Z}/5\mathbb{Z}$), never reaching $-1 = 4$ ([✓ perpetual_prime_mod5_orbit ✓](#)). These results combine into a clean structural statement:

Theorem 6.6 (Perpetual primality periodicity). Suppose $P(n) + 1$ is prime for all $n \geq N$. Then for every prime q :

1. The walk mod q satisfies the autonomous recurrence $w(n+1) = w(n) \cdot (w(n)+1)$ ([✓ perpetual_prime_autonomous_mod ✓](#)).
2. The walk is eventually periodic mod q with period $T \leq q$ ([✓ perpetual_prime_eventually_periodic ✓](#)).
3. For $q = 3$: the walk avoids $1 \bmod 3$, confined to $\{0, 2\}$ ([✓ perpetual_prime_excludes_mod3_one ✓](#)).
4. For $q = 5$: the orbit from $\{1, 2\}$ is a 2-cycle, never reaching $-1 = 4$ ([✓ perpetual_prime_mod5_orbit ✓](#)).
5. For $q = 11$: the orbit from $\{2, 6, 9\}$ is a 3-cycle, never reaching $-1 = 10$ ([✓ perpetual_prime_mod11_orbit ✓](#)).

6. For $q = 17$: the orbit from $\{2, 3, 4, 6, 8, 12\}$ is a closed set (tail + 2-cycle $\{3, 12\}$), never reaching $-1 = 16$ ([✓ perpetual_prime_mod17_orbit ✓](#)).
7. For $q = 23$: the orbit from $\{2, 6, 12, 18, 19, 20\}$ is a closed set (tail + 5-cycle), never reaching $-1 = 22$ ([✓ perpetual_prime_mod23_orbit ✓](#)).

The four-prime orbit exclusion ([✓ perpetual_primality_multi_exclusion ✓](#)) shows that under perpetual primality, primes 5, 11, 17, 23 are simultaneously blocked from entering the sequence.

Factor Confinement

The reachable set $R_\infty(q, m) = \bigcup_n R_n(q, m)$ cannot be confined to any proper coset $g \cdot H$ of $(\mathbb{Z}/q\mathbb{Z})^\times$: if $R_\infty \subseteq g \cdot H$, then all multiplier residues lie in H , contradicting PrimeResidueEscape ([✓ prime_residue_escape ✓](#)).

Theorem 6.7 ([✓ reachableEver_not_in_coset ✓](#)). For $q \geq 3$ prime and any proper subgroup $H < (\mathbb{Z}/q\mathbb{Z})^\times$, $R_\infty(q, 2) \not\subseteq g \cdot H$ for every g .

Coset impossibility constrains the geometry of R_∞ but does not force $R_\infty = (\mathbb{Z}/q\mathbb{Z})^\times$. A complementary approach attacks the factor side: if $R_\infty \subsetneq (\mathbb{Z}/q\mathbb{Z})^\times$, then the prime factors of every Euclid number $P(n)+1$ along every path are confined to a “forbidden” set determined by R_∞ ([✓ factor_confinement ✓](#)).

Property 6.8 (FactorEscapeHypothesis — [✓ FactorEscapeHypothesis](#)). For every prime $q \geq 3$, the Euclid numbers of $\mathcal{T}(2)$ eventually have a prime factor outside the forbidden set induced by any proper $R \subsetneq (\mathbb{Z}/q\mathbb{Z})^\times$.

Theorem 6.9 ([✓ factor_escape_implies_mixed_mc_at ✓](#)). $\text{FactorEscapeHypothesis}(q) \implies R_\infty(q, 2) = (\mathbb{Z}/q\mathbb{Z})^\times \implies$ every prime q appears in $\mathcal{T}(2)$.

The comparison between the standard walk and the factor tree is stark:

	Standard walk	Factor tree $\mathcal{T}(m)$
Factors used per step	1 (minFac)	All $\omega(P+1)$
Confinement forces	minFac $\in F$	Every factor $\in F$
Population analogy	DSL (orbit transfer)	PSCD (sieve theory)

For the standard walk, avoiding -1 confines only one factor per step; for the tree, it confines all $\omega(P+1)$ factors simultaneously—a much stronger sieve constraint that can be attacked with population methods.

6.3 Almost All GenMixedMC: A Density-1 Result

The factor confinement of §6.2 connects the hitting question to sieve theory over the population of starting points. For a squarefree m coprime to q , if $-1 \notin R_\infty(q, m)$, then R_∞ misses a nonzero element of $\mathbb{Z}/q\mathbb{Z}$. Moreover, the walk stays in units (each multiplier is prime and $\neq q$, since $q \mid P+1$ would place -1 in R_∞), so $0 \notin R_\infty$ as well ([✓ zero_not_reachable_of_coprime_trapped ✓](#)). At the root, all prime factors of $m+1$ have residues in the allowed factor set determined by R_∞ ([✓ unconditional_confinement ✓](#)). By pigeonhole over the finitely many proper finsets R , the “trapped” count is bounded by the sum of confined counts ([✓ trapped_le_sum_confined ✓](#)).

The open hypothesis PSCD (Population Sieve Confinement Decay) asserts that among squarefree

integers, the fraction confined to any finset R missing a nonzero element of $\mathbb{Z}/q\mathbb{Z}$ tends to zero. The restriction to R missing a nonzero element is essential— $R = (\mathbb{Z}/q\mathbb{Z})^\times$ is proper as a subset of $\mathbb{Z}/q\mathbb{Z}$ but confinement to it merely means $q \nmid m+1$, which has density $(q-1)/q > 0$.

Theorem 6.10 ([✓ pscd_implies_almost_all_mixed_hitting ✓](#)). $\text{PSCD}(q)$ implies: for a density-1 set of squarefree m coprime to q , every prime q appears in $\mathcal{T}(m)$.

Reduction to analytic number theory.

PSCD reduces to standard analytic number theory via two routes.

Route 1 (original, one open bridge). Three intermediate steps:

1. [✓ PrimesEquidistributedInAP \$\Rightarrow \sum_{p \equiv a} 1/p\$ diverges for every coprime class \$a\$ \(\[✓ peap_implies_fcd_proved\]\(#\)\)](#).
2. [SieveUpperBound](#): confined count $\leq C \cdot \text{sqfreeCount} \cdot \prod_{p \in F} (1 - 1/p)$ (fundamental lemma of sieve).
3. [✓ Divergent \$\sum 1/p\$ implies sieve product \$\rightarrow 0\$ \(\[✓ sieve_product_vanishing_proved\]\(#\)\)](#).

[SieveUpperBound](#)—the fundamental lemma of sieve theory—is open.

Route 2 (WeakFMCD, unconditional). The key insight is that FMCD’s exact constant is unnecessary: the sieve product $\prod_{p \in F} (1 - 1/p) \rightarrow 0$ absorbs any finite multiplicative constant. Define [WeakFMCD](#) ([✓ weak_fmcd_proved ✓](#)): for any fixed finite set of primes S ,

$$\text{sqfreeSievedCount}(S, X) \leq \left(4 \prod_{p \in S} (1 - 1/p) + \varepsilon \right) \cdot \text{sqfreeCount}(X).$$

The constant 4 comes from the unconditional bound $\text{sqfreeCount}(X) \geq X/4$ ([✓ sqfreeCount_ge_quarter_real ✓](#)), proved via $\sum_{d=2}^D 1/d^2 < 3/4$. WeakFMCD is then immediate from CRT residue class counting (each class has $\leq X/N + 2$ elements) and Euler’s product $\varphi(\prod S) / \prod S = \prod (1 - 1/p)$.

The central theorem ([✓ weak_fmcd_fcd_implies_pscd ✓](#)) proves [ForbiddenClassDivergent](#) alone $\Rightarrow \text{PSCD}$. The proof decomposes the confined count over residue classes mod q as before, but with $\varepsilon/(8q)$ for the sieve target and $\varepsilon/(2q)$ for the WeakFMCD tolerance, so $4 \cdot \varepsilon/(8q) + \varepsilon/(2q) = \varepsilon/q$; summing over $q-1$ classes gives $< \varepsilon$.

The composed chain ([✓ weak_fmcd_chain_implies_pscd ✓](#)):

$$\text{PrimesEquidistInAP} \xRightarrow{\checkmark} \text{PSCD}(q) \xRightarrow{\checkmark} \text{a.a. GenMixedMC}.$$

Every arrow is machine-checked. No open hypothesis remains: PEAP (Wiener–Ikehara) is the sole input, and the CRT bound is unconditional.

Dead End #117: MultCancelToWalkCancel

The standard ensemble approach (§6.5) proves almost-all multiplier cancellation for the deterministic walk, but transferring this to walk hitting is blocked by Dead End #117 (MultCancelToWalkCancel). The factor tree approach bypasses #117 entirely:

	Standard ensemble	Factor tree ensemble
Population result	a.a. char cancel	a.a. GenMixedMC
Transfer gap	#117 (cancel $\not\Rightarrow$ hitting)	None
Selection used	minFac only	All factors of $P+1$
Open hypothesis	StepDecorrelation	None (PEAP only)

This is the strongest density result in the formalization: under `PrimesEquidistributedInAP` alone (no additional open hypothesis), for a density-1 set of squarefree starting points, every prime appears in the factor tree.

Interpolation: tree reachability implies positive probability capture.

The density-1 tree result bridges to the $(1-\varepsilon)$ minFac + ε random process. Assign to each factor choice at accumulator P the weight $\varepsilon/\omega(P+1)$ —a lower bound on the probability of that choice under the interpolated process (where ω counts distinct prime factors). A finite path σ of length n has weight $\prod_{k < n} \varepsilon/\omega(P_\sigma(k)+1) > 0$ ([✓ pathWeightLB_pos ✓](#)).

If $-1 \in R_\infty(q, m)$ (tree reachability), a capturing path exists, and that path has positive weight under the $(1-\varepsilon)$ process for any $\varepsilon > 0$ ([✓ reachable_implies_positive_prob_capture ✓](#)). Under PEAP, almost all squarefree m have -1 reachable (WeakFMCD chain), so almost all have positive probability capture ([✓ almost_all_positive_prob_capture ✓](#)). No additional open hypothesis beyond PEAP is needed.

Layer 2: cofinal and iterated hitting via regeneration.

Layer 1 proves a single hit with positive weight. Stronger conclusions require [Regeneration](#) ([✓ Regeneration](#)): every accumulator encountered along any valid walk from m is good, i.e., every unit of $(\mathbb{Z}/q\mathbb{Z})^\times$ remains reachable. The evidence is strong: accumulators grow super-exponentially and are squarefree, so by CRT blindness they should behave generically mod q .

Under Regeneration, the formalization proves:

- Block coverage ([✓ block_coverage ✓](#)): from a good accumulator, there is a uniform depth bound N_0 such that every unit can be reached within N_0 steps.
- Cofinal hitting ([✓ regeneration_implies_cofinal_hitting ✓](#)): for every threshold K , there exists a valid path that hits $-1 \bmod q$ at some step $n \geq K$, with positive weight.
- Iterated hitting ([✓ regeneration_implies_iterated_hitting ✓](#)): for every target number of -1 hits, there exists a valid path achieving at least that many hits, each with positive weight. The proof proceeds by induction on the target, concatenating walks via [concatMixedSelection](#) and using Regeneration to obtain fresh good accumulators past each existing hit.

Regeneration is the sole open hypothesis in this layer; it is strictly weaker than DSL (which implies MC outright) and concerns the persistence of a qualitative property (full reachability) along the walk, not a quantitative equidistribution statement.

TreeSieveDecay: the sieve-theoretic bridge.

Regeneration asks that every intermediate accumulator is good. A weaker, sieve-flavored variant is [TreeSieveDecay](#) ([✓ TreeSieveDecay](#)): there exists a threshold P_0 such that every squarefree $P \geq P_0$ coprime to q is a good accumulator. TSD is a “large- P ” hypothesis: all sufficiently generic starting points support full reachability. The coprimality condition is necessary: when $q \mid P$,

absorption traps the walk at 0 permanently, so no unit is reachable. TSD bypasses Regeneration entirely—for a single hit (which suffices for MC), a cleaner dichotomy argument works. The formalization proves two bridges:

- Squarefree propagation ([✓ mixedWalkProd_squarefree ✓](#)): if the starting point is squarefree, every intermediate accumulator is squarefree (induction, using coprimality of factors).
- Coprimality from no death ([✓ mixedWalkProd_coprime_of_no_death ✓](#)): if the walk never hits $-1 \bmod q$, all intermediate accumulators stay coprime to q (if the factor equaled q , then $q \mid P+1$, so the walk position would be -1 —contradiction).

Together: under TSD, from any sufficiently large squarefree m , the walk either hits -1 (done) or stays coprime to q , in which case TSD applies to each intermediate accumulator directly ([✓ tsd_implies_neg_one_reachable ✓](#)).

A weaker variant [TreeSieveDecayHitting](#) ([✓ TreeSieveDecayHitting ✓](#)) requires only that -1 is reachable (not all units). TSD implies TSDH ([✓ tsd_implies_tsdh ✓](#)).

Orbit melting: backward propagation.

Two squarefree positive integers with the same prime factor set are equal ([✓ eq_of_same_primeFactors_squarefree ✓](#)). This “orbit melting” identity means the accumulator carries no hidden state beyond its prime factorization: same accumulator $\Rightarrow$ same reachable set ([✓ same_accumulator_same_future ✓](#)). More importantly, a good accumulator at any intermediate step propagates backward: if $P_\sigma(K)$ supports full reachability, then -1 is reachable from the starting point m by concatenating the prefix path with the witnessing path ([✓ good_accumulator_propagates_to_start ✓](#)).

TSD-Hitting(3): unconditional.

For $q = 3$, [TreeSieveDecayHitting](#) holds unconditionally—no open hypothesis at all. The argument is a mod-3 dichotomy on any squarefree P coprime to 3:

- If $P \equiv 2 \pmod{3}$: then $(P : \mathbb{Z}/3\mathbb{Z}) = -1$, so the walk is already at -1 at step 0.
- If $P \equiv 1 \pmod{3}$: then $P+1 \equiv 2 \pmod{3}$. A number ≥ 2 that is $\equiv 2 \pmod{3}$ and not divisible by 3 must have a prime factor $p \equiv 2 \pmod{3}$ ([✓ exists_factor_of_succ_mod3 ✓](#), by strong induction: if $\text{minFac} \equiv 1$, divide out and recurse). Selecting that factor gives $P \cdot p \equiv 1 \cdot 2 = -1 \pmod{3}$ at step 1.

Theorem 6.11 ([✓ tsd_hitting_three_unconditional](#)). [TreeSieveDecayHitting\(3\)](#) holds unconditionally: for every squarefree $P \geq 2$ coprime to 3, there exists a valid path in the factor tree that reaches $-1 \bmod 3$ within one step.

```
-- The hypothesis: -1 reachable from all large squarefree coprime P
def TreeSieveDecayHitting (q : Nat) : Prop :=
  exists P_0, forall P, P_0 ≤ P -> Squarefree P -> Nat.Coprime P q ->
    (-1 : ZMod q) in reachableEver q P

-- UNCONDITIONAL for q = 3 (zero open hypotheses)
theorem tsd_hitting_three_unconditional : TreeSieveDecayHitting 3 := by
  -- mod-3 dichotomy: P = 2 -> already at -1;
```

```
-- P = 1 -> P+1 = 2, so some prime factor p = 2 mod 3 exists
...
```

This is the first unconditional tree-level hitting result beyond the trivial $q = 3$ capture of §6.2. The previous $q=3$ result ([✓ mixed_capture_three ✓](#)) showed that 3 appears in $\mathcal{T}(2)$ from the fixed root $m=2$; TSD-Hitting(3) proves that -1 is reachable from every coprime squarefree accumulator—a uniform statement that applies at any intermediate point of the walk.

TSD $\Rightarrow$ MixedMC: composition.

The threshold condition in TSD (accumulator P must exceed a bound P_0) is eliminated by the exponential growth of the mixed walk: the accumulator $P_\sigma(k)$ satisfies $P_\sigma(k) \geq m \cdot 2^k$ ([✓ mixedWalkProd_exp_growth ✓](#)), since each multiplier is a prime ≥ 2 . For any starting accumulator $m \geq 2$, the walk eventually exceeds P_0 ; a death/no-death dichotomy then applies: either the walk already hit -1 (done), or all accumulators remain coprime to q and squarefree, so TSD applies at the first step past the threshold, and tail reachability (Part 9) propagates -1 back to the root. This gives the full chain:

Theorem 6.12 (TSD reduction chain). For each prime $q \geq 5$:

$$\text{TSD}(q) \implies \forall m \geq 2 \text{ sqfree}, \gcd(m, q) = 1: -1 \in R_\infty(q, m) \implies \text{MixedMC}(q).$$

The first arrow uses exponential growth (Theorem 6.4) to exceed the TSD threshold ([✓ tsd_implies_reachable_from_any ✓](#)); the second uses hit-implies-capture ([✓ hit_implies_capture' ✓](#)). Consequently ([✓ tsd_all_implies_mixed_mc ✓](#)):

$$(\forall q \geq 5, \text{TSD}(q)) \implies \text{MixedMullinConjecture}.$$

Small primes $q \in \{2, 3\}$ are handled by their existing unconditional results ([mixed_mc_two](#), [mixed_mc_three](#)).

Coset ambiguity: why TSD-Hitting(3) is free but $q \geq 5$ is not.

The structural reason that TSD-Hitting(q) closes unconditionally for $q = 3$ but requires a hypothesis for $q \geq 5$ is coset ambiguity. For $q = 3$, the group $(\mathbb{Z}/3\mathbb{Z})^\times = \{1, 2\}$ has a single non-identity element, which is $-1 = 2$: any prime factor escaping the trivial coset $\{1\}$ automatically lands on -1 ([✓ single_coset_implies_immediate_hit ✓](#)). For $q = 5$, the group $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$ has the subgroup $\{1, 4\} = \{1, -1\}$ and two other cosets $\{2\}$, $\{3\}$, neither equal to -1 . A concrete counterexample: $P = 2$, $P+1 = 3$, $(3 \bmod 5) = 3 \neq 4 = -1$ —the only available factor misses -1 ([✓ coset_ambiguity_counterexample ✓](#)).

Ensemble factor diversity and the stochastic EM process.

The ensemble-level structure of the prime factor sets $\text{primeFactors}(P_n(k) + 1)$ is formalized in [Advanced/FactorDiversity.lean](#) (346 lines, zero [sorry](#)).

Theorem 6.13 (Spectral contraction from factor diversity — [✓ factor_diversity_spectral_contraction ✓](#)). If the prime factor set of $P_n(k) + 1$ contains ≥ 2 distinct residue classes mod q , then the mean character value over the factor set satisfies $\|\bar{\chi}(F)\| < 1$ for every nontrivial χ . If such diverse steps occur infinitely often with a uniform gap $\delta > 0$, the averaged character product

tends to zero ([✓ diverse_steps_imply_vanishing ✓](#)).

The bridge from diversity to stochastic capture ([Advanced/DiverseStepsToCapture.lean](#), 281 lines) shows each prime factor of $P_2(k) + 1$ produces a distinct element of the reachable set $R_\infty(q, 2)$ ([✓ genFactor_in_reachableEver ✓](#)).

Theorem 6.14 (Integer stochastic MC — [✓ stochastic_mc_of_tsd ✓](#)). $\text{TSD}(q) \implies \text{StochasticMC}(q)$ for all $\varepsilon > 0$. The phase transition is sharp at $\varepsilon = 0$ ([✓ phase_transition_summary ✓](#)): character norms are exactly 1 (no contraction) for the deterministic walk, but strictly < 1 under any stochastic perturbation.

Probabilistic infrastructure.

The $\text{TSD} \rightarrow \text{MixedMC}$ chain is supported by three infrastructure modules. [Probability/TransitionKernel.lean](#) defines the epsilon-mixed walk’s step PMF and proves the weight lower bound $\varepsilon/\omega(P+1)$ bridges to InterpolationMC’s [stepWeightLB](#). [Probability/PathMeasure.lean](#) converts set-theoretic reachable sets to computable Finsets and formalizes branching distinctness: when P is a unit mod q and $P+1$ has distinct factor residues, each gives a distinct next walk position. [Probability/GeometricCapture.lean](#) provides the block-geometric engine: per-block positive capture weight decays non-capture probability as $(1-\delta)^K \rightarrow 0$, turning TSD into asymptotic capture certainty.

6.4 The $\{\text{minFac}, \text{secondMinFac}\}$ Variant

The two-point variant interpolates between the random and deterministic extremes. At each step, instead of a single deterministic choice or a fully random one, two specific factors are available: $\text{minFac}(P_2(n) + 1)$ and $\text{secondMinFac}(P_2(n) + 1)$. A decision sequence $\sigma : \mathbb{N} \rightarrow \text{Bool}$ selects between them; the all-true sequence recovers the standard EM walk.

The ε -walk.

The formalization defines a self-consistent ε -walk ([✓ epsWalkProd](#)) in which the chosen factor—either $\text{minFac}(P(n)+1)$ or $\text{secondMinFac}(P(n)+1)$ —becomes the actual multiplier, updating the accumulator and producing a new Euclid number at the next step. The function $\text{secondMinFac}(n)$ (second-smallest prime factor) is defined and proved prime, divisor of n , and $\geq \text{minFac}(n)$ ([✓ secondMinFac_prime](#), [✓ secondMinFac_dvd](#), [✓ minFac_le_secondMinFac](#)).

This self-consistent walk generates a binary branching tree of depth N with 2^N leaves, where each internal node has its own path-dependent factor set. The tree character sum ([✓ treeCharSum](#)), a weighted sum of $\chi(\text{leaf position})$ over all leaves, is proved ≤ 1 in norm. Unlike the product multiset of the stochastic MC, this sum does not factorize as $\prod_k \bar{\chi}(S_k)$: the left and right subtrees start from different accumulators, so their character sums differ. The open hypothesis [TreeContractionHypothesis](#) asserts that the tree character sum tends to zero for all nontrivial χ ; it is strictly weaker than DSL but strictly stronger than the product approximation.

Non-self-consistent variant MC.

An independent route bypasses the tree contraction problem entirely. At each EM step n , form the two-element unit set $\{\text{minFac}(P(n)+1), \text{secondMinFac}(P(n)+1)\} \bmod q$ in $(\mathbb{Z}/q\mathbb{Z})^\times$. When one factor equals q (a non-unit), fall back to [Finset.univ](#), whose mean character value is 0 by orthogonality—giving perfect contraction ([✓ meanCharValue_univ_eq_zero](#)). Under [UFDStrong](#)—spectral gaps of the padded unit sets are non-summable for every nontrivial χ —the

sparse product contraction machinery gives vanishing averaged products, and the Fourier counting argument yields path existence. Concretely:

Theorem 6.15 ([✓ variant_mc_from_ufd_strong_proved](#)). For every prime $q \geq 3$, if $\text{UFDStrong}(q)$ holds, then every element of $(\mathbb{Z}/q\mathbb{Z})^\times$ is representable as a product of selections from $\{\text{minFac}, \text{secondMinFac}\}$ of consecutive Euclid numbers. In particular, some selection path hits -1 .

The verified chain is:

$$\text{UFDStrong}(q) \implies \text{vanishing products} \implies \text{path existence} \implies \text{VariantMC}.$$

Every arrow is machine-checked with zero [sorry](#).

The gap: UFDStrong .

$\text{UFDStrong}(q)$ asserts: for each nontrivial character χ on $(\mathbb{Z}/q\mathbb{Z})^\times$, the spectral gaps $1 - \|\bar{\chi}(\text{paddedUnitSet}_n)\|$ are non-summable. The hypothesis splits into two cases:

1. Fallback case: if infinitely many steps have $\text{minFac} \equiv \text{secondMinFac} \pmod{q}$ (or one equals q), the padded set falls back to the full group, giving perfect contraction (gap = 1) at each such step. Non-summability is immediate. Proved: [✓ ufd_fallback_not_summable](#).
2. Diversity case: if cofinitely many steps have two distinct units mod q , one needs the character values at those units to differ infinitely often—i.e., the ratio $\text{minFac} / \text{secondMinFac}$ does not persistently lie in $\ker(\chi)$. This is a genuine number-theoretic question about Euclid number factorizations. Open: [UFDImpliesUFDStrong](#).

Variant MC does not imply standard MC.

The selection σ that achieves hitting may differ per prime q , and the virtual accumulator $\prod_k \text{factor}_{\sigma(k)}$ is not the EM accumulator $P(n)$. The gap is the same orbit-specificity barrier (Dead End #130, [✓ alternating_walk_misses_two](#)): generation does not imply coverage for deterministic walks.

Three routes to UFDStrong .

The formalization identifies a hierarchy of progressively weaker sufficient conditions for $\text{UFDStrong}(q)$:

1. Quantitative escape ([MinFacRatioEscape](#)): for each nontrivial χ , there exists $\delta > 0$ such that infinitely many steps have spectral gap $\geq \delta$. Since a non-summable sequence with a positive lower bound infinitely often is non-summable, this implies UFDStrong directly ([✓ ratio_escape_implies_ufdStrong](#)).
2. Qualitative escape ([MinFacRatioEscapeQual](#)): for each nontrivial χ , infinitely many steps have $\text{card}(\text{rawTwoPointUnitSet}) \geq 2$ and $\chi(\text{minFac}) \neq \chi(\text{secondMinFac})$. This implies quantitative escape via a finite-group pigeonhole argument: on a finite group, the gap function takes finitely many values, so being positive infinitely often gives a uniform positive lower bound ([✓ qual_implies_quant](#)).
3. Orbit-level minFac equidistribution ([OrbitMFRE](#)): the residues $\text{minFac}(P_2(n)+1) \pmod{q}$ are equidistributed among the units of $(\mathbb{Z}/q\mathbb{Z})^\times$ in Cesàro density. This is the weakest condition in the hierarchy: equidistributed residues prevent persistent chi-degeneracy. The bridge $\text{OrbitMFRE} \implies \text{MinFacRatioEscapeQual}$ is open ([OrbitMFREImpliesEscapeQual](#)).

The assembled chain is:

$$\underbrace{\text{OrbitMFRE}}_{\text{open}} \xrightarrow{\text{open bridge}} \text{EscapeQual} \xrightarrow{\text{proved}} \text{EscapeQuant} \xrightarrow{\text{proved}} \text{UFDStrong} \xrightarrow{\text{proved}} \text{VariantMC}.$$

Every arrow except the first is machine-checked with zero [sorry](#). The most concrete attack target is `MinFacRatioEscapeQual`: one needs to show that `minFac` and `secondMinFac` do not persistently agree modulo every character. `OrbitMFRE` faces the same orbit-specificity barrier as DSL for standard MC.

Stochastic Two-Point MC.

The non-self-consistent machinery yields a clean quantitative statement. Define the [StochasticTwoPointMC](#) problem: under `UFDStrong( $q$ )` with $q \geq 3$, does every element of $(\mathbb{Z}/q\mathbb{Z})^\times$ appear with positive count in the product multiset of padded unit sets?

[Theorem 6.16](#) ([✓ stochastic_two_point_mc_proved](#)). $\text{UFDStrong}(q) \implies \text{StochasticTwoPointMC}(q)$.

The product multiset grows exponentially ($\geq 2^N$ paths for $q \geq 3$, [✓ productMultiset_card_ge_two_pow](#)). A [fairCoin](#) schedule ($\varepsilon = 1/2$ at every step) gives the [fairTreeCharSum](#)—the tree character sum at constant $1/2$ —which is bounded by 1 in norm ([✓ fairTreeCharSum_norm_le_one](#)). `TreeContractionHypothesis` implies the weaker [TreeContractionAtHalf](#) ([✓ treeContraction_of_hypothesis](#)), and at $\varepsilon = 0$ the tree degenerates to the deterministic `minFac` path ([✓ treeCharSum_at_zero_step](#)).

The gap between the non-self-consistent result (proved) and a self-consistent stochastic MC is the selection coherence problem: the non-self-consistent product multiset uses factor sets from the actual EM walk (which always picks `minFac`), so the factor sets are fixed and the product factorizes; in the self-consistent tree, picking `secondMinFac` at step n changes the accumulator, hence the factor set at step $n+1$, preventing factorization.

Faithful character escape.

A character χ on $(\mathbb{Z}/q\mathbb{Z})^\times$ is [faithful](#) if it is injective (equivalently, $\ker(\chi) = \{1\}$). For faithful characters, whenever the raw two-point unit set has two distinct elements, their χ -values are automatically distinct—giving a positive spectral gap at every non-fallback step, with no number-theoretic hypothesis. Combined with the fallback case (gap = 1 at fallback steps, already proved), this yields unconditional non-summability:

[Theorem 6.17](#) ([✓ faithful_character_escape](#)). For every prime $q \geq 3$ and every faithful nontrivial character χ on $(\mathbb{Z}/q\mathbb{Z})^\times$:

$$\sum_n (1 - \|\tilde{\chi}(\text{paddedUnitSet}_n)\|) = +\infty.$$

Unconditional—zero open hypotheses.

For groups of prime order, every nontrivial character is faithful (by Lagrange: the kernel is a subgroup of a prime-order group, hence trivial or the whole group; [✓ prime_order_nontrivial_is_faithful](#)). Since $|(\mathbb{Z}/3\mathbb{Z})^\times| = 2$ is prime, `UFDStrong(3)` holds unconditionally ([✓ ufdStrong_three](#)), and the full chain closes:

[Theorem 6.18](#) ([✓ variant_mc_three_unconditional](#)). `StochasticTwoPointMC(3)` holds with zero open hypotheses: every element of $(\mathbb{Z}/3\mathbb{Z})^\times$ appears with positive count in the product multiset.

For primes $q \geq 5$ where $|(\mathbb{Z}/q\mathbb{Z})^\times| = q-1$ is composite, non-faithful nontrivial characters exist, and the sole remaining open hypothesis is [NonFaithfulCharacterEscape](#)—spectral gap non-summability for characters with nontrivial kernel. The faithful/non-faithful decomposition is clean:

Theorem 6.19 ([✓ nfce_implies_ufdStrong](#)). $\text{NonFaithfulCharacterEscape}(q) \implies \text{UFDStrong}(q)$ for all primes $q \geq 3$.

The faithful half is free; only the non-faithful half remains open.

NFCE infrastructure.

The formalization develops a precise characterization of what NFCE requires. Define the [factorRatio](#) at step n as $\text{liftToUnit}(\text{minFac}) \cdot \text{liftToUnit}(\text{secondMinFac})^{-1}$ in $(\mathbb{Z}/q\mathbb{Z})^\times$. At non-fallback steps (raw card ≥ 2), the spectral gap is zero if and only if this ratio lies in $\ker(\chi)$ ([✓ gap_zero_iff_ratio_in_ker](#)). NFCE thus reduces to: for each non-faithful nontrivial χ , the factor ratio does not eventually stay in $\ker(\chi)$.

If NFCE fails for a specific χ , the gaps are summable, which forces [KernelConfinement](#): the ratio is eventually confined to $\ker(\chi)$ ([✓ summable_implies_ratio_confined](#)). Since $\ker(\chi)$ has index ≥ 2 for nontrivial χ ([✓ ker_index_ge_two](#)), kernel confinement is a strong arithmetic constraint: $\text{minFac} / \text{secondMinFac} \bmod q$ eventually lies in a proper subgroup of the unit group.

NFCE(5): reduction to a single quadratic escape.

For $q = 5$, the unit group $(\mathbb{Z}/5\mathbb{Z})^\times$ has order 4 (cyclic, $\cong \mathbb{Z}/4\mathbb{Z}$). Its characters partition as: the trivial character, two faithful characters (order 4, handled unconditionally by [FaithfulCharacterEscape](#)), and one non-faithful nontrivial character (order 2, the unique quadratic character). Thus NFCE(5) reduces to a single escape condition: the factor ratio $\text{minFac} / \text{secondMinFac} \bmod 5$ must leave the unique index-2 subgroup of $(\mathbb{Z}/5\mathbb{Z})^\times$ infinitely often. The formalization proves that this [RatioKernelEscape](#) condition suffices for the full chain [✓ variant_mc_five_of_ratio_escape](#) [✓](#): $\text{RatioKernelEscape} \implies \text{NFCE}(5) \implies \text{UFDStrong}(5) \implies \text{StochasticTwoPointMC}(5)$.

Intersection kernel dichotomy.

When NFCE fails for all non-faithful nontrivial characters simultaneously ([TotalNFCEFailure](#)), the factor ratio is eventually confined to every non-faithful kernel (from the summability-to-confinement lemma of Part 25). The intersection of these kernels is constrained by a group-theoretic fact: in a finite group whose order has ≥ 2 distinct prime factors, there exist subgroups of coprime orders whose intersection is trivial ([✓ coprime_order_zpowers_inf_bot](#) [✓](#)). Combined with Cauchy's theorem, this yields [NonFaithfulCharSeparation](#) (NFCS): for every $g \neq 1$, some non-faithful nontrivial character has $\chi(g) \neq 1$. NFCS is proved for all groups whose order has ≥ 2 distinct prime factors ([✓ nonFaithfulCharSeparation_of_two_prime_factors](#) [✓](#)), which for $(\mathbb{Z}/q\mathbb{Z})^\times$ covers all primes except $q = 2, 3, 5$ and Fermat primes (where $q - 1$ is a power of 2). Under NFCS, [TotalNFCEFailure](#) forces $\text{minFac} / \text{secondMinFac} \equiv 1 \pmod{q}$ eventually ([✓ total_confinement_implies_ratio_one](#) [✓](#)). This yields:

Theorem 6.20 (Intersection kernel dichotomy). Let q be prime with $q - 1$ having ≥ 2 distinct prime factors. Then exactly one of the following holds:

1. $\text{NFCE}(q)$ holds, giving $\text{UFDStrong}(q)$ and $\text{StochasticTwoPointMC}(q)$; or
2. $\text{minFac}(P_2(n) + 1) \equiv \text{secondMinFac}(P_2(n) + 1) \pmod{q}$ for all sufficiently large n .

The second alternative is a strong arithmetic rigidity: the two smallest prime factors of each Euclid number are eventually indistinguishable mod q .

The proof combines NonFaithfulCharSeparation ([✓nonFaithfulCharSeparation_of_two_prime_factors ✓](#)) with the summability-to-confinement lemma ([✓total_confinement_implies_ratio_one ✓](#)). The condition “ $q-1$ has ≥ 2 distinct prime factors” covers all primes except $q = 2, 3, 5$ and Fermat primes (where $q-1$ is a power of 2).

Quotient character lift.

The quotient character infrastructure, formalized in Part 28 of [VanishingNoiseVariantD.lean](#) (173 lines), provides a structural reduction of NFCE. For a non-faithful nontrivial character χ , the kernel $\ker(\chi)$ is a proper nontrivial subgroup; by the universal property, χ factors as $\chi = \bar{\chi} \circ \pi$ where $\pi: G \rightarrow G/\ker(\chi)$ is the quotient map. The induced character $\bar{\chi}$ on the quotient group is faithful (injective) ([✓quotientChar_faithful ✓](#)), the quotient has order ≥ 2 ([✓quotient_card_ge_two ✓](#)), and KernelConfinement is equivalent to the quotient projection of the factor ratio being eventually trivial ([✓kernelConfinement_iff_quotient_eventually_one ✓](#)).

Theorem 6.21 (Quotient character lift). For any non-faithful nontrivial character χ on $(\mathbb{Z}/q\mathbb{Z})^\times$:

1. $\chi = \bar{\chi} \circ \pi$ where $\bar{\chi}$ is faithful on $G/\ker(\chi)$ and $|G/\ker(\chi)| \geq 2$.
2. $\text{KernelConfinement}(\chi) \iff$ the quotient projection of the factor ratio is eventually trivial.
3. $\text{RatioKernelEscape}(\chi) \iff$ the quotient projection is nontrivial infinitely often.

This reduces NFCE to: the image of factorRatio in each quotient $G/\ker(\chi)$ escapes $\{1\}$ infinitely often. Since $\bar{\chi}$ is faithful on the (smaller) quotient group, the unconditional FaithfulCharacterEscape result (Theorem 6.17) applies to $G/\ker(\chi)$, connecting the non-faithful case to the faithful framework.

Comparison of open hypotheses.

Standard MC requires DSL (population-to-orbit transfer for minFac residues). The $\{\text{minFac}, \text{secondMinFac}\}$ variant requires UFDStrong (spectral diversity of two-point factor sets), and for $q = 3$ this is now unconditional; for $q \geq 5$ the residual hypothesis is NonFaithfulCharacterEscape. Neither DSL nor UFDStrong implies the other: DSL is about the selected factor’s equidistribution; UFDStrong is about the pair’s spectral gap. But both are instances of the same meta-question: does the deterministic EM construction produce enough diversity in its factorizations to prevent persistent alignment with any fixed character?

Ensemble two-point: population-level reduction.

The ensemble framework of Section 4 can also be applied to the two-point variant. For generalized EM starting from squarefree n , the [genFactorRatio](#) at step k is the ratio $\text{minFac}/\text{secondMinFac} \bmod q$ in $(\mathbb{Z}/q\mathbb{Z})^\times$. For a nontrivial character χ , a step “escapes” if this ratio lies outside $\ker(\chi)$ —producing a spectral gap. The formalization ([Ensemble/Two-PointEnsemble.lean](#), 566 lines, zero [sorry](#)) develops the full first-moment chain:

$$\text{PRED} \xrightarrow[\text{Fubini}]{\checkmark} \text{linear first moment} \xrightarrow[\text{partition}]{\checkmark} \text{positive density of high escapers.}$$

Here PopulationRatioEscapeDensity (PRED) asserts that for each nontrivial χ and each step $k \geq 1$, a positive density $\delta > 0$ of squarefree starting points have factor ratio outside $\ker(\chi)$.

Under PRED, the ensemble average of cumulative escape counts grows linearly ([✓ escape_first_moment_linear ✓](#)), and the partition argument gives [✓ positive_density_high_escape ✓](#): for each threshold M , a positive density of starting points achieve $\geq M$ cumulative escapes within K steps. PRED itself is open; it should follow from MFRE (MinFacResidueEquidist) via CRT independence of the two smallest factors.

Self-consistent random walk: the Fourier bridge.

The stochastic two-point variant above fixes the factor sets $\{p_1(n), p_2(n)\}$ from the actual EM walk and randomizes the selection. A stronger variant makes the construction self-consistent: at each step, the accumulator determines the available factors, and the fair-coin choice feeds back into the next accumulator. Formally, define $\text{epsWalkProdFrom}(\text{acc}, \sigma, 0) = \text{acc}$, $\text{epsWalkProdFrom}(\text{acc}, \sigma, n+1) = P_n \cdot (\text{if } \sigma(n) \text{ then } \text{minFac}(P_n+1) \text{ else } \text{secondMinFac}(P_n+1))$ where $P_n = \text{epsWalkProdFrom}(\text{acc}, \sigma, n)$. The decision sequence $\sigma : \mathbb{N} \rightarrow \text{Bool}$ replaces the deterministic minFac rule with a random binary choice.

The central result is the Fourier bridge identity ([✓ fourier_bridge_identity_proved ✓](#)): for any accumulator $\text{acc} \geq 2$,

$$\text{treeCharSum}(q, \chi, N, \text{acc}, \tfrac{1}{2}) = \frac{1}{2^N} \sum_{\sigma \in \{0,1\}^N} \prod_{k < N} \text{chiAt}(q, \chi, \text{epsWalkFactorFrom}(\text{acc}, \sigma, k)).$$

The proof is by induction on N : at each step, the 2^N paths split by first bit into two halves corresponding to the left and right subtrees of the tree recursion; shift lemmas ([epsWalkProdFrom_shift](#), [epsWalkFactorFrom_shift](#)) connect the concatenated walk to the shifted walk at the new accumulator; the $\frac{1}{2}$ weights combine to match the tree recurrence. Specializing to $\text{acc} = 2$ gives the unconditional identity $\text{fairTreeCharSum} = \text{pathCharSum}$ ([✓ fairTreeCharSum_eq_pathCharSum ✓](#)).

[RandomTwoPointMC](#) ([✓ RandomTwoPointMC](#)) asks: for $q \geq 3$ and every unit $a \in (\mathbb{Z}/q\mathbb{Z})^\times$, does some binary decision sequence of finite length produce a walk from $\text{acc} = 2$ landing in residue class a ? This is the self-consistent analog of StochasticTwoPointMC. The Fourier bridge connects the tree to the path character sum. To connect path character sums to reached residue classes, one needs [ChiAtMultiplicativity](#): that $\text{chiAt}(\text{acc}) \cdot \prod_k \text{chiAt}(\text{factor}_k) = \text{chiAt}(\text{endpoint})$, provided q does not divide the endpoint. This is proved ([✓ chiAt_multiplicativity_proved ✓](#)) by induction on N with backward non-divisibility propagation: if $q \nmid P_N$, then $q \nmid P_k$ for all $k \leq N$ ([✓ not_dvd_epsWalkProdFrom_of_le ✓](#)), so every factor is coprime to q and chiAt is multiplicative.

A full Fourier counting infrastructure is built on top: character orthogonality on $(\mathbb{Z}/q\mathbb{Z})^\times$, death/survival partition ([✓ survival_plus_death ✓](#): $\text{survivalCount} + \text{deathCount} = 2^N$), the Fourier indicator formula $|G| \cdot \text{pathCount}(a) = \sum_\chi \overline{\chi(a)} \cdot U(\chi, N)$, and norm bounds for the unit endpoint character sum. The chain is:

$$\text{TCA} \xRightarrow{\checkmark} \text{vanishing pathCharSum} \xRightarrow{\text{ChiAtMult} \checkmark} \text{Fourier counting} \xRightarrow{?} \text{RandomTwoPointMC}.$$

The remaining gap is [PathSurvival](#) ([✓ PathSurvival](#)): the ratio $\text{survivalCount}(N)/\text{deathCount}(N)$ grows without bound. Without positive survival at the same N where pathCharSum is small, the Fourier inversion gives no information about unit classes.

For $q = 3$, the analysis reveals a structural obstruction. Since $\text{acc} = 2$ and $\text{minFac}(3) = \text{secondMinFac}(3) = 3$, the walk immediately dies: $3 \mid P_1 = 6$ on all paths, and absorption is

permanent. Thus `PathSurvival(3)` is false—but `TreeContractionAtHalf(3)` is also false (the tree character sum is identically 1 at every depth, because all post-death factors are $\equiv 1 \pmod{3}$), making `TreeContractionImpliesRandomMC(3)` vacuously true.

For $q = 3$, the zero-mean property gives maximal per-step contraction (`✓ mean_char_zero_at_diverse_step_three ✓`): $(1/2)\chi(a) + (1/2)\chi(b) = 0$ whenever $\chi(a) \neq \chi(b)$ for a nontrivial character on $(\mathbb{Z}/3\mathbb{Z})^\times$ (order 2, so χ maps to $\{1, -1\}$). Although this property is proved, the immediate death at $q = 3$ means the zero-mean steps are never reached from the standard starting point.

TCA + PathSurvival $\Rightarrow$ RandomTwoPointMC.

The conditional implication is now proved (`✓ tca_path_survival_implies_random_mc_proved ✓`). The proof (~ 400 lines) builds the full Fourier counting infrastructure on $(\mathbb{Z}/q\mathbb{Z})^\times$:

1. Path sum decomposition (`✓ path_sum_decomp ✓`): the unscaled path sum $2^N \cdot \text{pathCharSum}$ splits as $\text{chiAt}(2)^{-1} \cdot U(\chi, N) + E(\chi, N)$, where U sums over surviving paths via `ChiAt-Multiplicativity` and E sums over death paths.
2. Death error bound (`✓ death_error_norm_le ✓`): $\|E(\chi, N)\| \leq \text{deathCount}(N)$ (each death term has norm ≤ 1).
3. Unit endpoint bound (`✓ uecsi_norm_le ✓`): $\|U(\chi, N)\| \leq 2^N \|\text{pathCharSum}\| + \text{deathCount}$.
4. Fourier counting identity (`✓ fourier_counting_identity ✓`): $\sum_{\chi} \chi(a) \cdot U(\chi, N) = |G| \cdot \text{pathCount}(a, N)$.
5. Fourier lower bound (`✓ fourier_lower_bound ✓`): $(q-1) \cdot \text{pathCount}(a) \geq \text{survivalCount} - (q-2) \cdot B$ where B is the uniform bound on nontrivial $\|U(\chi)\|$.

Combining: TCA gives $\|\text{pathCharSum}\| < 1/(2q)$ for $N \geq N_0$; PathSurvival gives $\text{survivalCount} > 2q^2 \cdot \text{deathCount}$ for $N \geq N_1$. At $N = \max(N_0, N_1)$, the Fourier lower bound yields $\text{pathCount}(a) > 0$ for every unit a , via the algebraic identity $2q^3 + 2q^2 + 3q + 2 > 0$ (true for all $q \geq 5$). For $q = 3$, TCA is vacuously false, so the implication holds trivially. The self-consistent `RandomTwoPointMC` thus reduces to exactly two open hypotheses: `TreeContractionAtHalf` and `PathSurvival`.

Ensemble ε -walk.

The formalization (`Advanced/EpsilonWalk.lean`, 458 lines, zero `sorry`) develops an ensemble version: instead of fixing $\text{acc} = 2$, parametrize the ε -walk by squarefree starting point n . The `SigmaCRT-PropagationStep` hypothesis (open) asserts that CRT equidistribution propagates across steps for σ -walks, mirroring `CRTPropagationStep` for deterministic walks. Under `SRE + SigmaCRTPropagationStep`, the death density (fraction of squarefree n with $q \mid \text{epsWalkProdFrom}(n, \sigma, k) + 1$) converges to $1/(q-1)$ for every fixed k and σ (`✓ sigma_death_density_tendsto ✓`).

Deterministic coverage via Cauchy–Davenport.

An independent, purely group-theoretic route to coverage is formalized in `Advanced/IteratedProductCoverage.lean` (293 lines, zero `sorry`). Given a sequence $S_0, S_1, \dots$ of subsets of a finite commutative group G with $\text{minOrder}(G) = |G|$ (true when $|G|$ is prime), the iterated product $P_n = P_{n-1} \cdot S_{n-1}$ starting from $P_0 = \{1\}$ satisfies

$$|P_n| \geq \min\left(|G|, 1 + \sum_{k < n} (|S_k| - 1)\right)$$

by iterated application of the Cauchy–Davenport theorem (`✓ iteratedMulFinset_card_growth ✓`). If every $|S_k| \geq 2$, then after $|G| - 1$ steps the product is `Finset.univ` (`✓ iteratedMulFinset_eq_univ ✓`). For $(\mathbb{Z}/p\mathbb{Z})^\times$ with $p-1$ prime (safe primes: $p = 5, 7, 11, 23, \dots$), $\text{minOrder} = p-1 = |G|$,

and the coverage bound gives diameter $p - 2$ ([✓ safe_prime_coverage ✓](#)). This complements the probabilistic Fourier approach: if the ε -walk factor sets have $|S_k| \geq 2$ at every step (the diversity condition), then Cauchy–Davenport gives deterministic coverage in finitely many steps—no spectral hypothesis needed. The gap is ensuring diversity: when $\text{minFac} \equiv \text{secondMinFac} \pmod{q}$, the factor set has size 1, and the Cauchy–Davenport step gives no progress.

Dense capture and Baire category.

A topological perspective on the two-point variant is developed in [Advanced/DenseCapture.lean](#) (301 lines, zero [sorry](#)). The space of binary selection sequences $\sigma : \mathbb{N} \rightarrow \text{Bool}$ carries the product topology (Cantor space), which is compact Hausdorff and hence a Baire space. The [captureSet\(\$q, \text{acc}\$ \)](#)—the set of selection sequences that capture prime q starting from accumulator acc —is open, unconditionally ([✓ captureSet_isOpen ✓](#)): it equals $\bigcup_k \{\sigma \mid \text{factor}(k) = q\}$, and each level set is open because the factor at step k depends only on $\sigma(0), \dots, \sigma(k)$ (finitary dependence, [✓ epsWalkFactorFrom_depends_on_prefix ✓](#)).

The sole open hypothesis is [DenseCaptureHypothesis](#): for $\text{acc} \geq 2$, [captureSet\(\$q, \text{acc}\$ \)](#) is dense. Under this hypothesis, the Baire category theorem gives that the intersection $\bigcap_{q \text{ prime}} \text{captureSet}(q, \text{acc})$ is comeager (residual): a generic selection sequence captures every prime ([✓ fullCapture_residual ✓](#)). In particular, such a sequence exists ([✓ fullCapture_nonempty ✓](#)). A tail restart identity ([✓ epsWalkProdFrom_tail_restart ✓](#)) shows the walk from step K onward is a fresh walk from the accumulator at step K , connecting the density hypothesis to the ensemble framework.

6.5 Ensemble MC: Almost Every Starting Point

The variants above change the selection rule; here we change the starting point. The generalized EM construction (Section 4) parametrizes the sequence by its initial value n : $P_n(0) = n$, $\text{em}_n(k) = \text{minFac}(P_n(k) + 1)$, $P_n(k+1) = P_n(k) \cdot \text{em}_n(k)$. The standard orbit is $n = 2$.

Definition 6.22 ([✓ GenMullinConjecture](#)). $\text{GenMC}(n)$: every prime appears in the generalized EM sequence from n , i.e. $\forall q \text{ prime}, \exists k, \text{em}_n(k) = q$.

The reduction [✓ gen_hitting_implies_gen_mc_proved ✓](#) proves that cofinal walk hitting implies $\text{GenMC}(n)$ for every squarefree n , mirroring the standard inductive bootstrap. And [✓ gen_mc_two_implies_mc ✓](#) shows $\text{GenMC}(2) \Rightarrow \text{MC}$, so the generalized conjecture specializes to the standard one.

Bag arithmetic.

At each step k , the Euclid number $P_n(k) + 1$ has a “bag” of prime factors; the selected prime $\text{em}_n(k)$ is the minimum of this bag. The formalization ([Ensemble/BagArithmetic.lean](#), 224 lines, zero [sorry](#)) develops the bag-level quantities: [genEuclidOmega](#) $\omega(n, k) = |\text{primeFactors}(P_n(k) + 1)|$ (at least 1, [✓ genEuclidOmega_pos ✓](#)), [genBagDiversity](#) (number of distinct residue classes mod q among the factors, $\leq \omega$ and $\leq q$), [genFactorsInClass](#) F_a (factors in class a , with the partition identity $\sum_a |F_a| = \omega$, [✓ genFactorsInClass_card_sum ✓](#)), and [genEuclidCofactor](#) (the quotient $(P_n(k) + 1) / \text{em}_n(k)$, whose prime factors are a subset of the bag, [✓ cofactor_primeFactors_subset ✓](#)). For $k \geq 1$, all bag elements are ≥ 3 ([✓ primeFactors_ge_three_of_pos ✓](#)).

Density variants.

Instead of proving $\text{GenMC}(n)$ for the single orbit $n = 2$, one can ask how many starting points satisfy it. Two density levels are formalized:

- [PositiveDensityRSD](#) ([✓ PositiveDensityRSD](#)): a positive density $\delta > 0$ of squarefree n have

divergent reciprocal sums $\sum_k 1/\text{em}_n(k) = \infty$.

- AlmostAllSquarefreeRSD ([✓ AlmostAllSquarefreeRSD](#)): density 1 of squarefree n have divergent reciprocal sums.

The reciprocal sum is a natural proxy: if $\sum 1/\text{em}_n(k)$ diverges, then the primes appearing in the sequence from n cannot be confined to a thin set; although divergence alone does not formally imply $\text{GenMC}(n)$, it rules out the worst obstruction (a bounded sequence of captured primes).

The one-hypothesis route: LMG.

The proved chain [✓ lmg_implies_positive_density_rsd](#) [✓](#) shows:

$$\text{LMG} \implies \text{PositiveDensityRSD},$$

where LinearMeanGrowth (LMG) asks that $\mathbb{E}_n[S_K(n)] \geq \kappa K$ for some $\kappa > 0$ and all K large enough (here $S_K(n) = \sum_{k \leq K} 1/\text{em}_n(k)$ is the partial reciprocal sum, averaged over squarefree n). LMG is a population-level statement: it involves no specific orbit, bypassing the orbit-specificity barrier that obstructs DSL.

With the additional hypothesis PairwiseStepDecorrelation (PSD)—that $\text{Cov}[1/\text{em}_n(j), 1/\text{em}_n(k)] \rightarrow 0$ for $j \neq k$ —the stronger conclusion follows: $\text{PSD} + \text{LMG} \implies \text{AlmostAllSquarefreeRSD}$ ([✓ dead_end_125_pairwise_revival](#) [✓](#)). The bridges are all machine-verified: $\text{IVB}(1/4)$ [✓](#), $\text{PSD} + \text{IVB} \implies$ variance bound [✓](#), Chebyshev concentration [✓](#).

CRT propagation and the first moment.

LMG follows from FirstMomentStep (FMS): $\mathbb{E}_n[1/\text{em}_n(k)] \rightarrow \kappa$ as $X \rightarrow \infty$ for each k ([✓ FirstMomentStep](#)). FMS in turn follows from CRTPropagationStep ([✓ CRTPropagationStep](#)): the equidistribution of $P_n(k) \bmod q$ (among coprime residues) propagates from step k to step $k+1$. The chain is:

$$\text{SRE} + \text{CRTPropagationStep} \xRightarrow{\text{✓}} \text{AccumEquidist} \implies \text{FMS}(\kappa) \implies \text{LMG} \xRightarrow{\text{✓}} \text{PositiveDensityRSD}.$$

Here SRE (squarefree residue equidistribution) is the standard base case, and $\text{AccumEquidist} \implies \text{FMS}$ uses population equidistribution of minFac via the Alladi–Buchstab weights.

CRTPropagationStep now reduces to a cleaner open hypothesis: EnsembleTransitionApprox (ETA, [✓ EnsembleTransitionApprox](#)), which says that the empirical transition probability—the fraction of squarefree n with $P_n(k) \equiv c$ whose multiplier satisfies $\text{em}_n(k) \equiv b$ —converges to $1/(q-1)$ for all nonzero b, c . The proved reduction [✓ eta_implies_crt_propagation](#) [✓](#) shows $\text{ETA} \implies \text{CRTPropagationStep}$ via the backward density decomposition: the density at step $k+1$ is the matrix product of the density at step k with the transition matrix, and when both converge to $1/(q-1)$, so does the product. The full chain [✓ eta_sre_implies_prsd](#) [✓](#) composes this with the proved bridges to give $\text{ETA} + \text{SRE} \implies \text{PositiveDensityRSD}$.

The structural basis for ETA is CRT multiplier invariance ([✓ crt_multiplier_invariance](#) [✓](#)): $\text{minFac}(P_n(k) + 1)$ depends on the full integer, not just its residue mod q , so the factoring outcome is “ q -blind.”

Backward dynamics approach (Tao–Collatz analogy).

The backward dynamics infrastructure is formalized in [Ensemble/BackwardDynamics.lean](#) (577 lines). Joint counts, transition probabilities, their basic probability axioms (non-negativity, partition

identity, sum-to-one), and the backward density decomposition are all proved. The approach follows Tao's backward dynamics for the Collatz conjecture [20]: for a target residue $a \in (\mathbb{Z}/q\mathbb{Z})^\times$ at step $k+1$,

$$F_{k+1}(a) = \sum_{c \in (\mathbb{Z}/q\mathbb{Z})^\times} F_k(c) \cdot R(c, a, k, X),$$

where $R(c, a, k, X) = 1/(q-1) + O(1/\log X)$ by CRT-blindness. When F_k is already uniform, so is F_{k+1} , giving the induction step. However, the absorption mechanism (Dead End #138) invalidates this approach: the equidistribution target $r/(r^2-1)$ per coprime class (not $1/(r-1)$), which neglects the zero class of squarefree multiples of r is never sustained for $k \geq 1$, because death events drain nonzero-class mass permanently into the absorbing state $0 \bmod q$. The formal chain $\text{ETA} + \text{DCTA} + \text{SRE} \Rightarrow \text{PRSD}$ remains proved but is vacuously true.

The accumulator as a commutative state.

The EM accumulator $P_n(k) = \prod S_k$ is a squarefree integer determined by its prime set S_k . The dynamics $S \mapsto S \cup \{\min\text{Fac}(\prod S + 1)\}$ is a well-defined map on the lattice of finite prime sets, and the walk mod q depends on S only through $\prod S \bmod q$. Two orbits from different starting points that ever reach the same product mod q have approximately identical future walks—the construction history is forgotten.

This approximate Markov property, combined with the super-exponential growth $P_n(k) \geq 2^{2^k}$ (which makes CRT blindness increasingly accurate), suggests that the walk mod q is effectively ergodic from generic starting points. Making this rigorous—showing that the CRT error is summable along the orbit—would give a.a. GenMC for the standard sequence, without any modification of the selection rule.

Comparison with DSL.

The DSL route (Section 4) proves MC for the single orbit $n = 2$, contingent on transferring population statistics to one deterministic trajectory. The ensemble route here proves a density result for almost all starting points, contingent on CRTPropagationStep, which is a population-level hypothesis with no orbit-specificity. Neither subsumes the other: DSL would give MC directly; CRTPropagationStep gives PositiveDensityRSD (which does not formally imply MC for $n = 2$). However, the ensemble route faces a structurally easier barrier, since its open hypothesis is a statement about averages, not about one specific orbit.

6.6 Weak Mullin and Analytic Weakenings

The variants above change the selection rule; this subsection changes the conclusion. Instead of proving that every prime appears ($M = \emptyset$), one can ask that the set of missing primes is small, or that the EM reciprocal sum diverges.

Missing primes and Weak Mullin.

Define the set of missing primes

$$M = \{p \in \mathbb{P} : p \neq \text{em}_2(k) \text{ for all } k\}.$$

MC is the assertion $M = \emptyset$. A natural weakening asks only that M is small in an analytic sense:

Property 6.23 ([WeakMullin \(WM\)](#)). [✓ WeakMullin](#) The reciprocal sum over missing primes converges: $\sum_{q \in M} 1/q < \infty$.

This is strictly weaker than MC (which gives $M = \emptyset$ and hence a trivially convergent empty sum) but strong enough to have a nontrivial consequence:

Property 6.24 ([ReciprocalDivergence \(RD\)](#)). [✓ ReciprocalDivergence](#) The reciprocal sum over EM primes diverges: $\sum_{k=0}^{\infty} 1/\text{em}_2(k) = \infty$.

Theorem 6.25 ([✓ mc_implies_wm](#)). $\text{MC} \implies \text{WM}$.

Theorem 6.26 ([✓ wm_implies_rd](#)). $\text{WM} \implies \text{RD}$.

The proof partitions $\mathbb{P} = A \cup M$ (appeared vs. missing), notes that both Σ_A and Σ_M converge under $\text{WM} + \neg \text{RD}$, contradicting the divergence of $\sum_p 1/p$.

Product divergence.

The negation of WM has a quantitative consequence: $\neg \text{WM}$ gives $\sum_{q \in M} 1/(q-1) = \infty$, so the “safe fraction” product $\prod_{q \in F} (q-2)/(q-1) \leq e^{-\Sigma 1/(q-1)} \rightarrow 0$ and its reciprocal $\prod (q-1)/(q-2)$ exceeds any bound ([✓ not_wm_product_diverges](#)). The product $\prod (q-1)/(q-2)$ over missing primes measures the “confinement cost”: each missing prime q confines the walk to $q-2$ of the $q-1$ units in $(\mathbb{Z}/q\mathbb{Z})^\times$ (avoiding -1). If WM fails, this confinement cost is unbounded.

Confined energy.

For a missing prime $q \geq 3$, the walk avoids -1 entirely, so the N visits are distributed among only $q-2$ residues. A pigeonhole argument gives a confined-energy lower bound ([✓ per_prime_confined_energy_lb](#)): $N^2/(q-2) \leq (q-1) \sum_a V_N(a)^2$. The excess energy ratio $(q-1)/(q-2)$ compared to the equidistributed baseline $N^2/(q-1)$ is bounded away from 1. [JointSVE](#) ([✓ JointSVE](#))—joint subquadratic visit energy over any finite set of missing primes—implies MMCSB ([✓ jointSVE_implies_mmcsb](#)), connecting the “missing primes” picture to the spectral route.

Avoidance tube and tube collapse.

Define the tube ratio for a finite set F of primes ≥ 3 :

$$\tau(F) = \prod_{q \in F} \frac{q-2}{q-1}.$$

This measures the fraction of $(\mathbb{Z}/q\mathbb{Z})^\times$ classes that a walk avoiding -1 at each q is allowed to visit. The tube ratio is monotone decreasing in F and satisfies $0 \leq \tau(F) \leq 1$ ([✓ tubeRatio_monotone ✓](#), [✓ tubeRatio_le_one ✓](#)).

Theorem 6.27 (Tube collapse — [✓ tube_collapse ✓](#)). $\neg \text{WM} \implies$ for every $\varepsilon > 0$, there exists a finite set $F \subseteq M$ with $\tau(F) < \varepsilon$.

The proof uses the inequality $\tau(F) \leq \exp(-\sum_{q \in F} 1/(q-1))$ together with the divergence of $\sum_{q \in M} 1/(q-1)$ under $\neg \text{WM}$. For any unit a with $V_N(a) = 0$, the confined energy gives a rogue character:

Theorem 6.28 (Rogue character — [✓rogue_character_exists ✓](#)). If a unit $a \in (\mathbb{Z}/q\mathbb{Z})^\times$ has $V_N(a) = 0$ (zero visits), then there exists a nontrivial character χ with $\|S_\chi(N)\|^2 \geq N^2/(q-2)^2$. In particular, $\|S_\chi(N)\| \geq N/(q-2)$: the character sum grows linearly, contradicting SVE.

Theorem 6.29 (SVE contradicts avoidance — [✓sve_contradicts_avoidance ✓](#)). $\text{SVE} \implies$ for all large N , no unit a can have $V_N(a) = 0$.

Shielding lemma and spectral conspiracy.

The [shielding lemma](#) observes that $\text{em}_2(n+1)$ is always a term of the sequence, hence never a missing prime:

Theorem 6.30 (Shielding lemma — [✓shielding_lemma ✓](#)). For every n , $\text{em}_2(n+1) \notin M$.

When a missing prime q divides $P_2(n) + 1$, some strictly smaller appeared prime shields q from capture ([✓factor_dichotomy_strong ✓](#)). Under $\neg\text{WM}$, this shielding mechanism creates a spectral conspiracy:

Theorem 6.31 (Spectral conspiracy — [✓spectral_conspiracy_landscape ✓](#)). $\neg\text{WM}$ implies:

1. M is infinite ([✓not_wm_implies_missing_infinite ✓](#)).
2. The joint excess energy $\sum_{q \in M} \log \frac{q-1}{q-2}$ diverges.
3. For any finite “guardian pool” G of appeared primes, the demand $\sum_{q \in F} 1/(q-1)$ from a large enough subset $F \subseteq M$ exceeds $|G|$.
4. Arbitrarily many missing primes exist, each witnessing a nontrivial character with linear growth (Theorem 6.28).

The guardian exhaustion clause (3) formalizes the intuition that finitely many appeared primes cannot “explain away” the avoidance at infinitely many missing primes. The three files [Population/AvoidanceTube.lean](#) (516 lines), [Population/InfiniteM.lean](#) (303 lines), [Population/SpectralConspiracy.lean](#) (306 lines) total 1,125 lines, all zero [sorry](#).

The shifted squarefree population.

The EM accumulator $P_2(n)$ is squarefree ([✓prod_squarefree](#)), so every Euclid number $P_2(n) + 1$ belongs to the shifted squarefree population $\mathcal{S} = \{m \geq 2 : m-1 \text{ squarefree}\}$ (density $6/\pi^2$). This decomposes the orbit equidistribution problem into:

- Population Equidistribution (PE): among elements of $\mathcal{S}$, the smallest prime factor is equidistributed modulo any prime q (provable by Selberg sieve + Dirichlet).
- Population Transfer (PT): the EM trajectory samples $\mathcal{S}$ without introducing mod- q bias. This is the open content.

The chain $\text{PE} + \text{PT} + \text{EMDImpliesCME} \implies \text{MC}$ is proved ([✓pe_transfer_cme_implies_mc](#)).

Pairwise decorrelation and positive density.

Dead End #125 proved that pairwise cancellation does not imply k -wise, but variance bounds only use $k = 2$. The formalization proves the revival chain ([✓dead_end_125_pairwise_revival](#)): $\text{PairwiseStepDecorrelation} + \text{LinearMeanGrowth} + \text{ChebyshevConcentration} \implies \text{AlmostAllSquarefreeRSD}$. A further simplification drops PSD entirely:

Theorem 6.32 ([✓ lmg_implies_positive_density_rsd](#)). $\text{LinearMeanGrowth} \implies \text{PositiveDensityRSD}$.

This is a 1-hypothesis route: LMG is strictly weaker than CME—it requires only that the ensemble mean grows linearly, not conditional equidistribution.

6.7 The Function Field Analog

Section roadmap

Replacing $\mathbb{Z}$ by $\mathbb{F}_p[t]$ and primes by monic irreducible polynomials gives a clean test bed where the analytic number theory becomes free. The Weil bound—an open hypothesis over $\mathbb{Z}$ —is a theorem of Weil [25] over $\mathbb{F}_p[t]$. Yet the function-field Mullin Conjecture is still open, and the residual difficulty is identical to the integer case: the orbit-specificity barrier. This subsection develops the FF-EM sequence, proves an unconditional six-step sieve chain and the Φ_3 exclusion criterion, and presents three formally verified dead ends that delineate the boundary of current techniques. The function-field module of the formalization comprises 17 files and $\sim 7,372$ lines with zero `sorry`.

Setting and definitions.

Let p be a prime. The ring $\mathbb{F}_p[t]$ is a Euclidean domain, hence a UFD; monic irreducible polynomials play the role of prime numbers. We write $\pi_p(d)$ for the number of monic irreducibles of degree d over $\mathbb{F}_p$. For $p = 2$ the first values are $\pi_p(1) = 2$ (namely t and $t+1$), $\pi_p(2) = 1$ (namely t^2+t+1), $\pi_p(3) = 2$, $\pi_p(4) = 3$.

Definition 6.33 (FF-EM Sequence — [✓ FFEMData](#)). Fix a prime p . An **FF-EM Data** consists of: $\text{ffSeq}(0) = t$ (the smallest monic irreducible); $\text{ffProd}(0) = t$; $\text{ffSeq}(n+1)$ a monic irreducible factor of $\text{ffProd}(n) + 1$ of minimal degree; $\text{ffProd}(n+1) = \text{ffProd}(n) \cdot \text{ffSeq}(n+1)$.

Conjecture 6.34 (Function-Field Mullin Conjecture — [✓ FFMullinConjecture](#)). For every monic irreducible $Q \in \mathbb{F}_p[t]$, there exists n with $\text{ffSeq}(n) = Q$.

Two structural differences from the integer setting are immediate. First, monic irreducibles of the same degree have no canonical total order, so the “minimal-degree” rule may select among several competitors—the analog of the integer `minFac` rule loses uniqueness. Second, the unit group $(\mathbb{F}_p[t]/(Q))^\times \cong \mathbb{F}_{p^{\deg Q}}^\times$ is cyclic of order $p^{\deg Q} - 1$, providing richer group-theoretic structure than $(\mathbb{Z}/q\mathbb{Z})^\times$.

Why study the function-field case?

The Weil bound for character sums over $\mathbb{F}_p$ -curves is a theorem [25, 21], not a conjecture. This makes population equidistribution of monic irreducibles across residue classes unconditional, in contrast to the integer setting where the analogous result requires the Weighted PNT in arithmetic progressions. By isolating the orbit-specificity barrier—the gap between “population statistics” and “the specific walk $\text{ffProd}(n) \bmod Q$ visits the residue class -1 ”—we will show that this barrier is the sole remaining obstacle to FF-MC, and that it is structurally identical to the obstacle for MC over $\mathbb{Z}$.

Basic structural facts.

The following hold unconditionally for any FF-EM data:

Lemma 6.35 (Coprimality cascade — [✓ coprimality_cascade](#)). No monic irreducible Q divides both $\text{ffProd}(n)$ and $\text{ffProd}(n) + 1$.

Proof. If Q divides both, then $Q \mid 1$, contradicting $\deg Q \geq 1$. $\square$

Lemma 6.36 (Sequence injectivity — [✓ ffSeq_injective_proved](#)). The map $n \mapsto \text{ffSeq}(n)$ is injective: no irreducible appears twice.

Lemma 6.37 (Degree growth — [✓ ffProd_natDegree_strict_mono](#)). $\deg(\text{ffProd}(n + 1)) > \deg(\text{ffProd}(n))$ for all n .

Lemma 6.38 (Pool degree to infinity — [✓ ffSeq_degree_tendsto_atTop_unconditional](#)). $\deg(\text{ffProd}(n) + 1) \rightarrow \infty$ as $n \rightarrow \infty$.

The necklace identity.

The combinatorial heart of the unconditional ANT chain over $\mathbb{F}_p[t]$ is the following exact identity.

Theorem 6.39 (Necklace identity — [✓ necklace_identity_proved](#)). For every $n \geq 1$,

$$\sum_{d \mid n} d \cdot \pi_p(d) = p^n.$$

The name comes from the combinatorial interpretation: $\pi_p(d)$ counts aperiodic necklaces of length d . The identity is classical (Moreau, 1872), and is equivalent to the factorization $t^{p^n} - t = \prod_{\deg Q \mid n} Q$ over $\mathbb{F}_p[t]$.

Proof sketch. Both sides count the p^n elements of $\mathbb{F}_{p^n}$, viewed as roots of $t^{p^n} - t$. The forward inclusion uses $Q \mid t^{p^d} - t \mid t^{p^n} - t$ when $\deg Q \mid n$; the reverse uses the tower law on $[\mathbb{F}_p(\alpha) : \mathbb{F}_p]$ to bound the degree of any minimal polynomial. The Lean proof spans ~ 380 lines and exercises Mathlib's Galois-field infrastructure ([GaloisField.card](#), [minpoly.irreducible](#), [IntermediateField.firank_dvd_of_le_right](#), [FiniteField.X_pow_card_pow_sub_X_ne_zero](#)). $\square$

Corollary 6.40 (Density bounds — [✓ necklace_count_mul_le](#)). For every $n \geq 1$, $n \cdot \pi_p(n) \leq p^n$ and $\pi_p(n) \leq p^n/n$.

A finer Möbius inversion gives $\pi_p(n) = \frac{1}{n} \sum_{d \mid n} \mu(n/d) p^d \sim p^n/n + O(p^{n/2}/n)$; the crude bound suffices for our sieve chain.

The unconditional sieve chain.

The integer sieve approach to Mullin's Conjecture proceeds via

$$\text{Char. Cancel.} \xrightarrow{\text{GRH}} \text{PNT-AP} \xrightarrow{\text{S-W}} \text{FCD} \Rightarrow \text{SPV} \Rightarrow \text{PSCD} \Rightarrow \text{Almost-All MC.}$$

Over $\mathbb{F}_p[t]$, every arrow is unconditional.

Step	Over $\mathbb{Z}$	Over $\mathbb{F}_p[t]$
Char. cancellation	GRH	Exact (necklace)
PNT-in-APs	Siegel–Walfisz	Exact ($\pi_p(d) \geq 1$)
FCD	WPNT-in-APs	Linear supply
SPV	FCD + sparsity	Necklace bound
PSCD	FCD + SPV	Density bound
Almost-all MC	PSCD + pigeonhole	PNT + necklace

Step 1 — Character cancellation.

Theorem 6.41 (FF Character Sum Cancellation — [✓ff_char_sum_cancellation_proved](#)). The necklace identity (Theorem 6.39) holds unconditionally, giving exact cancellation rather than an asymptotic bound.

Over $\mathbb{Z}$, the analogous statement is the Riemann Hypothesis for Dirichlet L -functions; over $\mathbb{F}_p[t]$ it is an identity of natural numbers.

Step 2 — PNT in arithmetic progressions.

Theorem 6.42 (FF PNT-in-APs — [✓fflredCount_pos](#)). For every $d \geq 1$, $\pi_p(d) \geq 1$.

The proof uses the Galois-field construction: the minimal polynomial of a primitive element of $\mathbb{F}_{p^d}$ is monic irreducible of degree d . The reduction to character cancellation is automatic ([✓ff_char_cancel_implies_pnt](#)).

Step 3 — Forbidden class divergence.

Property 6.43 (FF-FCD — [✓FFFCD](#)). $\sum_{d=1}^D \pi_p(d) \geq D$ for all $D \geq 1$.

Theorem 6.44 (FF-FCD is unconditional — [✓ff_fcd_proved](#)). Each summand is ≥ 1 by Step 2; the bound is immediate ([✓ff_pnt_implies_fcd](#)).

Step 4 — Sieve product vanishing.

Property 6.45 (FF-SPV — [✓FFSieveProductVanishing](#)). $n \cdot \pi_p(n) \leq p^n$ for all $n \geq 1$.

Theorem 6.46 (FF-SPV is unconditional — [✓ff_spv_proved](#)). Immediate from Theorem 6.39: the term $n \cdot \pi_p(n)$ is one summand and the total equals p^n . Reduction to FCD: [✓ff_fcd_implies_spv](#).

Step 5 — Population sieve confinement decay.

Property 6.47 (FF-PSCD — [✓FFPSCD](#)). $\pi_p(n) \leq p^n/n$ for all $n \geq 1$.

Theorem 6.48 (FF-PSCD is unconditional — [✓ff_pscd_proved](#)). Divide both sides of Step 4 by n . Reduction: [✓ff_spv_implies_pscd](#).

Step 6 — Almost-all generalized mixed MC.

A mixed selection at each step picks any irreducible factor of $\text{ffProd}(n) + 1$ (not necessarily minimal-degree); the generalized variant starts from an arbitrary monic squarefree polynomial. “Almost-all” refers to density-one over starting points.

Theorem 6.49 (Almost-All FF-GenMixedMC is unconditional — [✓ ff_almost_all_unconditional](#)). For every prime p , the conjunction of FF-PNT-in-APs and FF-SPV holds unconditionally; in particular, a density-1 set of monic squarefree starting polynomials satisfies generalized mixed MC.

Full chain witness ([✓ ff_full_chain_unconditional](#)): the conjunction

Char. Cancel. $\wedge$ PNT-AP $\wedge$ FCD $\wedge$ SPV $\wedge$ PSCD $\wedge$ Almost-All FF-GenMixedMC

holds unconditionally for every prime p . Zero open hypotheses.

The factor tree and structural prerequisites.

Independent of the sieve chain, the factor-tree structure (any valid mixed selection) admits the following formally verified properties:

Theorem 6.50 (Factor-tree structure — [✓ theorem_A_factor_tree](#)). For any valid mixed selection from any monic squarefree start,

- (a) the walk explores infinitely many distinct monic irreducibles ([✓ ff_mixed_explores_infinitely_many](#));
- (b) the factor pool degree $\deg(\text{acc}(n) + 1) \rightarrow \infty$ ([✓ ff_factor_pool_degree_grows](#));
- (c) selection is injective ([✓ ffMixedSel_injective](#));
- (d) walk products are injective ([✓ ffMixedWalkProd_injective](#));
- (e) the start is never recapturable ([✓ start_not_capturable](#)).

Stochastic FF-MC and the phase transition.

Theorem 6.51 (Stochastic FF-MC — [✓ stochastic_mc_unconditional](#)). For any monic irreducible target Q and any valid mixed walk, the factor pool eventually has degree exceeding $\deg(Q)$; with any positive probability $\varepsilon > 0$ of non-greedy selection, every target is eventually captured.

Theorem 6.52 (Phase transition at $\varepsilon = 0$ — [✓ ff_phase_transition_unconditional](#)). For $\varepsilon > 0$, stochastic MC captures every monic irreducible. For $\varepsilon = 0$, the walk is the standard greedy FF-EM and capture of every irreducible is exactly FF-MC. The phase transition is sharp.

The bootstrap: from dynamical hitting to FF-MC.

Property 6.53 (FF Dynamical Hitting (FFDH) — [✓ FFDynamicalHitting](#)). For every FF-EM data and every monic irreducible Q not in the sequence, there exist cofinally many indices n at which $Q \mid \text{ffProd}(n) + 1$.

Theorem 6.54 (One-hypothesis reduction — [✓ ff_dh_implies_ffmc](#)). $\text{FF-DH} \implies \text{FF-MC}$ (using the unconditional finiteness of irreducibles per degree, [✓ ff_dh_implies_ffmc_unconditional](#)).

The proof is by strong induction on degree. At degree d_0 , the hypothesis gives MC below d_0 ; a uniform sieve gap exhausts all sub-degree irreducibles past some N_0 ; FFDH provides cofinally many hits past N_0 ; injectivity makes each hit consume a distinct competitor of the same degree; finiteness per degree then forces Q itself to be selected (competitor exhaustion, [✓ ff_element_capture](#)). This is the function-field analog of the integer bootstrap [dynamical_hitting_implies_mullin](#); the only difference is the final exhaustion step, which over $\mathbb{Z}$ collapses because primes are totally ordered.

The Φ_3 exclusion criterion.

We now develop a structural obstruction with no integer analog. When all irreducible factors of $\text{ffProd}(n) + 1$ have degree 1, the walk $\text{ffProd}(n) \bmod Q$ for a degree-one target $Q = (t - a)$ follows the autonomous map $f(w) = w(w + 1)$ on $\mathbb{F}_p$. The target Q divides $\text{ffProd}(n) + 1$ iff the walk position $w_n = -1$. Hence degree-one capture reduces to: does the orbit of f reach -1 ?

Definition 6.55 (Autonomous map — [✓ ffAutonomousMap](#)). For a prime p , $f: \mathbb{F}_p \rightarrow \mathbb{F}_p$ is given by $f(w) = w(w + 1)$. The orbit from $a \in \mathbb{F}_p$ is $a, f(a), f^2(a), \dots$

Lemma 6.56 (Absorption at zero — [✓ ffAutonomousMap_neg_one](#)). $f(-1) = 0$ and $f(0) = 0$; once the orbit hits 0 it stays there ([✓ ffAutonomousOrbit_absorbed](#)).

Lemma 6.57 (Φ_3 criterion — [✓ ffAutonomousMap_eq_neg_one_iff](#)). $f(w) = -1 \iff w^2 + w + 1 = 0$. The preimage of -1 is the root set of the third cyclotomic polynomial $\Phi_3(w)$.

Lemma 6.58 (Cube-root condition — [✓ pow_three_eq_one_of_phi3](#)). If $w^2 + w + 1 = 0$ in $\mathbb{F}_p$, then $w^3 = 1$.

Theorem 6.59 (Φ_3 has no roots when $p \equiv 2 \pmod{3}$ — [✓ phi3_no_roots](#)). For $p \equiv 2 \pmod{3}$ with $p \geq 5$, $w^2 + w + 1 \neq 0$ for all $w \in \mathbb{F}_p$.

Proof. Suppose $w^2 + w + 1 = 0$ in $\mathbb{F}_p$. Then $w^3 = 1$, and $w \neq 0, 1$ (each substitution gives a contradiction with $p \geq 5$). Hence $\text{ord}(w) = 3$ in $\mathbb{F}_p^\times$, so $3 \mid p - 1$ by Lagrange. But $p \equiv 2 \pmod{3}$ forces $p - 1 \equiv 1 \pmod{3}$, contradiction. $\square$

Theorem 6.60 (Unreachability of -1 — [✓ ff_neg_one_unreachable](#)). For $p \equiv 2 \pmod{3}$ with $p \geq 5$ and any starting value $a \neq -1$ in $\mathbb{F}_p$, the orbit $f^n(a)$ never equals -1 .

Corollary 6.61 (Visited at most once — [✓ ff_neg_one_visited_at_most_once](#)). For the same primes, -1 is visited only at step 0 (and only if $a = -1$); for all $n \geq 1$, the orbit value is never -1 .

Remark 6.62. The criterion fails to apply for $p = 2$ (where $\Phi_3 = t^2 + t + 1$ is itself irreducible) and $p = 3$ (where Φ_3 has the double root $w = 1$). Primes $p \equiv 2 \pmod{3}$ with $p \geq 5$ include 5, 11, 17, 23, 29, 41, 47, $\dots$

Integer non-analog.

Over $\mathbb{Z}$ the multiplier $a_{n+1} \bmod q$ is not determined by $P(n) \bmod q$ alone—it depends on the global factorization of $P(n) + 1$. No autonomous map exists, so the Φ_3 criterion has no direct integer analog.

The cyclotomic structure (Dead End #129).

Over $\mathbb{F}_2$, the product of all monic irreducibles of degree dividing $d = 2$ is $t(t+1)(t^2+t+1) = t^4 + t$, and adding 1 gives $\Phi_5(t) = t^4 + t + 1$. Its Galois group is $\text{Gal}(\mathbb{F}_{2^4}/\mathbb{F}_2) \cong \mathbb{Z}/4\mathbb{Z}$ —abelian, not S_4 . More generally, the classical identity $\prod_{\deg Q|d} Q = t^{p^d} - t$ forces the greedy FF-EM to build products dividing $t^{p^d} - t$ for some d , so $\text{ffProd}(n) + 1$ has splitting field inside $\mathbb{F}_{p^N}$ for some N and the Galois group is cyclic, generated by Frobenius. Thus the “large monodromy” hypothesis is structurally false for the greedy FF-EM, witnessed by [✓ff_cyclotomic_dead_end](#).

The orbit-specificity boundary.

We now establish that the orbit-specificity barrier is the sole remaining obstacle to FF-MC, and that it is structurally identical to the barrier over $\mathbb{Z}$. The central dichotomy is between two types of equidistribution:

- Population equidistribution: among all monic irreducibles of degree d , residues mod Q are equidistributed across $|\mathbb{F}_{p^{\deg Q}}^\times|$ classes, with error $O(p^{d/2})$ from the Weil bound. Unconditional over $\mathbb{F}_p[t]$.
- Orbit equidistribution: the specific walk $\text{ffProd}(n) \bmod Q$ visits each residue class with frequency $1/(p^{\deg Q} - 1)$ (the function-field DSL). Open.

Theorem 6.63 (Population CCSB from Weil — [✓weil_implies_population_mult_ccsb](#)). The Weil bound proves population-level multiplier CCSB unconditionally; [✓selection_bias_chain](#) formalizes the population $\rightarrow$ fiber $\rightarrow$ selection $\rightarrow$ orbit hierarchy.

Dead End #127 — Weil bound insufficient.

Theorem 6.64 (Weil does not close the gap — [✓dead_end_127_weil_not_sufficient](#)). The Weil bound implies population equidistribution but does not imply FF-DH, because: (i) Weil controls $\sum_{f \text{ monic}, \deg f=d} \chi(f \bmod Q)$, i.e. averages over all polynomials of degree d ; (ii) FF-DH requires $\sum_{n < N} \chi(\text{ffProd}(n) \bmod Q)$, a sum along a specific deterministic orbit; (iii) $\text{ffProd}(n) + 1$ is the unique element of its congruence class, not a random draw.

Dead End #129 — Large monodromy structurally false.

Property 6.65 (FF Large Monodromy (FFLM) — [✓FFLargeMonodromy](#)). The Galois group of $\text{ffProd}(n) + 1$ eventually contains the alternating group: $|\text{Gal}| \geq (\deg \text{ffProd}(n))!/2$ for all sufficiently large n .

Theorem 6.66 (FFLM is structurally false — [✓dead_end_129_monodromy_dead](#)). The greedy FF-EM produces products dividing $t^{p^d} - t$. Adding 1 gives a polynomial whose splitting field lies in $\mathbb{F}_{p^N}$ for some N , hence Galois group $\cong \mathbb{Z}/N\mathbb{Z}$, generated by Frobenius—abelian, not A_d . The monodromy approach reduces to abelian Galois theory = Dirichlet characters = Weil =

population equidistribution; the route is circular.

Dead End #130 — Generation $\neq$ coverage.

A natural hope is that subgroup escape (multipliers generate the full group) might imply dynamical hitting (the walk visits every element). Both PRE (over $\mathbb{Z}$) and Weil (over $\mathbb{F}_p[t]$) prove subgroup escape. The following counterexample shows this is insufficient.

Theorem 6.67 (Generation $\neq$ coverage — [✓ dead_end_130_generation_ne_coverage](#)). On $\mathbb{Z}/4\mathbb{Z}$, the steps $\{1, 3\}$ generate the full group, yet the alternating walk $0, 1, 0, 1, \dots$ visits only $\{0, 1\}$, missing $\{2, 3\}$. Witnessed in [✓ alternating_walk_misses_two](#); the converse direction (cancellation $\Rightarrow$ coverage via Weyl) is [✓ walk_cancel_implies_covers](#).

The barrier theses.

Theorem 6.68 (Orbit barrier thesis — [✓ orbit_barrier_thesis](#)). The orbit-specificity barrier has the following formally verified properties:

- (i) All structural prerequisites for FF-MC are unconditional: injectivity, degree growth, finiteness per degree, coprimality cascade.
- (ii) FF-DH $\Rightarrow$ FF-MC (Theorem 6.54).
- (iii) FF-DH is the sole remaining hypothesis.
- (iv) The barrier is not closed by population statistics (Theorem 6.64).
- (v) The barrier is not closed by Galois theory (Theorem 6.66).
- (vi) The barrier is not closed by algebraic generation (Theorem 6.67).
- (vii) The barrier dissolves under stochastic perturbation ($\varepsilon > 0$, Theorem 6.51).
- (viii) The barrier is sharp at $\varepsilon = 0$ (Theorem 6.52).

Theorem 6.69 (Setting independence — [✓ barrier_setting_independent](#)). (i) Over $\mathbb{F}_p[t]$: ANT is free (Weil $\Rightarrow$ PE).
 (ii) Over $\mathbb{F}_p[t]$: orbit specificity remains (FF-DH open; FF-DH $\Rightarrow$ FF-MC).
 (iii) Over $\mathbb{F}_p[t]$: stochastic perturbation dissolves the barrier; the deterministic limit does not.

The function-field setting eliminates the “easy” obstacle (ANT for PE) and adds new structural information (Φ_3 exclusion), but the “hard” obstacle—orbit specificity = DSL—is identical to the integer case. This proves that the difficulty of Mullin’s Conjecture lies entirely in the orbit-specificity barrier and not in the analytic number theory.

Side-by-side summary.

Aspect	Over $\mathbb{Z}$	Over $\mathbb{F}_p[t]$
Population equidist. (PE)	Open (WPNT)	Free (Weil)
Subgroup escape (SE)	Proved (PRE)	Proved (Weil)
Coprimality cascade	Proved	Proved
CME $\Rightarrow$ MC chain	Proved	Proved
Orbit specificity (DSL)	Open	Open
Walk character sums	Not standard	Not standard
Stochastic MC ($\varepsilon > 0$)	Proved	Proved
Φ_3 exclusion	—	Proved

Open problems specific to FF-MC.

We close with a few open problems particular to the function-field analog (the integer counterparts are catalogued in §5).

Deterministic FF-MC. Does every monic irreducible over $\mathbb{F}_p[t]$ eventually appear in the greedy FF-EM sequence? Equivalently, does FF-DH hold? Our work shows that FF-DH is the sole remaining hypothesis: the stochastic version ($\varepsilon > 0$) is proved (Theorem 6.51), and the open content is exactly the $\varepsilon = 0$ boundary.

Controlling the $\varepsilon \rightarrow 0$ limit. Stochastic FF-MC is proved for all $\varepsilon > 0$. Can the limit $\varepsilon \rightarrow 0$ be controlled to deduce deterministic FF-MC? Specifically: does the capture time $T_Q(\varepsilon)$ for a target Q remain bounded as $\varepsilon \rightarrow 0$? The phase transition is sharp—the proof structure for $\varepsilon > 0$ relies on non-greedy exploration, which is absent at $\varepsilon = 0$.

Large monodromy beyond cyclotomic starts. Dead End #129 shows the greedy FF-EM produces cyclotomic (hence abelian) products. Does the obstruction persist for non-greedy starting points? Cohen’s theorem [26] shows a random monic polynomial of degree d has Galois group S_d with probability tending to 1. Is the recursive FF-EM structure forcing cyclotomic products an artifact of the greedy minimal-degree selection?

Integer analog of the Φ_3 criterion. The Φ_3 exclusion (Theorem 6.60) uses the absence of cube roots of unity in $\mathbb{F}_p$. Over $\mathbb{Z}$ the walk does not follow an autonomous map—can systematic exclusion criteria be developed nonetheless for specific (q, a) pairs? Cox–van der Poorten [13] suggested structural reasons for the difficulty of the prime 41; the function-field Φ_3 result is a formal analog.

Selection bias neutrality (✓ [SelectionBiasNeutral](#)). Does greedy minimal-degree selection preserve character-sum cancellation? A positive answer would close the orbit-specificity

gap for both FF-MC and MC simultaneously, since the question concerns the selection mechanism, not the ambient ring. This is the cleanest formulation of the barrier; the integer analog is DSL.

Headline summary.

The function-field portion of the formalization is collected in the landscape theorem [✓ ff_unconditional_grand_summary](#), which assembles theorems A, B, C, D ([✓ theorem_A_factor_tree](#), [✓ theorem_B_stochastic_mc](#), [✓ theorem_C_phase_transition](#), [✓ theorem_D_sieve_chain](#)) into a single conjunction with zero open hypotheses. Together they witness:

- the unconditional six-step sieve chain (Theorem [6.49](#));
- stochastic FF-MC and the $\varepsilon = 0$ phase transition (Theorems [6.51](#), [6.52](#));
- the necklace identity (Theorem [6.39](#));
- the Φ_3 exclusion criterion (Theorem [6.60](#));
- the orbit-barrier and setting-independence theses (Theorems [6.68](#), [6.69](#)).

The function-field treatment thus delivers the cleanest available diagnosis: the difficulty of Mullin’s Conjecture is the orbit-specificity barrier, not the analytic number theory.

A Appendix

A.1 History and Computational Status

Mullin posed the conjecture in 1963 [\[1\]](#). In over sixty years, no proof has been found, and no theoretical approach has come close. The problem sits in an unusual position: it is elementary to state, each individual step is deterministic, yet the global behavior of the sequence appears completely intractable.

The largest-factor variant.

Cox and van der Poorten [\[13\]](#) showed that the related sequence using the largest prime factor—where each term is $\text{gpf}(P_2(n) + 1)$ instead of minFac —provably misses infinitely many primes (for instance, 5 never appears). By always jumping to the largest factor of $P_2(n) + 1$, the sequence leaps past small primes and can never return to them, since each Euclid number is coprime to every earlier term. This refuted the natural strengthening that surjectivity holds regardless of the factor-selection rule, and showed that the minFac rule is essential to the conjecture.

The two selection rules are also structurally different. The largest prime factor $P^+(n) = \text{gpf}(n)$ has rich algebraic structure: the sets $\{n : P^+(n) \leq y\}$ (smooth numbers) are controlled by the Dickman function, and for polynomial sequences $f(n) = P_2(n) + 1$ one can sometimes exploit algebraic curves and Hasse–Weil bounds. In contrast, the smallest prime factor $P^-(n) = \text{minFac}(n)$ is purely a sieve object: controlling $P^-(n)$ requires excluding small prime divisors one at a time (CRT + Mertens), and no algebraic-geometric handle exists. This gpf/minFac asymmetry is one reason the largest-factor variant admits a short proof while the smallest-factor conjecture remains open.

The walk heuristic.

Shanks [\[4\]](#) was the first to articulate the heuristic argument underlying the conjecture: if q is the smallest prime that never appears, the walk $P_2(n) \bmod q$ in $(\mathbb{Z}/q\mathbb{Z})^\times$ is well-defined for all n ,

and under a pseudo-randomness assumption it must eventually hit -1 . The formalization’s CME, CCSB, and DSL (Sections 3–4) are rigorous formulations of this heuristic: each specifies the precise sense in which “the walk behaves pseudo-randomly” would suffice.

Variants and surveys.

Booker [3] showed that a carefully chosen variant of the Euclid–Mullin sequence does contain every prime: by selecting a specific (not necessarily smallest) prime factor at each step, one can steer the sequence to hit every prime. This demonstrates that the conjecture is delicate: the surjectivity depends on the precise rule, not just the Euclidean structure. Pollack and Treviño [7] surveyed the problem’s place in the broader landscape of Euclid-inspired sequences, and studied distributional properties of primes “forgotten” by Euclid-type constructions. Most recently, Booker and Simon [5] generalized the framework to primes in arithmetic progressions via cyclotomic polynomials $\Phi_m(cx)$, proving (under ERH, and unconditionally for $m \leq 18$) that the maxFac variant omits infinitely many primes $p \equiv 1 \pmod{m}$.

Computational status.

The sequence has been extended through a series of large-scale factoring efforts:

- Wagstaff (1993) computed through the 43rd term.
- In 2010, the 180-digit number $P_2(43) + 1$ was factored via GNFS (General Number Field Sieve), yielding a 68-digit prime as $a(44)$. Terms $a(45)$ – $a(47)$ followed.
- In 2012, Propper factored the 256-digit number $P_2(47) + 1$ by ECM (Elliptic Curve Method), discovering a 75-digit factor and extending the sequence to 51 terms (Booker–Irvine [2]).
- Finding $a(52)$ requires factoring a 335-digit number. No factorization is known as of 2025.

After 51 terms, the smallest primes not yet observed are 41 and 47. Note that 31—a smaller prime—does not appear until position 50.

Brute force and the complexity of factoring.

Brute-force computation cannot attack the conjecture. Each new term requires factoring a number whose digit count grows roughly linearly with the number of terms, quickly exceeding the reach of any known algorithm. But even if we could compute millions of terms, the conjecture is a $\forall$ -statement over all primes, and no finite prefix can rule out a prime that first appears at an astronomically large index.

The deeper issue is a possible link to the computational complexity of integer factorization. Factoring is believed to be slightly less than hard: neither polynomial (as it would be if $P = NP$) nor fully exponential, but subexponential—the best known algorithms (number field sieve) run in time $L_N(1/3, c) = \exp(c(\ln N)^{1/3}(\ln \ln N)^{2/3})$. This intermediate complexity may be reflected in the conjecture itself. Each term $a(n+1) = \min\text{Fac}(P_2(n) + 1)$ depends on the complete factorization of a number encoding all previous terms; changing a single early term alters every subsequent one. If factoring were easy, the sequence would be transparently computable and the conjecture might yield to direct analysis. If factoring were maximally hard (truly exponential), the sequence would behave like a random walk and the conjecture could be studied probabilistically. The conjectured intermediate complexity of factoring places the Euclid–Mullin sequence in a regime where it has enough algebraic structure to support a proof (it is not random), yet resists brute computation—precisely the regime addressed by the structural approach in the body of this paper.

A.2 Analogies and Context

Mullin’s Conjecture has no known applications: if proved tomorrow, no other theorem in number theory would follow from it. The value of the problem lies instead in what it is an instance of—and in the methods its resolution would require. The orbit-specificity barrier identified by this formalization appears, in recognizable form, across several active areas of mathematics.

Artin’s conjecture and the orbit-specificity gap.

The closest structural analogue to MC is Artin’s conjecture on primitive roots [18]: for any integer $a \neq -1$ that is not a perfect square, the group $(\mathbb{Z}/p\mathbb{Z})^\times$ is generated by a for infinitely many primes p . The parallel is precise:

- Artin asks whether the orbit of a fixed generator a under repeated multiplication fills $(\mathbb{Z}/p\mathbb{Z})^\times$.
- Mullin asks whether the orbit of 2 under multiplication by successive EM primes in $(\mathbb{Z}/q\mathbb{Z})^\times$ hits the single element -1 .

Both conjectures are blocked by the same fundamental obstacle: transferring averaged equidistribution results (which hold for most moduli or most generators) to one specific deterministic orbit. Hooley [9] proved Artin’s conjecture conditional on GRH for Dedekind zeta functions of Kummer extensions—because GRH provides the uniformity across individual characters needed to control a single orbit. In our setting, this uniformity is exactly what CCSB demands.

The analogy is not merely structural. The Kummer extensions $\mathbb{Q}(\zeta_\ell, a^{1/\ell})$ that appear in Hooley’s proof are the same extensions that arise in the EffectiveKummerEscape approach to SubgroupEscape (Section 5). The elementary PRE lemma (Theorem 2.16) sidesteps this Chebotarev machinery entirely for the algebraic component, but the dynamical component—does the walk hit -1 , not merely generate the full group?—remains exactly the orbit-specificity gap that GRH closes for Artin and that no known tool closes for Mullin.

Multiplicative walks on finite groups.

The walk reformulation (Section 2) places MC in the framework of random walks on finite groups, studied systematically by Diaconis and others since the 1980s [16]. The Diaconis–Shahshahani upper bound lemma [15] shows that a random walk on a finite group G driven by i.i.d. multipliers from a conjugation-invariant distribution mixes in $O(\log |G|)$ steps, with mixing measured by character sums.

The EM walk has the same algebraic structure—a multiplicative walk on the cyclic group $(\mathbb{Z}/q\mathbb{Z})^\times$ —but violates every assumption of the classical mixing theory. The multipliers are deterministic, not random; they are not identically distributed; and most critically, the multiplier at step n depends on the walk position at step n , creating exactly the position-multiplier correlation that the Diaconis–Shahshahani framework assumes away. CCSB is the precise derandomization of the mixing-time bound: it asks that the Fourier coefficients of the walk’s occupation measure tend to zero for every non-trivial character.

Smallest prime factor distribution.

The SieveTransfer hypothesis (Section 5) connects MC to the distribution of the smallest prime factor function $P^-(n) = \min\{p : p \mid n\}$, studied by Alladi [11], Hildebrand [12], and others. For generic integers in an arithmetic progression $n \equiv a \pmod{q}$, the distribution of $P^-(n)$ is controlled by the Dickman function and CRT-based equidistribution results. MC asks whether

this equidistribution transfers from generic integers to the specific subsequence $\{P_2(n) + 1\}_{n \geq 0}$. The same orbit-specificity transfer problem arises for Mersenne numbers $2^p - 1$, Fibonacci numbers, and polynomial iterates $f^{\text{on}}(a)$ in arithmetic dynamics.

The marginal/joint barrier and Sarnak’s conjecture.

The formalization identifies a precise meta-obstacle (Section 5): marginal equidistribution of the multiplier residues is provable (the EM primes are equidistributed in residue classes, by Dirichlet’s theorem); what MC requires is joint equidistribution of the pair (walk position, multiplier), conditioned on the walk’s history. This barrier is an instance of a broader phenomenon. Sarnak’s conjecture [17] asserts that the Möbius function $\mu(n)$ is orthogonal to every bounded deterministic sequence: $\frac{1}{N} \sum_{n \leq N} \mu(n) a_n \rightarrow 0$. CCSB is a Möbius-orthogonality-type statement for the EM sequence, placing MC squarely within the Sarnak program’s conceptual framework, even though the EM sequence falls outside the technical scope of existing results (which require zero topological entropy).

Greedy sieves and orbit-hitting.

MC is the simplest nontrivial instance of a broader question: does a greedy, deterministic prime-selection process eventually cover all primes? The Cox–van der Poorten result [13] shows that this determinism is fragile: choosing the largest factor instead provably misses primes. More generally, MC belongs to the family of orbit-hitting problems in arithmetic dynamics: given a map T on a space X and a target set $S \subset X$, does the orbit eventually enter S ? Unlike Artin (where the map $x \mapsto ax$ is the same at every step) or Collatz (where the map depends only on the current state), the EM map varies at each step, determined by the factorization of a number that depends on the entire orbit history. This is the accumulator coupling of Section 1: the running product $P_2(n)$ is a cumulative digest of the full orbit, and the walk–multiplier framework makes it precise. The formalization shows it is the sole source of difficulty: once the coupling is controlled (via CCSB or CME), MC follows by machine-checked deduction.

A.3 Additional Sieve and Spectral Routes

This appendix collects the sieve and spectral-energy routes to MC that complement the three principal reductions (DH, CCSB, BV) presented in the body. All reduction arrows are machine-verified; the sole open content in each route is the orbit-specificity transfer.

Arithmetic Large Sieve Route

Theorem A.1 ([✓ arith_ls_chain_mc](#)). $\text{ArithLS} + \text{ArithLSImpliesMMCSB} \implies \text{MC}$.

The arithmetic large sieve gives character sum bounds for Dirichlet characters (a known result, not in Mathlib). The transfer $\text{ArithLSImpliesMMCSB}$ is open and is in fact a dead end: universal coefficient bounds cannot distinguish equidistributed walks from clumped walks.

Analytic Large Sieve Route

The most developed route connects the analytic large sieve to MC via Gauss sum inversion.

Property A.2 ([AnalyticLargeSieve \(ALS\)](#)). For well-separated points $\{\alpha_r\} \subset \mathbb{R}/\mathbb{Z}$ with

$\min_{r \neq s} \|\alpha_r - \alpha_s\| \geq \delta$:

$$\sum_r \left\| \sum_{n < N} a_n e(n\alpha_r) \right\|^2 \leq (N - 1 + \delta^{-1}) \sum_{n < N} \|a_n\|^2.$$

Theorem A.3 ([✓ weak_als_from_card_bound](#)). A weak version with constant $N \cdot (\delta^{-1} + 1)$ is proved (the optimal constant is $N - 1 + \delta^{-1}$, but the difference is immaterial since MMCSB requires only $o(N)$).

The key bridge is [Gauss sum inversion](#): a Gauss sum $\tau(\chi) = \sum_a \chi(a) e(a/p)$ intertwines multiplication and addition on $\mathbb{Z}/p\mathbb{Z}$, converting character sums to exponential sums.

Theorem A.4 ([✓ char_sum_to_exp_sum](#)). For a non-trivial character $\chi \bmod p$ prime: $\sum_n f(n) \chi(n) = \tau^{-1} \sum_{b=1}^{p-1} \chi^{-1}(b) \sum_n f(n) \psi(bn)$.

The GaussConductorTransfer composes eight internal lemmas (all proved, §56–§62) into the bridge from ALS to the prime arithmetic large sieve:

Theorem A.5 ([✓ als_implies_prime_arith_ls](#)). $\text{AnalyticLargeSieve} \implies \text{PrimeArithLS}$.

Theorem A.6 ([✓ als_prime_arith_ls_chain_mc](#)). $\text{ALS} + \text{PrimeArithLSImpliesMMCSB} \implies \text{MC}$.

A deeper decomposition of ALS itself is architected in [IKCh7.lean](#): the Hilbert chain

$$\text{HilbertInequality} + \text{CscPartialFraction} \xrightarrow{\text{open}} \text{CscBilinearBound} \xrightarrow{\text{open}} \text{GramOffDiagBilinearBound} \xrightarrow{\text{proved}} \text{ALS}$$

([✓ hilbert_chain_als](#)). [HilbertInequality](#) is the Montgomery–Vaughan generalized Hilbert inequality (IK Lemma 7.8); [CscPartialFraction](#) is the Mittag-Leffler expansion $\pi / \sin(\pi\theta) = 1/\theta + \sum_{n \geq 1} (-1)^n (1/(\theta+n) + 1/(\theta-n))$, derivable from Mathlib’s [cot_series_rep](#) via $\csc x = \cot(x/2) - \cot x$. The infrastructure for CscPartialFraction (the identity $\pi \cot(\pi\theta/2) - \pi \cot(\pi\theta) = \pi / \sin(\pi\theta)$ and both Mathlib cot-series specializations) is proved; the final [Has-Sum](#) assembly is complete (proved via [mittag_leffler_csc_proved](#)). Two bridges ([HilbertCscBilinearBridge](#), [CscBilinearImpliesGramOffDiag](#)) and the terminal arrow ([gram_offdiag_bilinear_implies_als](#)) complete the chain.

The remaining open content is PrimeArithLSImpliesMMCSB: transferring prime-modulus arithmetic large sieve bounds to multi-modular character sum bounds for the specific EM orbit.

The Spectral Energy Route

Instead of individual character sums, this route examines the total energy of the walk occupation measure $V_N(a) = |\{n < N : P_{2,q}(n) = a\}|$.

Theorem A.7 ([✓ walk_energy_parseval](#)). $\sum_\chi \|S_\chi(N)\|^2 = (q-1) \sum_{a \in (\mathbb{Z}/q\mathbb{Z})^\times} V_N(a)^2$ (Parseval).

The excess energy $E(N) = \sum_{\chi \neq 1} \|S_\chi(N)\|^2$. If $E(N) = o(N^2)$, then every non-trivial character sum is individually $o(N)$, which is CCSB.

Property A.8 ([SubquadraticVisitEnergy \(SVE\)](#)). For every missing prime q and $\varepsilon > 0$, there exists N_0 such that for $N \geq N_0$: $E(N) \leq \varepsilon N^2$.

Theorem A.9 ([sve_implies_mmcsb](#)). $\text{SVE} \implies \text{MMCSB} \implies \text{MC}$.

SVE bounds the total energy $\sum_{\chi \neq 1} \|S_\chi(N)\|^2 = o(N^2)$; since each non-trivial character sum is a non-negative summand, every individual $\|S_\chi(N)\|^2 = o(N^2)$, giving CCSB for each prime individually and MMCSB jointly.

Van der Corput and higher-order decorrelation.

The van der Corput inequality bounds $|\sum z_n|$ via autocorrelations:

Theorem A.10 ([van_der_corput_bound](#)). $\|\sum_{n \leq N} z_n\|^2 \leq \frac{N+H}{H+1} (N + 2 \sum_{h=1}^H |\text{Re} \sum_{n \leq N-h} z_n \overline{z_{n+h}}|)$.

For the EM walk, lag- h autocorrelations involve h -step multiplier products. At $h = 1$, the autocorrelation equals the multiplier character sum (Theorem 3.23), so VdC with a single shift gives only $O(N)$. Higher lags may decorrelate:

Property A.11 ([HigherOrderDecorrelation \(HOD\)](#)). For every missing prime q , non-trivial χ , and $\varepsilon > 0$: there exists H_0 such that for $H \geq H_0$, N_0 such that for $N \geq N_0$ and all $1 \leq h \leq H$: $\|R_h(N)\| \leq \varepsilon N$.

Theorem A.12 ([hod_implies_ccsb](#)). $\text{HOD} \implies \text{CCSB} \implies \text{MC}$.

Conditional multiplier equidistribution.

Property A.13 ([ConditionalMultiplierEquidist \(CME\)](#)). For every missing prime q , non-trivial χ , $\varepsilon > 0$, N_0 such that for $N \geq N_0$ and every $c \in (\mathbb{Z}/q\mathbb{Z})^\times$: $\|\sum_{\substack{n \leq N \\ w(n)=c}} \chi(m(n))\| \leq \varepsilon N$.

Theorem A.14 ([cme_implies_dec](#)). $\text{CME} \implies \text{DecorrelationHypothesis}$.

Theorem A.15 ([cme_implies_ccsb](#)). $\text{CME} \implies \text{CCSB}$.

Proof sketch. The walk telescoping identity gives $\sum \chi(w(n)) = \sum \chi(w(n))\chi(m(n)) - (\chi(w(N)) - \chi(w(0)))$. The product sum decomposes by fiber: $\sum \chi(w(n))\chi(m(n)) = \sum_a \chi(a) \cdot \sum_{w(n)=a} \chi(m(n))$. CME bounds each fiber sum by $\varepsilon' N$; the triangle inequality sums over at most $|(\mathbb{Z}/q\mathbb{Z})^\times|$ fibers; the boundary term $\chi(w(N)) - \chi(w(0))$ has norm ≤ 2 and is absorbed for large N . $\square$

This is the key reduction that bypasses PED, BRE, and the $d \geq 3$ barrier. The proof works for all character orders because it uses only the fiber decomposition and telescoping—no block rotation estimate is needed.

Theorem A.16 ([cme_implies_mc](#)). $\text{CME} \implies \text{MC}$.

Proof. Compose [cme_implies_ccsb](#) with [complex_ccsb_mc](#). $\square$

Theorem A.17 ([cme_chain_mc](#)). $\text{CME} + \text{PEDImpliesCSB} \implies \text{MC}$.

This older route through the $\text{Dec} \rightarrow \text{PED} \rightarrow \text{CCSB}$ chain is superseded by the direct $\text{CME} \rightarrow \text{CCSB}$ reduction above, which requires no additional hypotheses.

The Complete Hypothesis Hierarchy

$$\text{PED} < \text{Dec} < \text{CME} \xrightarrow{\text{proved}} \text{CCSB} \approx \text{HOD} \approx \text{SVE},$$

where “ $<$ ” means strictly weaker (proved implication, known not to reverse) and “ $\approx$ ” means equivalent. $\text{HOD} \Leftrightarrow \text{CCSB}$ via van der Corput; $\text{SVE} \Leftrightarrow \text{CCSB}$ via Parseval; $\text{CME} \Rightarrow \text{CCSB}$ via telescoping + fiber decomposition (Theorem A.15).

Every hypothesis implies MC. The PED route has an open BRE bridge for $d \geq 3$ characters, but this is now bypassed: the direct $\text{CME} \rightarrow \text{CCSB}$ arrow is proved for all character orders. CME is the sharpest sufficient condition—the weakest hypothesis known to imply MC.

A.4 Dead Ends Catalogue

137 potential approaches were explored and found to be dead ends (catalogued in full in [Meta/DeadEnds.lean](#), with 14 formal Lean witnesses). Each elimination is informative: it narrows the space of viable strategies. Eight dead ends have weak-MC revival score ≥ 2 , meaning they contribute to weaker forms of the conjecture (see §5). A representative selection, grouped by the type of obstruction:

Dead end	Why it fails
Ensemble-to-orbit transfer:	tool applies to generic sequences, not the specific EM orbit
BV for EM subsequence	BV applies to all primes in APs, not to a greedy subsequence.
Furstenberg / ergodic theory	Standard ergodic methods assume classical multiplicativity; the EM sequence is recursive and non-multiplicative.
Diaconis–Shahshahani lemma	Requires i.i.d. random steps; inapplicable to the deterministic EM walk.
Independence / linearity violated: tool requires additive or independent structure the walk lacks	
Large sieve for partial products	The large sieve handles linear sums, not multiplicative walks.
Abel summation	Converts multiplier decorrelation to walk-sum bounds, but the summation weights amplify rather than cancel.
Self-avoidance $\Rightarrow$ CCSB	Self-avoidance (no repeated $\mathbb{Z}$ positions) is invisible to characters, which see only residues.
Wrong algebraic structure: the group or decomposition has no room for the desired bound	
Non-abelian / representation	$(\mathbb{Z}/q\mathbb{Z})^\times$ is cyclic; all irreps are 1-d characters. No higher-dimensional structure to exploit.
CRT product group	Reformulating on $\prod_{q \leq Q} (\mathbb{Z}/q\mathbb{Z})^\times$ makes the problem harder: the product group is exponentially large.
NoLongRuns + PED $\Rightarrow$ BRE ($d \geq 3$)	Variable block lengths align adversarially with character phases.
DPED $\Rightarrow$ CCSB ($d \geq 3$)	Alternating ω, ω^2 rotations satisfy DPED yet produce $\Theta(N)$ walk sums. All PED-to-CCSB intermediates ruled out.
Reduces to single-modulus CCSB: no genuine simplification	

Dead end	Why it fails
Multi-modular approaches	All variants (BV + threshold, CRT, death coupling) collapse to single-modulus CCSB.
Death set coupling across moduli	Death sets $\{m : \text{minFac}(m) \equiv -c^{-1}\}$ vary per step; no uniform coupling bound exists.
Spectral gap (deterministic walk)	Spectral gap theory applies to probability measures on groups (convergence of random sampling); the EM walk is a single deterministic path.
Information-theoretic bounds	Category error: entropy/mutual-information tools assume a random variable; the EM sequence is deterministic with zero entropy.
Multiplier-to-walk transfer: multiplier cancellation does not force walk cancellation	
MultCancelToWalkCancel	Multiplier energy $o(K)$ does not force walk energy $o(K)$. Counterexample: alternating multipliers $\{2, 3\} \bmod 5$ with $\chi(2) = i$ give multiplier cancellation yet $ W_K = \Theta(K)$. Compatible with all EM structural properties (squarefree accumulator, coprime cascade, CRT invariance). Dead End #117.
Cofactor identity	The “+1 shift” identity $w(n)+1 = m(n) \cdot \text{cofZ}(n)$ (Theorem 3.24) factorizes one correlated variable into two; the cofactor remains a joint quantity, so the Marginal/Joint Barrier persists. Dead End #115.
SelfCorrectingDrift	The Lyapunov function $L(N) = \sum_c (V_c(N) - N/(q-1))^2$ telescopes to $L(N) = 2R(N) + O(N)$, so SCD ($R(N) = o(N^2)$) is algebraically equivalent to SVE ($L(N) = o(N^2)$). SCD provides no new proof leverage; it is a dynamical reformulation, not a reduction. Dead End #120.
SMSB + SE per-class escape	SecondMomentSquaredBound gives a global bad-set density bound; combining with SubgroupEscape to get per-class escape requires BadSetEquidistribution (bad steps don’t concentrate in one class). BSE is a consequence of CME: going from marginal to conditional bad-density requires joint (position, badness) independence, which IS the orbit-specificity gap. Pigeonhole gives “some class” but not the specific class -1 needed for MC. Dead End #121.
Dobrushin coefficient	The Dobrushin coefficient of ergodicity equals 1 for every step of the deterministic EM walk (each transition is a point mass; disjoint support gives overlap = 0). Dobrushin’s theorem requires $\sum(1 - \alpha) = \infty$, but $\sum 0 = 0$. The weaker MUB hypothesis is genuinely weaker than CME (CME $\Rightarrow$ MUB proved) but useless without Dobrushin. Dead End #131.
L-function factorization	The factorization $L(s, \chi) = L_{\text{EM}}(s, \chi) \cdot L_{\text{non-EM}}(s, \chi)$ is circular: proving $L_{\text{non-EM}} \neq 0$ on $\Re(s) = 1$ requires knowing which primes are non-EM, i.e., requires MC itself. Dead End #132.

Dead end	Why it fails
Self-similar functional equation	The tail identity gives $L_{\text{EM}} = \text{head} + L_{\text{EM,from } P(M)}$, but the tail orbit starts from $P(M) \neq 2$, so it is not a scalar multiple of L_{EM} . The Lapidus–van Frankenhuysen fractal zeta framework requires scaling ratios $\zeta(s) = \sum r_j^s \zeta(s) + f(s)$; the EM self-similarity does not match this form. Dead End #133.
No Tauberian lever	The super-exponential growth of $P(n)$ makes $\sum \text{prod}(n)^{-s}$ converge for all $\Re(s) > 0$ (proved: accum_reciprocal_summable). The resulting Dirichlet series is entire in $\Re(s) > 0$ — no pole at $s = 1$ to exploit via Tauberian methods. The standard PNT argument uses the pole of $\zeta(s)$; L_{EM} has no such pole because EM primes are too sparse. Dead End #134.
Konyagin–Shparlinski	Character sums $\sum \chi(g^n)$ cancel when $\text{ord}(g)$ is large (Konyagin–Shparlinski [22]). The EM walk position $P_2(n) \bmod q = 2 \cdot \text{em}_2(1) \cdots \text{em}_2(n) \bmod q$ looks like g^n when $g = 2$, and 2 has large order mod most primes (Artin’s conjecture). Fatal gap: the walk is $2 \cdot \prod \text{em}_2(k)$, not 2^n . The varying multipliers destroy the geometric progression structure that makes the Konyagin–Shparlinski bound work. Dead End #136.
AEP absorption	AccumulatorEquidistPropagation (AEP) is heuristically false at all fixed primes q . At $q = 3$, death ($P_n(k) \equiv -1 \bmod 3$) deterministically produces $\text{em}_n(k) = 3$ (genSeq_eq_three_of_genProd_mod3), sending the orbit to the absorbing state $0 \bmod 3$ forever. The nonzero-class densities $F_k(a)$ decay exponentially; the correct equidistribution limit $r/(r^2-1)$ is never reached for $k \geq 1$. CRTPropagationStep inherits the falsity. The backward dynamics chain $\text{ETA} + \text{DCTA} + \text{SRE} \Rightarrow \text{PRSD}$ is vacuously true. Dead End #138.

A.5 Methodology: Human–AI Collaboration

This work was produced through a sustained collaboration between a human author and an AI system (Claude, Anthropic) across 150+ sessions. The human author directed the mathematical strategy—proof architecture, dead-end identification, and editorial control—while the AI system handled Lean 4 formalization, Mathlib API search, literature scouting, and exploration of candidate proof strategies.

The interaction was organized at scale via an agent swarm: a multi-agent system built on the Claude Agent SDK. The swarm comprises seven specialized agents, each with its own system prompt, tool access, and model:

- A coordinator that reads the current proof state, selects the most promising action, dispatches specialists, and updates shared state files.
- A formalizer that writes and compiles Lean code in rapid iteration cycles.
- A literature scout that searches papers and Mathlib for relevant results.
- Four attack vector specialists focused on analytic, algebraic, combinatorial, and information-theoretic approaches.
- A paper writer that maintains this document (that is nevertheless often revised and maintained by the author).
- A code stylist whose goal is to improve the code quality and trying to enforce Mathlib conventions.

Agent prompts are self-evolving: after each session the coordinator updates them to record dead ends, new Mathlib discoveries, and shifted priorities. This prevents agents from rediscovering settled territory. All agent state (progress, strategy log, findings) is stored as git-tracked markdown, making the exploration history fully reproducible.

The division of labor between human and AI was sharp:

- Human: mathematical direction, proof strategy, identification of dead ends, evaluation of intermediate results, architectural decisions on the reduction hierarchy, and editorial control over the final formalization and paper.
- AI (Claude): Lean 4 formalization using Mathlib, Mathlib API search, literature scouting, exploration of candidate proof strategies, and drafting of this paper.

The human author guided the proof effort across 150+ sessions, suggesting attack vectors (algebraic, analytic, combinatorial, sieve-theoretic), identifying when an approach had reached a dead end, and pushing toward the sharpest possible reductions. The AI agents wrote all Lean code, searched Mathlib for relevant lemmas, explored dozens of proof strategies to completion or refutation, and maintained the evolving paper.

The swarm is optimized for formalization and reduction, not mathematical discovery. The next breakthrough, if it comes, will probably be a human insight about the structure of minFac on EM products—not something an agent finds by systematic search.

A.6 Glossary of Definitions and Hypotheses

The table below collects every named definition, hypothesis, and key theorem introduced in this paper, with abbreviations and the section where each is defined.

Abbr.	Name	Meaning	Ref.
Core sequence and walk			
—	Walk / Multiplier	$P_{2,q}(n) = P_2(n) \bmod q$; $m_{2,q}(n) = em_2(n+1) \bmod q$	Def. 2.3
MC	MullinConjecture	Every prime appears in the Euclid–Mullin sequence	Conj. 1.1
ANT	—	Standard analytic number theory (Dirichlet + sieve $\Rightarrow$ MFRE); assumed, not formalized	§4.1
MFRE	MinFacResidueEquidist	minFac residues equidist. mod q among q -rough sqfree ints; MFRE $\Rightarrow$ PE (proved)	§4.1
Algebraic hypotheses (§2–§2)			
SE	SubgroupEscape	No proper subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$ contains all multipliers	Def. 2.14
HH	HittingHypothesis	The walk reaches -1 cofinally: $\forall N, \exists n \geq N, q \mid P_2(n) + 1$	Def. 2.21
DH	DynamicalHitting	$SE(q) \Rightarrow HH(q)$ for every missing prime q	Def. 2.21
SHH	SingleHitHypothesis	$MC(< q) + SE(q) + q \text{ missing} \Rightarrow \exists n \geq N_0 \text{ with } q \mid P_2(n) + 1$	Def. 2.19
PRE	PrimeResidueEscape	Every proper subgroup of $(\mathbb{Z}/p\mathbb{Z})^\times$ is escaped by some odd prime $< p$	Thm. 2.16
PRE_ℓ	PowerResidueEscape	Multipliers escape the index- ℓ subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$	§2
—	ThresholdHitting	DH restricted to primes $q \geq B$	§3.5
Character-analytic hypotheses (§3)			
CCSB	ComplexCharSumBound	Walk char sums $S_\chi(N) = o(N)$ for all non-trivial χ	§3
MMCSB	MultiModularCSB	CCSB simultaneously for all primes q in a range	§3
ALS	AnalyticLargeSieve	Large sieve inequality adapted to EM walk	§3

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Abbr.	Name	Meaning	Ref.
—	HilbertInequality	Montgomery–Vaughan generalized Hilbert inequality (IK Lemma 7.8)	App. A.3
—	CscPartialFraction	Mittag-Leffler expansion of $\pi \csc(\pi\theta)$ (proved)	App. A.3
PED	PositiveEscapeDensity	Positive density of n with $m_{2,q}(n) \notin H$, for every proper H	§3
—	DecorrelationHypothesis	$m_{2,q}(n)$ and $m_{2,q}(n+1)$ are asymptotically independent	§3
BRE	BlockRotationEstimate	Cancellation in block sums of characters applied to walk	§3
SVE	SubquadraticVisitEnergy	$\sum_a \{n \leq N : P_{2,q}(n) = a\} ^2 = o(N^2/(q-1))$	§3
SCD	SelfCorrectingDrift	$R(N) = o(N^2)$; $\equiv$ SVE via Lyapunov telescope (Dead End #120)	§4
TWD	TailWindowDecorrelation	Avg. cross term between windows vanishes; TWD $\Rightarrow$ CCSB $\Rightarrow$ MC	§4.7
HOD	HigherOrderDecorrelation	Higher-order correlation bounds for walk increments	§3
CME	ConditionalMultiplierEquidist	Conditional equidist. of multipliers given walk state; implies CCSB (proved)	§3
—	EMDirichlet	EM primes equidistributed in residue classes; alias for Dec	App. A.3
—	EMDImpliesCME	Open: unconditional equidist. $\Rightarrow$ conditional equidist.	App. A.3
WM	WeakMullin	$\sum_{q \in M} 1/q < \infty$; MC $\Rightarrow$ WM $\Rightarrow$ RD (proved)	§6.6
RD	ReciprocalDivergence	$\sum_k 1/\text{em}_2(k) = \infty$; WM $\Rightarrow$ RD (proved)	§6.6
EMBV	EM Bombieri–Vinogradov	Multi-modular char sum bound (alias for MMCSB); EMBV $\Rightarrow$ MC	§6.6
—	JointSVE	Joint subquadratic visit energy; JointSVE $\Rightarrow$ MMCSB (proved)	§6.6
—	PopulationEquidist	minFac equidist mod q in shifted squarefree population (provable by sieve)	§6.6
—	PopulationTransfer	PE $\Rightarrow$ EMDirichlet (open = unbiased sampling); PE+PT $\Rightarrow$ MC	§6.6
—	ExcursionIndependence	Excursion char sums approximately independent; EIP + BME $\Rightarrow$ MC	App. A.3
Ensemble framework (§4)			
DSL	DeterministicStabilityLemma	Ensemble char energy concentrates over sqfree starting points; DSL $\Rightarrow$ MC	§4
—	DSLHitting	Weaker: walk hits -1 for positive density of starting points; PE+DSLHitting $\Rightarrow$ MC	§4
—	genProd / genSeq	Generalized EM from arbitrary starting point n ; $\text{genProd}(2, k) = P_2(k)$	§4
—	EnsembleConcentration	Char energy density for bad starting points $\rightarrow 0$ as $X \rightarrow \infty$	§4
—	CharSumVarianceBound	Ensemble average of char energy $\leq C \cdot K$ for all $X \geq X_0$	§4
—	FiberEnergyBound	$\sum_a \ F(a, N)\ ^2 \leq \varepsilon N^2$; FEB $\Rightarrow$ CME (proved)	§3.4
—	MinFacResidueEquidist	minFac residues equidist for q -rough sqfree ints; MFRE $\Rightarrow$ PE (proved)	§4
—	JointStepEquidist	Joint distribution of consecutive genSeq equidist over sqfree n (open)	§4
—	MultCancelToWalkCancel	Mult. char cancel $\Rightarrow$ walk char cancel; $\equiv$ CCSB (Dead End #117)	§4
—	WeylHittingBridge	Walk char cancel $\Rightarrow$ cofinal hitting; proved	§4

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Abbr.	Name	Meaning	Ref.
—	Cofactor Identity	$w(n)+1 = m(n) \cdot \text{cofZ}(n)$; proved but doesn't escape correlation	§3.5
Function field analog (§6.7)			
FF-MC	FFMullinConjecture	Every monic irreducible appears in FF-EM sequence	§6.7
—	FFDynamicalHitting	Cofinal $Q \mid F(n)+1$ for every monic irred. Q ; FFDH $\Rightarrow$ FF-MC (proved)	§6.7
—	SelectionBiasNeutral	Min-degree selection preserves character cancellation (open; = Dead End #90)	§6.7
—	FFSubgroupEscape	SE for $p \geq (d-1)^2+1$ via Weil contradiction (proved)	§6.7
SDDS framework			
—	SDDS	Abstract sieve-defined dynamical system $s_{n+1} = s_n \cdot \Phi(s_n + 1)$	App. A.3
—	NoAlgebraicObstruction	Multiplier subgroup generates full unit group; SE $\Rightarrow$ NAO (proved)	App. A.3
—	StrongSME	Walk hits every unit cofinally; StrongSME $\Rightarrow$ MC (proved)	App. A.3
Named theorems			
—	Confinement	If SE fails, the walk is confined to a proper coset	Thm. 2.13
—	Walk-Divisibility Bridge	$P_{2,q}(n) = -1 \Leftrightarrow q \mid P_2(n) + 1$	Thm. 2.5
—	One-prime gap	$\text{MC}(<q) + \text{cofinal hit} \Rightarrow \text{MC}(q)$	Thm. 2.29
—	Squarefree Accumulator	$P_2(n)$ is squarefree (product of distinct primes); proved	§6.6
—	QR Obstruction	SE failure has density $O(2^{-k})$ by CRT + quadratic reciprocity	§2
—	Gauss sum inversion	Character sums $\Leftrightarrow$ exponential sums via Gauss sums	§3

A.7 The Lean Formalization

Having developed the mathematical reduction and identified the structural obstacles, we describe the Lean 4 formalization that certifies these results.

Codebase Structure

The formalization uses Lean 4 with Mathlib v4.29.0-rc1 across 148 files totaling $\sim 78,000$ lines. The dependency chain is linear, with three leaf modules:

File	Content	Lines
Core sequence and walk		
Core/Euclid.lean	Constructive Euclid's theorem	422
Core/Defs.lean	<code>seq</code> , <code>prod</code> , <code>aux</code> , identities	527
Core/Conjectures.lean	MC, Conjecture A (FALSE), HH	490
Core/DWH.lean	DivisorWalkHypothesis (leaf)	547
Core/ResidueWalk.lean	WalkCoverage, residue walk, concrete MC	603
Group theory and algebraic framework		
Group/Core.lean	walkZ, multZ, confinement, SE	422
Group/Escape.lean	Escape lemmas, 8-element argument	634
Group/SEInstances.lean	29 concrete SE instances ($q \leq 157$)	364

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File	Content	Lines
Group/Pumping.lean	Gordon sequenceability [6] (leaf)	343
Group/QR.lean	QR obstruction ($\leq 1.6\%$) (leaf)	684
Group/CRT.lean	CRT multiplier invariance, walk recurrence	160
Group/DepartureGraph.lean	Departure graph, infinite recurrence, safe prime lattice	641
Group/RotorRouter.lean	Scheduled walk coverage (standalone)	455
Group/Bridge.lean	EMPR + SE $\Rightarrow$ MC bridge	87
Equidistribution and character sums		
Equidist/Preamble.lean	PE $\Rightarrow$ MC, bootstrapping	234
Equidist/Sieve.lean	Sieve, WHP $\Leftrightarrow$ HH, forbidden multiplier, Möbius death	755
Equidist/SelfAvoidance.lean	Self-avoidance, periodicity	450
Equidist/CharPRE.lean	Character non-vanishing, PRE $\Leftrightarrow$ SE	807
Equidist/Bootstrap.lean	Inductive bootstrap, DH $\Rightarrow$ MC, first passage	694
Equidist/Threshold.lean	Sieve gap, one-prime gap, cofinal pair avoidance	321
Equidist/CofinalOrbit.lean	Cofinal orbit expansion, quotient walk, cofinal escape	667
Equidist/OrbitAnalysis.lean	TailSE, sieve reduction, oracle barrier, selectability	778
Equidist/Fourier[B].lean	Character sums, Fourier bridge	1257
Equidist/SelfCorrecting.lean	Decorrelation, BRE, telescoping, kernel	1163
Equidist/SieveTransfer.lean	Sieve transfer, coprimality, neg-inv involution	615
Equidist/WalkDecomposition.lean	Quadratic/general walk sum decomposition, escape decorrelation	863
CME decomposition and spectral analysis		
CME/Variants.lean	CME weaker variants (CME_d, CME_avg, CME_subseq, CME_target)	96
CME/Decomposition.lean	CME = A + B decomposition, EMD $\Rightarrow$ CME	355
LargeSieve/Basic.lean	BV, ALS, ArithLS, MMCSB, Farey, GCT chains	855
LargeSieve/Structural.lean	Coprimality, divisibility, BV decomposition, sieve bridge	945
LargeSieve/Harmonic.lean	Parseval, Gauss sums, DFT, kernel	899
LargeSieve/Analytic.lean	Gauss inversion, WeakALS, GCT chain	1149
LargeSieve/PrimeArithLS.lean	Prime arithmetic large sieve, walk as partial product, dead ends	561
LargeSieve/Spectral.lean	Walk energy, SVE, Weyl criterion, HOD, VdC	1071
CME/Reduction.lean	CME definition, CME $\Rightarrow$ CCSB $\Rightarrow$ MC, FEB, hierarchy	627
LargeSieve/WalkAnalysis.lean	VCB, energy increments, fiber decomposition, transition matrix	1006
SDDS and independence		
SDDS/Dynamics.lean	Abstract SDDS framework, factoring rules	195
SDDS/Bridge.lean	SDDS $\Leftrightarrow$ EM bridge, orbit/walk/mult correspondence	375
SDDS/Reduction.lean	StrongSME $\Rightarrow$ MC via sieve-map equidist.	126
Transfer/Excursion.lean	Walk excursions, char. sum independence	461
Transfer/CRTFiber.lean	SE $\Rightarrow$ NAO, CRT independence, death channel	255
Population and weak variants		
Population/WeakMullin.lean	Missing primes, WM $\Rightarrow$ RD, EMBV $\Rightarrow$ MC	362
Population/AvoidanceTube.lean	Tube ratio, confined energy, rogue characters, SVE $\Rightarrow$ coverage	516
Population/InfiniteM.lean	Shielding lemma, factor dichotomy, spectral conspiracy	303
Population/SpectralConspiracy.lean	$\neg$ WM $\Rightarrow$ infinite missing, guardian exhaustion, tube squeeze	306
Population/WeakErgodicity.lean	Squarefree accumulator, PE+PT+EMDImpliesCME	160
Population/Proof.lean	MinFacResidueEquidist $\Rightarrow$ PE (analytic)	187
Population/ReciprocalSum.lean	Reciprocal sum divergence, PSD chain, positive density route	1260
Population/TransferStrategy.lean	PE + DSL $\Rightarrow$ MC, ensemble concentration	273
Population/Tauberian.lean	One-sided Tauberian, L-series bounds, prime power stripping	557

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File	Content	Lines
Population/AbelChain.lean	Abel summation chain: $(\log p)/p \rightarrow 1/p$ infrastructure	641
Population/AlladiDensity.lean	Alladi chain scaffolding, WPNT+DSL $\Rightarrow$ MC	267
Ensemble framework (new)		
Ensemble/Structure.lean	Generalized EM: genProd, genSeq, injectivity	269
Ensemble/EM.lean	Generalized walk/multiplier, bridge to standard EM	114
Ensemble/CRT.lean	CRT equidistribution, death density bridge, absorption mechanism, CRT blindness, Dead End #137	1101
Ensemble/BackwardDynamics.lean	Backward dynamics: joint counts, transition probs, ETA $\Rightarrow$ CRTPropagationStep	577
Ensemble/Decorrelation.lean	CRT-based decorrelation, Markov density bounds	431
Ensemble/FirstMoment[B].lean	First moment, parity, SMLB chain, $k=0/k=1/k=2$	1488
Ensemble/BagArithmetic.lean	CRT, sieve density, divergence hierarchy	
	Prime factor bag structure: ω , residue diversity, class partition, cofactor, landscape	224
Ensemble/PT.lean	Variance bounds, joint equidist, CRT chain, SD $\Rightarrow$ cancellation	1094
Ensemble/WeylChain.lean	Generalized MC, Weyl hitting bridge, cancel_weyl_implies_mc	834
Ensemble/CRTFreedom.lean	SCRTI + SRE $\Rightarrow$ AEP, CRT bootstrap	631
Ensemble/BaseCase.lean	JSE base case, tower rule, uniform witness bounds	377
Ensemble/FiberAutonomy.lean	CRT fiber determines multiplier dynamics, walk read-out	255
Ensemble/TwoPointEnsemble.lean	Ensemble two-point: ratio escape counting, first moment, partition argument, PRED $\Rightarrow$ positive density	566
Ensemble/MixedEnsemble.lean	PSCD $\Rightarrow$ almost-all mixed hitting, PEAP chain, sieve product, WeakFMCD (unconditional), sqfreeCount $\geq X/4$, pigeonhole	1958
DSL and master reduction (new)		
Reduction/DSLInfra.lean	Energy evolution, fiber analysis, FEB $\Rightarrow$ CME	627
Reduction/ShiftedDensity.lean	Sieve density $g(r)$, algebraic properties	251
Reduction/VisitEquidist.lean	ExcessEnergy = visit deviation, SVE $\leftrightarrow$ VE	187
Reduction/SelfCorrecting.lean	Lyapunov identity, SCD $\equiv$ SVE, Dead End #120	624
Reduction/WeakRecurrence.lean	LinearDrift $\Rightarrow$ MC, VisitGrowth $\Rightarrow$ HH $\Rightarrow$ MC (threshold-free)	425
Reduction/SMSB.lean	Per-class bad set analysis, Dead End #121	459
Reduction/Master.lean	All reduction chains assembled, DSL $\Rightarrow$ MC	210
Reduction/DSLVariance.lean	Population variance route to DSL, second moment	365
Transfer/CRTPointwise.lean	OCE = CME, all routes to MC via CRT	260
Transfer/Substitution.lean	SP = CME, fiber return visits, 4-route witness	290
Reduction/TailIdentity.lean	Tail energy identities, 4th moment, Chebyshev bridge	433
Reduction/TailWindow.lean	Window decomposition, variance identity, TWD $\Rightarrow$ MC	477
Transfer/IntegerDioph.lean	T -iteration $T(n)=(n-1) \cdot \text{minFac}(n)+1$, orbit, cofactor	340
Transfer/SieveConstraint.lean	Prime support S_n , sieve exclusion, $P(n)+1 \equiv 1 \pmod{S_n}$	261
Adelic and profinite framework (new)		
Adelic/Equidist.lean	Adelic decomposition: CME $\Leftrightarrow$ MWI + MME, Fourier bridge, CPD	656
Adelic/Profinite.lean	Profinite equidistribution, UPE $\Rightarrow$ CCSB $\Rightarrow$ MC, Weyl criterion	182
Adelic/UniformConductor.lean	UCE hypothesis, conductor counting, UCE+DSL $\Rightarrow$ MC	301
Adelic/ProfiniteGeneration.lean	Primes generate $(\mathbb{Z}/N\mathbb{Z})^\times$; MC $\Rightarrow$ full generation	176
FunctionField/Analog.lean	FF-EM over $\mathbb{F}_p[t]$, Galois/monodromy, cyclotomic obstruction, Dead Ends #127, #129	952
Advanced/Diamonds.lean	p -adic geometric perspective, Hecke obstruction, Dead End #128	340

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File	Content	Lines
Advanced/MarkovSieve.lean	Doebelin/sieve collapse certificates, Artin analogy, Dead Ends confirmed	538
Advanced/Dobrushin.lean	Dobrushin coefficient = 1, MUB $\neq$ CME, stopping time, Dead End #131	704
Advanced/LFunction.lean	L_{EM} Dirichlet series, self-similar decomposition, Dead Ends #132–#134	448
Advanced/VanishingNoise[B,C].lean	Spectral gap, stochastic MC, path existence, sparse contraction, phase transition	2104
Advanced/VanishingNoiseVariant[B–D].lean	ε -walk with <code>secondMinFac</code> , variant MC under UFD-Strong, stochastic two-point MC, routes to UFDStrong, faithful char escape, unconditional $q=3$, NFCE infrastructure, NFCE(5) reduction, intersection kernel dichotomy, quotient character lift	2805
Advanced/RandomTwoPointMC[B].lean	Self-consistent random two-point MC: Fourier bridge identity, ChiAtMultiplicativity, death/survival partition, path sum decomposition, Fourier counting identity, TCA + PathSurvival $\Rightarrow$ RandomTwoPointMC (proved)	1579
Advanced/EpsilonWalk.lean	Ensemble ε -walk: sigma-walk equidist, death fiber density, SigmaCRTPropagationStep (open), 8-clause landscape	435
Advanced/EpsilonRandomMC.lean	Mixed $(1-\varepsilon) \cdot \text{minFac} + \varepsilon \cdot \text{random}$ variant MC, reachable sets, coset impossibility, factor confinement	1967
Advanced/InterpolationMC.lean	Positive probability capture, block coverage, TreeSieveDecay, orbit melting, TSD-Hitting(3), TSD $\rightarrow$ MixedMC composition, coset ambiguity gap, perpetual primality multi-exclusion ($q=5, 11, 17, 23$), reachable set monotonicity	1798
Probability/TransitionKernel.lean	Epsilon-mixed walk step PMF, weight lower bounds	267
Probability/PathMeasure.lean	Reachable Finsets, branching distinctness, factor residues	333
Probability/GeometricCapture.lean	Block capture weight, geometric decay, TSD $\rightarrow$ PositiveProbCapture	378
Advanced/RandomFactorMC.lean	Pure random-factor variant MC, purification lemma, MC $\Rightarrow$ PureRandomMC	375
Advanced/IteratedProductCoverage.lean	Cauchy–Davenport iterated product coverage: card growth, diameter $ G -1$, safe prime instantiation	420
Advanced/FactorDiversity.lean	Ensemble factor diversity: genFactorSet, spectral contraction from ≥ 2 distinct residues, vanishing	346
Advanced/DiverseStepsToCapture.lean	Factor diversity $\rightarrow$ reachable set growth, DiversityImpliesReachable, TSD bridge	281
Advanced/StochasticEM.lean	Stochastic EM process: StochasticMC, phase transition at $\varepsilon = 0$, geometric capture	347
Advanced/DenseCapture.lean	Topological density of capturing selection sequences: captureSet open (unconditional), Baire category $\Rightarrow$ comeager full capture, DenseCaptureHypothesis (open)	301
FunctionField/Bootstrap.lean	FF-MC $\Leftarrow$ FFDynamicalHitting, competitor exhaustion, injectivity	438
FunctionField/SubgroupEscape.lean	SE for large p via Weil contradiction, threshold $(d-1)^2+1$	552
FunctionField/CyclicWalkCoverage.lean	Weyl criterion for walks on finite groups, $\mathbb{Z}/4\mathbb{Z}$ counterexample	549
FunctionField/MultiplierCCSB.lean	4 routes to FF-MC, population/fiber/selection hierarchy	763
FunctionField/WeakMC.lean	Degree escape, pool partition, capture exhaustion (Dead End #129 revival)	471

continued on next page

File	Content	Lines
FunctionField/FactorTree.lean	FF factor tree: mixed selection, coprimality cascade, injectivity, start not capturable	455
FunctionField/PopulationEquidist.lean	Weil $\Rightarrow$ population diversity, degree growth $\rightarrow \infty$, weak first moment	370
FunctionField/StochasticMC.lean	FF stochastic MC, phase transition ($\epsilon = 0$ critical), pigeonhole capture	431
FunctionField/Finiteness.lean	FFFiniteIrreduciblesPerDegree PROVED (unconditional), coefficient injection	188
FunctionField/NecklaceFormula.lean	Necklace identity $\sum_{d n} d \pi_p(d) = p^n$ PROVED (Galois theory)	446
FunctionField/FFCharacterSums.lean	Character cancellation, PNT-in-APs, FCD (all unconditional)	235
FunctionField/FFSieve.lean	Sieve chain: SPV $\rightarrow$ PSCD $\rightarrow$ almost-all GenMixedMC (zero hypotheses)	327
FunctionField/Master.lean	Theorems A/B/C/D assembly: factor tree + stochastic MC + sieve chain	306
FunctionField/AutonomousMap.lean	Φ_3 criterion: -1 unreachable for $p \equiv 2 \pmod{3}$, orbit dynamics	225
GaussEM/GaussEMDefs.lean	Gaussian EM over $\mathbb{Z}[i]$: sequence, irreducibility, coprimality	290
GaussEM/GaussWalkStructure.lean	Gaussian walk: split/inert classification, norm kernel, orbit barrier	331
GaussEM/GaussConfinement.lean	Integer walk confined to real axis, norm closure, Dead End #135	347
Cross-check infrastructure		
Meta/Blueprint.lean	LeanArchitect [24] annotations for 20 key theorems	229
Meta/Registry.lean	Content-addressed registry, @[publish]/@[open_point] annotations	189
Meta/DeadEnds.lean	Dead end registry: 134 documented, 13 witnessed, 8 revivable	220
Iwaniec–Kowalski infrastructure		
IK/Ch1.lean	Iwaniec–Kowalski [14] Ch. 1: arithmetic functions	434
IK/Ch2.lean	Iwaniec–Kowalski Ch. 2: summation formulas	270
IK/Ch3.lean	Iwaniec–Kowalski Ch. 3: combinatorial sieve	557
IK/Ch4.lean	Iwaniec–Kowalski Ch. 4: summation formulas	598
IK/Ch5.lean	Iwaniec–Kowalski Ch. 5: Kloosterman sums	877
IK/Ch7Foundations.lean	Iwaniec–Kowalski Ch. 7: foundations, norm bounds	328
IK/Ch7AdditiveLS.lean	Iwaniec–Kowalski Ch. 7: additive large sieve	989
IK/Ch7MultiplicativeLS.lean	Iwaniec–Kowalski Ch. 7: multiplicative large sieve	462
IK/Ch7SieveApplications.lean	Iwaniec–Kowalski Ch. 7: sieve applications (prime case)	949
IK/Ch7SieveCRT.lean	Iwaniec–Kowalski Ch. 7: Linnik, CRT general square-free	609
IK/Ch7Hilbert[B].lean	Iwaniec–Kowalski Ch. 7: Hilbert rescaling, Mittag-Leffler, Gram	1282
IK/Ch7CesaroChain.lean	Iwaniec–Kowalski Ch. 7: Cesàro convergence, circular ALS chain	1040

A.7.1 Axiom Usage: What’s Constructive

The core definitions (`seq`, `prod`, `aux`) and their basic properties (`seq_isPrime`, `seq_injective`) are fully constructive: they use only `propext` and `Quot.sound` (no `Classical.choice`, no `Decidable` instances beyond $\mathbb{N}$). Euclid’s theorem itself (`Core/Euclid.lean`) is constructive.

Classical reasoning enters at the reduction level:

- The $\text{HH} \Rightarrow \text{MC}$ proof uses well-founded induction (strong induction on $\mathbb{N}$), which in Lean 4 is constructive but relies on `Classical.choice` for the cofinal-implies-hit argument.

- Character theory (orthogonality, Fourier inversion) is inherently classical via [open Classical](#).
- All open hypotheses are stated as `def ...: Prop`, never as `sorry`'d theorems. The type-checker guarantees that no proof obligation is silently assumed.

A.7.2 Mathlib Dependencies

The formalization draws on several Mathlib libraries:

- Group theory: [Subgroup](#), [QuotientGroup](#), [orderOf](#), cyclic group structure, maximal subgroups ([Subgroup.IsCoatom](#)).
- Number theory: [Nat.minFac](#), Legendre symbols, quadratic reciprocity, [ZMod](#), Dirichlet characters, Gauss sums.
- Character theory: [DirichletCharacter.Orthogonality](#), roots of unity in algebraically closed fields, character bounds, [MulChar.sum_eq_zero_of_ne_one](#).
- Analysis: [norm_sum_le](#), complex norms, [IsOfFinOrder.norm_eq_one](#), Fourier analysis on $\mathbb{Z}/n\mathbb{Z}$ ([ZMod.dft](#), discrete Fourier transform).
- Dirichlet's theorem: [Nat.infinite_setOf_prime_and_eq_mod](#) (primes in arithmetic progressions, via L -series).
- Harmonic analysis: Parseval's theorem for finite abelian groups, trigonometric exponentials, geometric series identities.

A.7.3 ANT Formalization Status

The sole unformalized input to the master reduction is ANT (§4.1): the standard analytic number theory implying MFRE. The chain is

$$\underbrace{\text{Dirichlet} + \text{Sieve}}_{\text{ANT}} \xrightarrow[\text{open}]{\text{W-I}} \text{WeightedPNTinAP} \xrightarrow[\checkmark]{\text{strip}} \text{PrimeLogSum} \xrightarrow[\checkmark]{\text{Abel}} \text{PrimesEquidist} \xrightarrow[\text{open}]{\text{Alladi}} \text{MFRE} \xrightarrow[\checkmark]{} \text{PE}$$

where “ $\checkmark$ ” denotes machine-verified arrows and “open” denotes unformalized steps. Three files provide the ANT infrastructure: [Population/Tauberian.lean](#) (557 lines, zero `sorry`) proves the Tauberian and stripping steps; [Population/AbelChain.lean](#) (641 lines, zero `sorry`) proves the full discrete Abel summation chain converting log-weighted prime sums into reciprocal-weighted sums; and [Population/AlladiDensity.lean](#) (267 lines, zero `sorry`) defines intermediate hypotheses for the Alladi chain and proves the master reduction $\checkmark \text{wpnt_dsl_implies_mc}$: $\text{WeightedPNTinAP} + \text{Alladi} + \text{CRT} + \text{DSL} \Rightarrow \text{MC}$.

Proved steps.

- One-sided Tauberian lemma ($\checkmark \text{one_sided_tauberian_upper}$): for nonnegative $b_n \geq 0$, $\sum_{n \leq N} b_n \leq N^\epsilon \cdot \sum_n b_n / n^\epsilon$. Elementary proof via $n^\epsilon \leq N^\epsilon$. This gives $O(N^\epsilon)$ partial-sum bounds from Dirichlet series behavior, without requiring the full Wiener–Ikehara theorem.
- L-series two-sided bounds ($\checkmark \text{residueClass_tsum_both_bounds}$): Mathlib's [PrimesInAP.lean](#) proves a lower bound on $\sum \Lambda(n) \mathbf{1}_{n \equiv a} / n^s$ near $s = 1$ via [LSeries_residueClass_lower_bound](#), but does not export the corresponding upper bound. We extract the identity $\text{tsum} = \text{Aux}(a, s) + (\varphi(q))^{-1} / (s-1)$ from Mathlib's proof and derive both bounds.
- Prime power stripping ($\checkmark \text{prime_power_stripping_proved}$): $\text{WeightedPNTinAP} \Rightarrow \text{PrimeLogSumEquidist}$. The von Mangoldt function Λ counts prime powers p^k with weight $\log p$; the contribution from $k \geq 2$ is uniformly bounded by $\sum_p 2 \log p / p^2 < \infty$. This 250-line proof strips the higher prime powers and converts the Λ -weighted sum into a $(\log p)/p$ sum over primes only.

- Abel summation chain ([✓ prime_log_to_reciprocal_proved](#), 641 lines): $\text{PrimeLogSumEquidist} \Rightarrow \text{PrimesEquidistributedInAP}$. Discrete Abel summation (summation by parts) converts the log-weighted prime sum $\sum (\log p)/p$ into the reciprocal-weighted sum $\sum 1/p$. The proof includes derivative identities $(\log \log t)' = 1/(t \log t)$ and $(-1/\log t)' = 1/(t(\log t)^2)$; integral evaluations via FTC; the sandwich bound $|\sum_{k=2}^{m-1} (\log(k+1) - \log k)/\log(k+1) - (\log \log m - \log \log 2)| \leq 1/(\log 2)^2$; and the discrete Fubini step (double-sum reindexing).
- Full composed chain ([✓ wpnt_implies_primes_eqidist_proved](#)): $\text{WeightedPNTinAP} \Rightarrow \text{PrimesEquidistributedInAP}$, composing prime power stripping with discrete Abel summation.
- Dirichlet’s theorem ([✓ dirichlet_primes_in_ap](#)): imported directly from Mathlib’s `Nat.infinite_setOf_prime_and_eq_mod`.

Remaining gaps.

Mathlib v4.29.0 already provides $\sim 92\%$ of the required infrastructure: qualitative Dirichlet (via L -functions and non-vanishing on $\text{Re}(s) \geq 1$), Abel summation ([AbelSummation.lean](#)), Chebyshev bounds, Dirichlet character orthogonality, and the von Mangoldt function. Two gaps remain:

1. Wiener–Ikehara bridge (sole blocker, ~ 300 – 500 lines): convert $\sum \Lambda(n) \mathbf{1}_{n \equiv a} / n^{1+\varepsilon} = (\varphi(q))^{-1}/\varepsilon + O(1)$ into the partial-sum asymptotic $\sum_{n \leq x} \Lambda(n) \mathbf{1}_{n \equiv a} / n = \log x / \varphi(q) + O(1)$. Because $\Lambda \geq 0$, a one-sided monotone argument suffices: the one-sided Tauberian lemma (proved above) already gives $O(N^\varepsilon)$ bounds; what remains is the asymptotic, which requires a quantitative version of the abelian Tauberian theorem. As of March 2026, no external Lean 4 project provides a usable Wiener–Ikehara theorem: the PNT+ project (Kontorovich–Tao et al.) includes W-I in its blueprint but has not implemented it in Lean, and the Gauss project (Math, Inc.) uses a different method (zero-free regions). If a W-I formalization lands in Mathlib, the gap closes immediately via the L-series identity above.
2. Alladi chain ($\sim 1,000$ – $1,400$ lines remaining): Alladi’s formula [11] for the density of $\{m : \text{minFac}(m) = p\}$ combined with CRT sieving gives $\text{PrimesEquidistInAP} \Rightarrow \text{MFRE}$. The reduction scaffolding is now formalized in [Population/AlladiDensity.lean](#) (267 lines, zero sorry), which decomposes the chain into two precisely stated open hypotheses: [PrimesEquidistImpliesRoughLPF](#) (Alladi density + weight independence, ~ 600 – 800 lines) and [RoughLPFImpliesMFRE](#) (CRT sieve transfer from generic q -rough integers to shifted squarefree integers, ~ 400 – 600 lines). The master reduction [✓ wpnt_dsl_implies_mc](#) composes these with the proved ANT chain and DSL to give MC. The sieve density infrastructure already exists ([Reduction/ShiftedDensity.lean](#)). The key observation is that the Alladi weight $h(p) = (1/p) \cdot \prod_{q < r < p} (1 - 1/r)$ depends on p only through $|p|$, not through $p \bmod q$, so weighted Dirichlet equidistribution transfers to minFac residues. No prior formalization of Alladi’s density formula or the Buchstab identity exists in any proof assistant.

We estimate $\sim 1,300$ – $2,300$ lines to close the full chain. The Abel summation step is complete ([Population/AbelChain.lean](#)), the Alladi chain scaffolding with master reduction is formalized ([Population/AlladiDensity.lean](#)), and the Wiener–Ikehara bridge and Alladi density content are the two remaining components. Formalizing ANT is feasible future work but does not affect the paper’s contribution, since the reduction $\text{DSL} \Rightarrow \text{MC}$ is the novel content.

Lines of Lean code	~68,500
Files	125
Theorems/lemmas	~2,000
Definitions	~800
<code>sorry</code> occurrences	0
Open hypotheses (stated as <code>def</code>)	~50
Mathlib version	v4.29.0-rc1

A.7.4 Summary of Formally Verified Results

The formalization contains ~78,000 lines across 148 files with zero `sorry`. The table below is a curated selection of the most important results; the full inventory is in the repository. Status markers: `[proved]` = machine-checked in Lean 4; `[open]` = open hypothesis (sole unproved content); Dead = false or blocked (formally witnessed). For declarations registered with the CA content-addressing framework [23], the L1 column shows the first 8 hex digits of the SHA-256 type hash (L0, name-based), providing a stable identifier independent of human naming.

Result	Status	Lean identifier	L1
Principal reductions to MC			
DSL $\Rightarrow$ MC (sole hypothesis, full chain)	<code>[proved]</code>	<code>full_chain_dsl</code>	7948b5ed
CME $\Rightarrow$ CCSB $\Rightarrow$ MC (sharpest)	<code>[proved]</code>	<code>cme_implies_mc</code>	0168917b
DH $\Rightarrow$ MC (cofinal hitting)	<code>[proved]</code>	<code>dynamical_hitting_implies_mullin</code>	b069682d
CCSB $\Rightarrow$ MC (char sums)	<code>[proved]</code>	<code>complex_csb_mc'</code>	2fb09cde
MMCSB $\Rightarrow$ MC (multi-modular)	<code>[proved]</code>	<code>mmcsb_implies_mc</code>	a4e5b5f2
VCB + PED $\Rightarrow$ MC	<code>[proved]</code>	<code>vcb_ped_implies_mc</code>	d24ad157
Inductive bootstrap and walk dynamics			
PrimeResidueEscape (elementary, no Burgess)	<code>[proved]</code>	<code>prime_residue_escape</code>	5f0da0ec
Single Hit Theorem (SHH $\Rightarrow$ MC)	<code>[proved]</code>	<code>single_hit_implies_mc</code>	ed2a8b99
Walk-divisibility bridge ($w = -1 \Leftrightarrow q \mid P+1$)	<code>[proved]</code>	<code>walkZ_eq_neg_one_iff</code>	19e01a7e
Confinement Theorem	<code>[proved]</code>	<code>confinement_forward</code>	e1f5ff43
Fourier bridge (char sums $\rightarrow$ hit count)	<code>[proved]</code>	<code>walk_hit_count_fourier_step</code>	695a2823
Telescoping identity & norm ≤ 2	<code>[proved]</code>	<code>walk_telescope_identity</code>	b716b929
Shift-one autocorrelation = conjugate mult sum	<code>[proved]</code>	<code>walk_shift_one_correlation</code>	1c240c93
$P_2(n)$ squarefree	<code>[proved]</code>	<code>prod_squarefree</code>	06e3904a
Character sum chain			

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Result	Status	Lean identifier	L1
$\text{CME} \Rightarrow \text{CCSB}$ (all character orders)	[proved]	<code>cme_implies_ccsb</code>	c761d4e9
$\text{CME} = \text{VCB} + \text{Dec}$ (decomposition)	[proved]	<code>cme_implies_vcb</code>	0112fe00
$\text{Dec} \Rightarrow \text{PED}$; $\text{NoLongRuns}(L) \Rightarrow \text{PED}$	[proved]	<code>decorrelation_implies_ped</code>	dbc9644c
$\text{FEB} \Rightarrow \text{CME}$ (fiber energy)	[proved]	<code>feb_implies_cme</code>	4299967b
$E_\chi(N) = N + 2 \sum \text{cross}$ (energy identity)	[proved]	<code>char_sum_energy_eq_N_plus_cross</code>	6abba0fb
Cofactor identity: $w + 1 = m \cdot \text{cof}$	[proved]	<code>shifted_walk_eq_mult_mul_cof</code>	79af39f7
Ensemble and DSL framework			
$\text{SD} \Rightarrow \text{CSV}(C=2) \Rightarrow \text{concentration} \Rightarrow \text{cancellation}$	[proved]	<code>sd_implies_cancellation</code>	4e941a85
WeylHittingBridge (char cancel $\Rightarrow$ hitting)	[proved]	<code>weyl_hitting_bridge_proved</code>	07796c68
$\text{LMG} \Rightarrow \text{PositiveDensityRSD}$ (1 hypothesis)	[proved]	<code>lmg_implies_positive_density_rsd</code>	dc23143a
$\text{genSeq}(n, k) \geq 3$ for $k \geq 1$ (parity)	[proved]	<code>genSeq_ge_three</code>	a453ba31
$\mathbb{E}[1/\text{genSeq}(\cdot, 0)] \geq 1/4$ (unconditional)	[proved]	<code>ensembleAvg_k0_ge_quarter</code>	9733c692
Death density bridge ($q \geq 3$)	[proved]	<code>ensembleAvg_ge_death_density</code>	2f5da145
Absorption: $\text{death} \Rightarrow \text{permanent}$ $0 \bmod q$	[proved]	<code>death_then_never_death_again</code>	e37a2056
$\text{ETA} + \text{SRE} \Rightarrow \text{PositiveDensityRSD}$	[proved]	<code>eta_sre_implies_prsd</code>	af923ac4
Spectral and large sieve routes			
$\text{SVE} \Leftrightarrow \text{VE} \Leftrightarrow \text{CCSB}$ (Parseval)	[proved]	<code>sve_iff_ve</code>	
Van der Corput bound	[proved]	<code>van_der_corput_bound</code>	3c0a4bd8
$\text{HOD} \Rightarrow \text{CCSB}$	[proved]	<code>hod_implies_ccsb</code>	82e08ba3
Weak ALS (with $N(\delta^{-1}+1)$ constant)	[proved]	<code>weak_als_from_card_bound</code>	f7a8cd3a
$\text{ALS} \Rightarrow \text{PrimeArithLS}$ (Gauss inversion, 8 lemmas)	[proved]	<code>als_implies_prime_arith_ls</code>	803082cc
MittagLefflerCsc (Mittag-Leffler for csc)	[proved]	<code>mittag_leffler_csc_proved</code>	3bfddc38
$\text{WPNT} \Rightarrow \text{PrimesEquidistInAP}$ (Abel chain)	[proved]	<code>wpnt_implies_primes_equidist_proved</code>	ae92835c

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Result	Status	Lean identifier	L1
Adelic and CRT framework			
$\text{CME} \Leftrightarrow \text{MWI} + \text{MME}$ (adelic decomposition)	[proved]	cme_iff_adelic	e3d49fb8
$\text{CRTFiber} + \text{MME} \Rightarrow \text{CME}$	[proved]	crt_fiber_mme_implies_cme	2fc3bae2
$\text{UPE} \Rightarrow \text{MC}$ (profinite)	[proved]	uniform_profinite_implies_mc	474313ac
Primes $< N$ generate $(\mathbb{Z}/N\mathbb{Z})^\times$	[proved]	primeUnitsBelow_generate	85c155d7
$\text{MC} \Rightarrow \text{EM}$ multipliers generate $(\mathbb{Z}/N\mathbb{Z})^\times$	[proved]	mc_implies_full_generation	7b29549c
Variant frameworks			
$\text{UFDStrong} \Rightarrow \text{VariantMC}$ (two-point)	[proved]	variant_mc_from_ufd_strong_proved	4b2fb306
FaithfulCharacterEscape (unconditional, all $q \geq 3$)	[proved]	faithful_character_escape	4bfec51b
$\text{UFDStrong}(3) / \text{StochasticTwo-PointMC}(3)$	[proved]	variant_mc_three_unconditional	79493410
NFCS for $ G $ with ≥ 2 prime factors	[proved]	nonFaithfulCharSeparation_of_two_prime_factors	ab2af646
Quotient char lift: $\bar{\chi}$ faithful on $G/\ker \chi$	[proved]	quotientChar_faithful	f9f9cadf
$\text{TCA} + \text{PathSurvival} \Rightarrow \text{RandomTwo-PointMC}$	[proved]	tca_path_survival_implies_random_mc_proved	3f4515d9
$\text{MixedDiversityWeak} + \text{MixedDiversity} \Rightarrow \text{MixedMC}$	[proved]	mixed_diversity_weak_implies_mixed_mc	b4e99dea
Mixed capture of $q = 3$ (unconditional)	[proved]	mixed_capture_three	e3b7121f
$\text{MixedHitting} + \text{MixedDiversity} \Rightarrow \text{MixedMC}$	[proved]	mixed_hitting_diversity_implies_mc	e71086df
Two-point $\hookrightarrow$ Mixed (embedding)	[proved]	embed_walk_agreement	dec2cc13
Hit $\Rightarrow$ Capture (walk surgery)	[proved]	hit_implies_capture'	68895c87
Hitting $\Leftrightarrow -1$ reachable (BFS)	[proved]	mixed_hitting_iff_neg_one_reachable	f6c253c7
Perpetual primality $\Rightarrow$ walk $\not\equiv 1 \pmod{3}$	[proved]	perpetual_prime_excludes_mod3_one	16b451bc
Factor branching grows R_{n+1}	[proved]	reachableAt_from_factor	b8330c76
Composite $P+1 \Rightarrow$ two positions in R_{n+1}	[proved]	reachable_composite_branch	12909295

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Result	Status	Lean identifier	L1
R_∞ not in any proper coset (PRE)	[proved]	reachableEver_not_in_coset	403b9151
Factor confinement from incomplete R_∞	[proved]	factor_confinement	6bc4a711
FactorEscape $\Rightarrow R_\infty$ full $\Rightarrow$ Mixed MC	[proved]	factor_escape_implies_mixed_mc_at	f0e544c3
PureRandomMC $\Leftrightarrow$ MixedMC (purification)	[proved]	pure_random_mc_iff_mixed_mc	9234ef21
MC $\Rightarrow$ PureRandomMC	[proved]	standard_mc_implies_pure_random	a1249f53
PSCD $\Rightarrow$ almost-all mixed hitting	[proved]	pscd_implies_almost_all_mixed_hitting	08f5e9ea
Trapped $\leq \Sigma$ confined (pigeonhole)	[proved]	trapped_le_sum_confined	b412fa7b
$0 \notin R_\infty$ for trapped coprime m	[proved]	zero_not_reachable_of_coprime_trapped	d59dfd35
PEAP + 3 bridges $\Rightarrow$ almost-all hitting	[proved]	peap_chain_implies_almost_all_mixed_hitting	c3eda394
$(\forall q \geq 5, \text{TSD}(q)) \Rightarrow \text{MixedMC}$	[proved]	tsd_all_implies_mixed_mc	d97b4938
Coset ambiguity: $q=3$ free, $q=5$ blocked	[proved]	coset_ambiguity_landscape	4cdfef34
Geometric capture: $(1-\delta)^K \rightarrow 0$	[proved]	product_failure_tendsto_zero	40ceab63
TSD $\Rightarrow$ PositiveProbCapture	[proved]	tsd_positive_capture	5266fc90
Perpetual primality $\Rightarrow$ eventually periodic mod q	[proved]	perpetual_prime_eventually_periodic	9c1903bf
Mod-5 orbit $\{1,2\}$ avoids -1 under perp. primality	[proved]	perpetual_prime_mod5_orbit	e3e54697
Reachable set monotone along walk prefixes	[proved]	reachableEver_mono_along_walk	aed1f58e
Function field analog			
FFDH $\Rightarrow$ FF-MC (competitor exhaustion)	[proved]	ff_dh_implies_ffmc	29679e3a
Weil $\Rightarrow$ FF-SE for $p \geq (d-1)^2+1$	[proved]	weil_implies_ff_se	90bbd755
$\mathbb{Z}/4\mathbb{Z}$: generation $\neq$ coverage (Dead End #130)	[proved]	alternating_walk_misses_two	4ddc83b6
FFLM structurally false (cyclotomic, Dead End #129)	[proved]	ff_cyclotomic_dead_end	8337b7dc

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Result	Status	Lean identifier	L1
FF mixed selection injective (no irred twice)	[proved]	ffMixedSel_injective	20c8914b
FF coprimality cascade (mixed walks)	[proved]	ffMixedWalkProd_coprime_succ	b14cdca2
FF start not capturable	[proved]	start_not_capturable	b91e4671
FF factor pool degree $\rightarrow \infty$	[proved]	ff_factor_pool_degree_grows	45cdf92d
FFFiniteIrreduciblesPerDegree (unconditional)	[proved]	ffFiniteIrreduciblesPerDegree_proved	17c25a13
FFDH alone $\Rightarrow$ FF-MC (finiteness eliminated)	[proved]	ff_dh_implies_ffmc_unconditional	ca639028
FFStochasticMC ($\varepsilon > 0 \Rightarrow$ capture)	[proved]	stochastic_mc_unconditional	677bbd33
FF phase transition (sharp at $\varepsilon = 0$)	[proved]	ff_phase_transition_unconditional	354d8033
Factor diversity $\Rightarrow$ spectral contraction	[proved]	factor_diversity_spectral_contraction	024381d7
Diverse steps $\Rightarrow$ char product vanishes	[proved]	diverse_steps_imply_vanishing	5ba9dece
TSD $\Rightarrow$ StochasticMC (integer side)	[proved]	stochastic_mc_of_tsd	89f5d828
Phase transition summary ($\varepsilon = 0$ critical)	[proved]	phase_transition_summary	028f0a20
Perpetual primality multi-exclusion ($q = 5, 11, 17, 23$)	[proved]	perpetual_primality_multi_exclusion	64d0302c
FF Φ_3 exclusion (-1 unreachable, $p \equiv 2 \pmod{3}$)	[proved]	ff_neg_one_unreachable	1efdbf06
Necklace identity ($\sum_{d n} d\pi_p(d) = p^n$)	[proved]	necklace_identity_proved	2d6542a5
FF almost-all GenMixedMC (zero hypotheses)	[proved]	ff_almost_all_unconditional	3b954837
Tube collapse ($\neg \text{WM} \Rightarrow$ tube ratio $\rightarrow 0$)	[proved]	tube_collapse	6230ae9a
Shielding lemma ($\text{em}_2(n+1) \notin M$)	[proved]	shielding_lemma	90208737
Rogue character exists (avoidance $\Rightarrow$ linear char growth)	[proved]	rogue_character_exists	37229a82
SVE contradicts avoidance	[proved]	sve_contradicts_avoidance	6043a2ab
$\neg \text{WM} \Rightarrow$ missing primes infinite	[proved]	not_wm_implies_missing_infinite	43a0b692

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Result	Status	Lean identifier	L1
Spectral conspiracy landscape (7-clause conjunction)	[proved]	spectral_conspiracy_landscape	2bdee217
Open hypotheses — live targets			
DeterministicStabilityLemma (PE $\Rightarrow$ CME)	[open]	DeterministicStabilityLemma	d5e0c044
DynamicalHitting (SE $\Rightarrow$ cofinal -1)	[open]	DynamicalHitting	e004d177
ComplexCharSumBound ($S_\chi(N) = o(N)$)	[open]	ComplexCharSumBound	d3bb3ee3
BVImpliesMMCSB (BV $\rightarrow$ orbit)	[open]	BVImpliesMMCSB	ca998ffc
SieveTransfer (population $\rightarrow$ orbit)	[open]	SieveTransfer	c8dd7777
StepDecorrelation (ensemble cross-terms)	[open]	StepDecorrelation	d1df5f92
LinearMeanGrowth ($\mathbb{E}[S_K] \geq \kappa K$)	[open]	LinearMeanGrowth	52e9cd90
EnsembleTransitionApprox (CRT blindness)	[open]	EnsembleTransitionApprox	9540d74e
NonFaithfulCharacterEscape ($q \geq 5$, two-point)	[open]	NonFaithfulCharacterEscape	
MixedDiversity (composite $P+1$ cofinally)	[open]	MixedDiversity	c9f2c638
MixedDiversityWeak (steering from diversity)	[open]	MixedDiversityWeak	a6f323bb
MixedHitting (-1 reachable under diversity)	[open]	MixedHitting	15b0a649
UFDStrongImpliesMixedMC (self-consistency bridge)	[open]	UFDStrongImpliesMixedMC	4708535a
FactorEscapeHypothesis (Euclid numbers escape confinement)	[open]	FactorEscapeHypothesis	
PSCD (population sieve confinement decay; follows from PEAP via WeakFMCD chain)	[proved]	weak_fmcd_chain_implies_pscd	fe852147
PEAPImpliesFCD (PEAP $\Rightarrow$ class divergence)	[proved]	peap_implies_fcd_proved	0de93587
SieveUpperBound (fundamental lemma of sieve; superseded by WeakFMCD)	[superseded]	SieveUpperBound	
SieveProductVanishing ($\sum 1/p$ div. $\Rightarrow \prod(1-1/p) \rightarrow 0$)	[proved]	sieve_product_vanishing_proved	cd6248d5
$\text{sqfreeCount} \geq X/4$ (unconditional)	[proved]	sqfreeCount_ge_quarter_real	0e4aaf02
WeakFMCD (unconditional CRT bound, constant 4)	[proved]	weak_fmcd_proved	
WeakFMCD + FCD $\Rightarrow$ PSCD (no open hypothesis)	[proved]	weak_fmcd_fcd_implies_pscd	

continued on next page

Result	Status	Lean identifier	L1
PEAP alone $\Rightarrow$ PSCD (FMCD eliminated)	[proved]	weak_fmcd_chain_implies_pscd	fe852147
PEAP alone $\Rightarrow$ a.a. mixed hitting	[proved]	weak_fmcd_chain_implies_almost_all	
Reachable $\Rightarrow$ positive prob. capture ($\forall \epsilon > 0$)	[proved]	reachable_implies_positive_prob_capture	
PEAP $\Rightarrow$ a.a. positive prob. capture	[proved]	almost_all_positive_prob_capture	
Block coverage (uniform depth for all units)	[proved]	block_coverage	
Regeneration $\Rightarrow$ cofinal hitting	[proved]	regeneration_implies_cofinal_hitting	
Regeneration $\Rightarrow$ iterated hitting	[proved]	regeneration_implies_iterated_hitting	
Regeneration (walk preserves goodness)	[open]	Regeneration	
FFDynamicalHitting (function field)	[open]	FFDynamicalHitting	
SelectionBiasNeutral (min-degree preserves cancel.)	[open]	SelectionBiasNeutral	
Dead ends — false or blocked (representative)			
MultCSBImpliesMMCSB (partial products don't cancel)	Dead	MultCSBImpliesMMCSB	
BRE for $d \geq 3$ (adversarial block alignment)	Dead	BlockRotationEstimate	
SCD $\equiv$ SVE (no new leverage, #120)	Dead	sve_implies_scd_above_threshold	
Dobrushin coeff = 1 (deterministic walks, #131)	Dead	dobrushin_coefficient_one	
L -function factorization circular (#132)	Dead	dead_end_132_witness	
No Tauberian lever for L_{EM} (#134)	Dead	tauberian_lever_absent_witness	
Cauchy–Davenport minOrder = 2 (#137)	vacuous: Dead	minOrder_units_zmod_eq_two	
AEP false: absorption drains nonzero classes (#138)	Dead	death_then_never_death_again	e37a2056
138 total dead ends, 15 formally witnessed, 8 revivable	[proved]	dead_end_registry	

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